SoFAME (Soil-Forage-Animal Management and Economics): Soil Organic Carbon in Grazing Decisions and Policy Incentives

Yining Wu[1,2,*], Dale Manning[1,2], Yao Zhang[3], Frank Lupi[4], Nathan DeLay[5], Dabit Bista[4]

[1]Department of Agricultural & Resource Economics, University of Tennessee
[2]Baker School of Public Policy, University of Tennessee
[3]Department of Soil and Crop Sciences, Colorado State University
[4]Department of Agricultural, Food, and Resource Economics, Michigan State University
[5]Department of Agricultural and Resource Economics, Colorado State University

[*]Corresponding author: ywu96@utk.edu

This version: October 25, 2026.
Click here for the latest version

Abstract. Economic models of soil organic carbon (SOC) on grazing land rely on soil responses fitted offline and miss the daily feedback among management, forage, animals, and soil. We develop SoFAME (Soil-Forage-Animal Management and Economics), an integrated cow-calf farm model in which a neural surrogate of a biogeochemical process-based model (MEMS) updates each patch's soil and forage daily. After ten years of recursive forecasts, the mean absolute error of the surrogate's SOC is 0.4 percent of the MEMS stock, and the surrogate simulates ten farm years in 80 to 85 seconds, against 3.1 hours with MEMS. We apply SoFAME to the profit maximization problem of a representative Michigan cow-calf farm. Five management histories over the past 100 years set the starting soil states, and for each we simulate 5,601 management systems, from no herd to 3.2 livestock units per hectare, each a combination of planned stocking rate, paddock count, grazing period, and feed-gap supplementation. Profit and SOC move in opposite directions up to the most profitable stocking rate, and both fall beyond it. Taking the most profitable management system as the farm's current management, we simulate three policies: an outcome payment per Mg C, and a practice payment and a calf-price premium for rotation. At prices of $44 per Mg C or more, within the observed range, an outcome payment against this baseline moves every farm to a management system that adds SOC at the least private cost. As the price rises, each farm lowers its stocking rate in steps, and at $102 to $120 per Mg C, still within the observed range, three of the five farms keep no herd. The SOC that a payment buys therefore depends on each farm's soil history and on the whole management system that the farm chooses without the payment.

Funding for this research was provided by the Foundation for Food and Agriculture Research (FFAR grant number: DSnew-0000000028) and The Noble Research Institute.

1. Introduction

Grazing land feeds livestock and stores soil organic carbon (SOC), and the way it is grazed changes both (Briske et al. 2023; Teague et al. 2016). Across 115 studies of improved grassland management and conversion to grassland, SOC rose by 0.54 Mg C per hectare per year on average (Conant, Paustian, and Elliott 2001). Programs now pay cow-calf producers to change how they graze. The Environmental Quality Incentives Program (EQIP) and the Conservation Stewardship Program (CSP) pay per hectare for prescribed grazing (USDA-NRCS 2024b,a), carbon protocols credit SOC gains against a modeled baseline (Verra 2025; Climate Action Reserve 2022), grazing labels set standards for qualifying beef (American Grassfed Association 2024; National Audubon Society 2023), and calves from verified programs sell at a premium (Zimmerman et al. 2012; Odde et al. 2019). These designs pay for a practice, a measured outcome, or a label. Whether a design changes what a producer does, and how much SOC it buys, depends on how the farm responds.

Economic analyses have represented the soil and forage process in two ways. The older approach writes it as one or two stocks governed by a few simple equations. In grazing models, forage grows logistically and the herd eats part of it (Noy-Meir 1975; Huffaker and Cooper 1995). Torell, Lyon, and Godfrey (1991) and Ritten et al. (2010) chose the annual stocking rate given a forage state that follows one equation, and Wang et al. (2018) followed two grasses and steered weight each day at constant stocking rates. In soil-capital models, soil is a stock with a shadow value (Burt 1981; McConnell 1983). In Berazneva et al. (2019), SOC changes each year by a residue input less a first-order loss, with two parameters fitted to equilibria of the RothC model, and an estimated yield function links SOC to profit. These models are fast and transparent, but a few parameters stand in for the plant and soil processes, the soil models treat a field as one uniform stock, and the grazing models carry no soil.

A second approach runs a process model offline and carries its output into the economic model. In the econometric-process approach, a biophysical model maps management to soil carbon, and an economic model chooses the management (Antle and Capalbo 2001). Antle et al. (2003) used carbon rates from the Century model to compare contracts per hectare and per tonne, and Antle et al. (2007) built supply curves of soil carbon for the central United States. Learned surrogates now emulate process models at low cost (Reichstein et al. 2019; Liu et al. 2024). Nguyen, Nong, and Paustian (2019) trained a neural network on output of the DayCent model to map fixed annual rates of nitrogen, irrigation, and residue removal to SOC after 30 years and to mean corn grain and stover yields, and used the network to maximize farm profit subject to SOC constraints. This approach brings process science into the decision, but it fixes the response before the decision problem is solved, at an annual step or for the whole horizon. On a grazing farm, the feedback among management, forage, animals, and soil works at the daily scale: the

herd moves to the forage it prefers, hay fills the daily gap between forage and demand, cow condition sets conception, and local grazing changes the soil of each patch. A fixed response averages these feedbacks out. Process models are also slow, so studies vary a few decisions and hold the others fixed, and grazing studies usually vary the stocking rate or the rotation. A cow-calf producer decides jointly on the stocking rate, the paddocks, the grazing period, and the share of the grazing-season feed gap to fill with supplemental hay, and the value of each choice depends on the others.

We ask whether a soil process model can run inside the daily decision loop of a farm cheaply enough to evaluate whole management systems and, if so, which decisions matter for profit and SOC and which payment designs change the system that a producer chooses.

We develop SoFAME (Soil-Forage-Animal Management and Economics), an integrated model of a cow-calf farm. In SoFAME, a recurrent neural surrogate of the Microbial Efficiency-Matrix Stabilization (MEMS) model, a biogeochemical process-based model of plant and soil carbon and nitrogen (Zhang et al. 2024; Santos et al. 2024), advances the soil and forage state of each of 12 patches every day. Each day, the farm model moves the herd by a calendar rule and by the forage that the animals prefer, and it fills a chosen share of the feed gap, the herd demand minus the grazeable forage, with supplemental hay on pasture. We call this share the feed-gap supplementation. The surrogate takes the local grazing density and the hay of each patch and returns the next soil and forage state, and the forage then sets the energy balance, body weight, and conception of each animal class. On 18 test simulations, each for a combination of site and start history held out from training, the mean absolute error of total SOC after ten years of recursive forecasts is 0.40 percent of the MEMS stock. In the farm loop, a ten-year farm simulation runs 130 to 140 times faster with the surrogate than with MEMS, because MEMS cannot save and resume its state and a daily loop must restart it.

We apply the model to a 60-hectare Michigan farm with five start histories, which come from a century of different grazing simulated in MEMS. For each history, we evaluate 5,601 management systems: every combination of 16 planned stocking rates from 0.2 to 3.2 livestock units (LSU) per hectare, 70 paddock designs, and 5 levels of feed-gap supplementation, plus one system without a herd. The planned stocking rate sets the size of the year-round herd, and a paddock design pairs the paddock count with the grazing period. The surrogate runs all 28,005 simulations in 340 CPU-hours, against about 88,000 CPU-hours with MEMS. We then add an outcome payment per Mg C, a practice payment per hectare, and a calf-price premium, each at observed rates.

The experiment yields five main findings. First, the start history, which the past 100 years of grazing set, determines the path that SOC follows without a herd, and management determines how far below that path the farm ends. The heavy continuous history gains SOC under every management system and the ungrazed history loses SOC under every

system with a herd, while in the other three histories the sign of the change depends on management. Second, the decisions interact: the best paddock design depends on the stocking rate, with continuous grazing earning the most at low stocking rates and rotation at high ones. Third, the least costly SOC gains come from changing the whole management system. For the first 150 Mg C on a 60-hectare farm, moving all decisions together gives up $14 to $50 of net present value per Mg C, depending on the start history, while lowering the stocking rate alone gives up as much in the low-pressure continuous history and 1.4 to 3.0 times as much in the other four histories. Fourth, the qualifying rule determines what a payment buys. A practice payment buys SOC only when the profit optimum of the farm does not already use the practice. On this farm, the profit optimum of every start history already rotates, so a payment for rotation alone buys no SOC, and a calf premium for rotation raises the stocking rate of three farms. An outcome payment against the farm's own baseline moves each farm at $14 to $44 per Mg C, and a practice payment with a stocking cap of 1.6 LSU per hectare moves each farm at $22 to $58 per hectare per year. Both designs move each farm along its profit and SOC frontier. Fifth, at high observed prices, an outcome payment ends beef production on the farm. As the price rises, each farm lowers its stocking rate in steps and keeps no herd from $102 to $160 per Mg C, depending on the start history. Three of the five farms reach this point within the observed range of $28 to $128 per Mg C, so part of the SOC that such a payment buys comes with calves that the farm no longer sells and that other farms may supply.

The paper contributes to three literatures. First, it adds a method to the bioeconomic modeling of soil (Antle and Capalbo 2001; Berazneva et al. 2019). A learned emulator of a process model replaces the fitted soil response and advances the daily soil state in step with the decisions, while herd movement, nutrition, and accounting stay in the economic model, where a researcher can change them. Earlier learned surrogates served a static optimization (Nguyen, Nong, and Paustian 2019) or a regional estimate (Liu et al. 2024), or emulated the farm model itself (Shang et al. 2024). Whole-farm simulators such as APEX (Gassman et al. 2010) and GrassGro (Moore, Donnelly, and Freer 1997) also advance soil, pasture, and livestock together each day, but with their own built-in soil and pasture models. Through its surrogate, SoFAME brings a detailed soil biogeochemical model into the daily loop and compares every management system of a grid under several payment designs. Second, it adds to grazing economics by treating management as a system of interacting decisions. Earlier studies varied the stocking rate (Torell, Lyon, and Godfrey 1991; Ritten et al. 2010) or compared rotation with continuous grazing (Briske et al. 2008; Wang et al. 2018). We show that the value of rotation depends on the stocking rate and that the least costly SOC comes from moving several decisions together. Third, it adds to the design of payments for ecosystem services (Engel, Pagiola, and Wunder 2008; Ferraro and Pattanayak 2006) by showing that additionality depends on where the private optimum

lies among the management systems. A payment for a practice that the optimum already uses becomes the windfall that Chabé-Ferret and Subervie (2013) measured for cover crops, and a premium on output can lower the SOC that it targets. As the offset literature predicts (Mason and Plantinga 2013), the baseline of an outcome payment determines who is paid.

We study one site, one weather path, and a ten-year horizon. The producer chooses among fixed management rules that do not respond to SOC, and the surrogate inherits the structure and the errors of MEMS. Section 2 states the producer's problem and the payment incentives, Section 3 describes the surrogate in the farm loop and the experiment, Section 4 reports its accuracy and speed, Section 5 reports the results, and Section 6 concludes.

2. The cow-calf producer's problem and payment incentives

We model the cow-calf producer's ten-year choice of a management system and add three payment designs to derive when a payment changes that choice and what the change buys.

2.1. The producer's problem

A cow-calf producer keeps beef cows and sells their calves at weaning weight. The farm has $A = 60$ hectares in $K = 12$ patches of $A_k = 5$ hectares, and years $t = 0, \ldots, 9$ run from 2026 to 2035. A start history is the state of the site at the end of 2025 after one of five grazing histories simulated in MEMS. We treat each start history as one farm and drop its index.

At the start, the producer chooses one management system $a = (\mathrm{SR}, N, \mathrm{gp}, \phi)$ from the grid of 5,601 systems in Section 3 and keeps it for ten years. The planned stocking rate SR is the year-round herd that the producer plans to keep, in LSU per hectare. A cow counts as 1 LSU, a heifer as 0.7, and a calf as 0.2, so with fixed herd shares of 0.50, 0.10, and 0.40, SR sets the cow target $0.50\,\mathrm{SR}\,A/0.65$. The paddock design pairs the paddock count N, where one paddock is continuous grazing, with the grazing period gp, the number of days that the herd stays in one paddock. The feed-gap supplementation $\phi \in [0, 1]$ is the share of the daily feed gap that the producer fills with supplemental hay on pasture. At SR = 0, the farm keeps no herd, so it has no paddock design, no hay, no sales, and no feed, herd, labor, or temporary-fence cost. It pays only for the permanent perimeter fence, which every management system pays. The model has no other use of the land, such as rent or a hay crop.

During the grazing season, from day of year 120 to 270, the daily feed gap is the herd demand $D_{t,d}$ minus the forage $F_{k_{t,d},t,d-1}$ above a residual of 20 grams of dry matter (DM) per square meter on the occupied patch $k_{t,d}$. Herd demand follows the current body weights, at 15 kilograms of DM per day for each demand-equivalent LSU, $L_{t,d} = D_{t,d}/15$.

The producer feeds $h_{t,d} = \phi \max\{0, D_{t,d} - F_{k_{t,d},t,d-1}\}$ of hay on pasture, the herd eats forage first and then hay, and unmet demand makes the animals lose weight. Outside the grazing season, the herd eats a fixed ration of hay and corn grain off pasture, the same in every management system.

A simulation runs one management system from one start history. A management system is complete when the herd in its simulation runs the ten years without reaching an animal weight floor, and the set $\mathcal{M}$ contains the complete systems.

The states are the soil and forage of each patch, $\mathbf{X}_t$, including its SOC, and the count and body weight of each animal class, $\mathbf{n}_t$. Farm SOC, SOC_t, is the mean of the patch values in Mg C per hectare. Given the weather W_t, the management system advances both states,

$$\mathbf{X}_{t+1} = G_N(\mathbf{X}_t, \mathbf{n}_t, a, W_t), \qquad \mathbf{n}_{t+1} = \mathcal{H}(\mathbf{X}_t, \mathbf{n}_t, a, W_t). \tag{1}$$

Inside G_N, the recurrent surrogate of MEMS advances each patch every day.

The producer values a management system by its ten-year net present value (NPV),

$$\mathrm{NPV}(a) = \sum_{t=0}^{9} \delta^t \pi_t(a) + \delta^9 p_{\mathrm{cull}} \sum_{g} n_g(a)\left[w_g(a) - w_g^5\right], \qquad \delta = \frac{1}{1.05}. \tag{2}$$

The first year is undiscounted, and the terminal herd value prices the weight of each cow cohort g at the end of 2035 above or below its weight at body condition score (BCS) 5, at the cull price p_{cull}. Annual profit is

$$\pi_t(a) = R_{\mathrm{calf},t} + R_{\mathrm{cull},t} - C_{\mathrm{feed},t} - C_{\mathrm{herd},t} - C_{\mathrm{min},t} - C_{\mathrm{labor},t}(N, \mathrm{gp}) - C_{\mathrm{fence}}(N) - C_{\mathrm{repl},t}. \tag{3}$$

Feed cost covers the hay on pasture and the off-season hay and corn, labor cost includes the scheduled moves, and the fence cost is an annual equivalent.

The profit optimum $a^0 = \arg\max_{a \in \mathcal{M}} \mathrm{NPV}(a)$ is the choice without a payment. Let $\Delta\mathrm{SOC}(a) = \mathrm{SOC}_{10}(a) - \mathrm{SOC}_0$ be the change in farm SOC from 1 January 2026 to 31 December 2035. The least-loss management system is the system in $\mathcal{M}$ with the largest ΔSOC. The profit and SOC frontier is the set of complete systems for which no other complete system has both a higher NPV and a larger ΔSOC, and it runs from the profit optimum to the least-loss management system (Figure 5). Each management system is a fixed rule that does not respond to SOC or the herd, and SOC left in the soil at the end of year 10 has no value in the objective.

2.2. How SOC reaches profit

SOC has no price in equation (2) and affects profit only through the forage, whose quantity and quality can rise with SOC in MEMS. More forage on the occupied patch narrows the

feed gap, so the producer buys less hay when $\phi > 0$ and C_{feed} falls. More and better forage also keeps the cows in condition, and because the BCS of a cow cohort on the calving date sets its conception rate, $\rho_g = \rho(\text{BCS}_g)$ with ρ increasing, better condition raises the next calf crop and R_{calf}. Feed-gap supplementation buys this condition with hay. Grazing and the excreta from forage and hay then change SOC and forage on each patch in later years.

2.3. The calendar rule

Under rotation ($N \geq 2$), the herd stays gp days in one paddock and then moves to the next in a fixed order, and within the open paddock it grazes the patch with the highest grazing value. On a grazing day,

$$\text{rest} = (N-1)\,\text{gp}, \qquad u_{k,t,d} = \mathbf{1}\{k = k_{t,d}\}\,\frac{L_{t,d}}{A_k}. \tag{4}$$

Each paddock rests $(N-1)$gp days between visits, and the occupied patch carries the whole herd at the local density $u_{k,t,d}$, in LSU per hectare, while the other patches carry none. More paddocks force the herd onto more of the patches and spread grazing across the farm. Under continuous grazing, every patch is open all season, and the grazing value alone selects the patch. Appendix B gives the selection rule, the herd demand, and the surrogate inputs.

2.4. Payment incentives

A payment adds to NPV the present value $P(a)$ of a transfer under a contract that covers 2026 to 2035. Profit is computed after the simulation, so a payment does not change the herd or SOC path of a simulation. A payment for SOC uses the annual change in farm SOC, $\Delta_t(a) = \text{SOC}_{t+1}(a) - \text{SOC}_t(a)$, times the 60 hectares.

The outcome payment pays p dollars per Mg C. In the main case, its baseline is the farm's own no-payment path:

$$P^{\text{out}}(a) = p\,S(a), \qquad S(a) = A \sum_{t=0}^{9} \delta^t \big[\Delta_t(a) - \Delta_t(a^0)\big]. \tag{5}$$

The payment is two-sided, and the profit optimum earns zero. This baseline follows VM0042 and the Soil Enrichment Protocol (Verra 2025; Climate Action Reserve 2022), with the no-payment path known exactly. Appendix B.13 defines two contrast cases against the start stock and a threshold variant.

The practice payment pays r dollars per hectare per year to a management system

that meets the qualifying rule $\mathcal{Q}$:

$$P^{\mathrm{pr}}(a) = r\,A\,\Gamma\,\mathbf{1}\{a \in \mathcal{Q}\}, \qquad \Gamma = \sum_{t=0}^{9} \delta^t = 8.11. \tag{6}$$

The qualifying rule is rotation, $N \geq 2$, or rotation with a stocking cap $\mathrm{SR} \leq \overline{\mathrm{SR}}$ of 2.0 or 1.6 LSU per hectare, modeled on the Prescribed Grazing standard (USDA-NRCS 2017). The calf premium, which the buyer of the calves pays, raises the calf price of a qualifying management system by the share q:

$$P^{\mathrm{prem}}(a) = q\,\overline{R}_{\mathrm{calf}}(a)\,\mathbf{1}\{a \in \mathcal{Q}\}, \qquad \overline{R}_{\mathrm{calf}}(a) = \sum_{t=0}^{9} \delta^t R_{\mathrm{calf},t}(a). \tag{7}$$

Its qualifying rule is rotation, rotation with a cap, or the outcome-verified rule $\Delta\mathrm{SOC}(a) \geq \Delta\mathrm{SOC}(a^0)$. The management system without a herd has one paddock and sells no calves, so only the outcome payment can pay it.

With a payment, the producer chooses

$$a^* = \arg\max_{a \in \mathcal{M}} \ \mathrm{NPV}(a) + P(a). \tag{8}$$

When a^0 is among the maximizers, the farm keeps a^0. The additional SOC is $A[\Delta\mathrm{SOC}(a^*) - \Delta\mathrm{SOC}(a^0)]$ in Mg C per farm, and the transfer $P(a^*)$ is the program cost. The windfall $P(a^0)$, the payment for no change, lies between 0 and $P(a^*)$. The private cost is $\mathrm{NPV}(a^0) - \mathrm{NPV}(a^*)$, and the switch level is the lowest payment level that moves the farm. For the outcome payment, we call it the switch price. The cost per additional Mg C is the transfer divided by the additional SOC.

2.5. Implications for payment design

The results of this subsection depend only on the NPV and SOC of the complete management systems, and Section 5.4 applies each of them to the experiment.

PROPOSITION 1 (A payment for a practice already in use). *If the profit optimum meets the qualifying rule, a practice payment at any rate leaves the farm at a^0, so it adds no SOC and the whole transfer is windfall.*

ARGUMENT. The payment adds the same amount to every qualifying management system, and a system outside $\mathcal{Q}$ receives nothing and has no more NPV than a^0. □

PROPOSITION 2 (Switch price of an outcome payment). *Against the farm's own baseline,*

the outcome payment moves the farm when p exceeds

$$\hat{p} = \min_{a \in \mathcal{M}:\, S(a) > 0} \frac{\mathrm{NPV}(a^0) - \mathrm{NPV}(a)}{S(a)}, \tag{9}$$

the least NPV given up per discounted Mg C gained. No complete management system then beats the new choice on both NPV and S.

ARGUMENT. Below $\hat{p}$, every management system with $S(a) > 0$ loses more NPV than it earns in payment, and a system with $S(a) \le 0$ has no more NPV than a^0 and earns no positive payment. A system that another system beats on both NPV and S cannot maximize $\mathrm{NPV} + pS$. □

As p rises above $\hat{p}$, the discounted SOC gain $S(a^*)$ of the chosen system never falls, because a higher price puts more weight on S in $\mathrm{NPV} + pS$. At a high enough price, the farm chooses the system with the largest $S(a)$, and when that system keeps no herd, the payment ends beef production on the farm.

PROPOSITION 3 (A premium for a practice already in use). *If the profit optimum qualifies and the calf premium moves the farm, the new management system earns more discounted calf revenue than a^0.*

ARGUMENT. The new management system qualifies, because a system outside $\mathcal{Q}$ receives no premium and has no more NPV than a^0, which earns a positive premium. The farm leaves a^0 only for a strictly higher objective, so the inequalities

$$\mathrm{NPV}(a^*) + q\overline{R}_{\mathrm{calf}}(a^*) > \mathrm{NPV}(a^0) + q\overline{R}_{\mathrm{calf}}(a^0) \quad \text{and} \quad \mathrm{NPV}(a^*) \le \mathrm{NPV}(a^0)$$

then give $\overline{R}_{\mathrm{calf}}(a^*) > \overline{R}_{\mathrm{calf}}(a^0)$. □

The premium thus rewards more calves, and it can lower SOC unless the rule caps the stocking rate.

PROPOSITION 4 (A stocking cap reaches the frontier). *Suppose that the capped rule excludes a^0, and let $a^{\mathcal{Q}}$ be the complete management system with the highest NPV under the cap. The capped practice payment then either leaves the farm at a^0 or moves it to $a^{\mathcal{Q}}$. If $a^{\mathcal{Q}}$ lies on the profit and SOC frontier, no system gains as much SOC at a lower private cost, and all designs that move the farm to $a^{\mathcal{Q}}$ differ only in their transfers.*

ARGUMENT. The payment is the same for every qualifying management system, so it selects the qualifying system with the highest NPV, and outside $\mathcal{Q}$ the best system is a^0. On the frontier, no system with at least the SOC of $a^{\mathcal{Q}}$ has a higher NPV. □

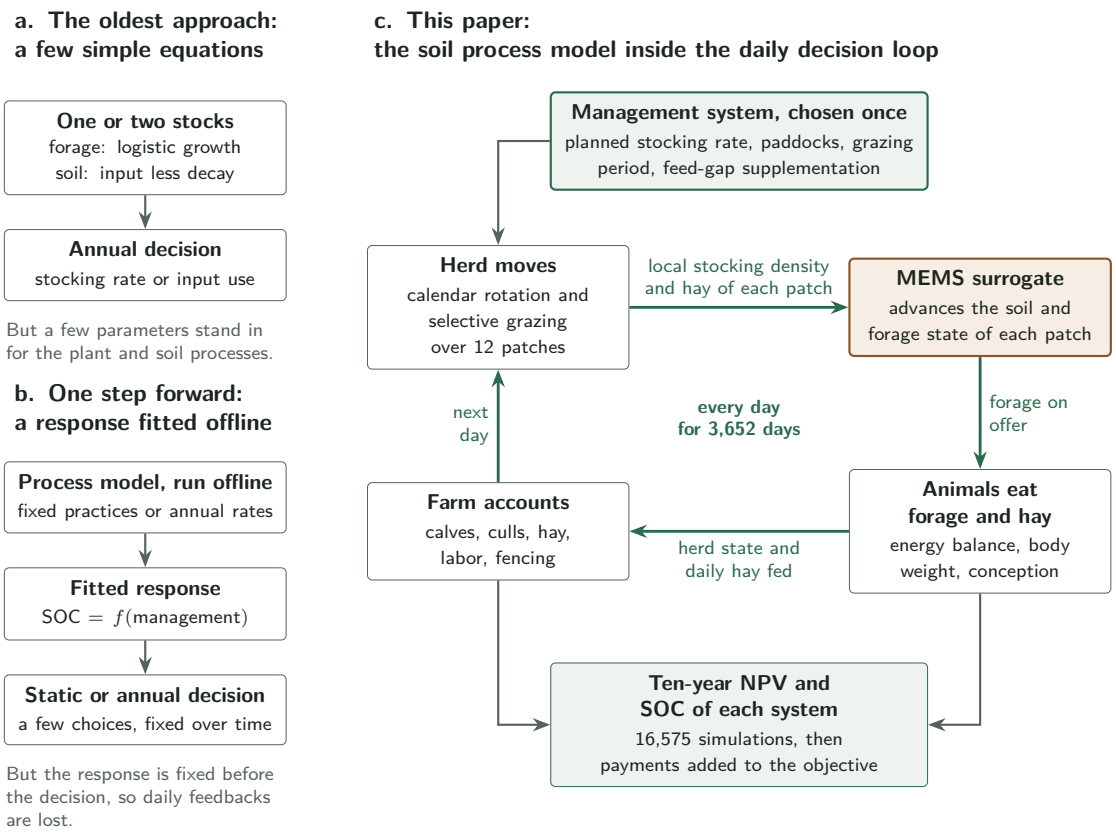


FIGURE 1. **Three ways to represent the soil process in an economic model.** (a) The oldest approach represents forage or soil as one or two stocks that follow a few simple equations, for example logistic forage growth or an SOC input less a first-order loss (Noy-Meir 1975; Ritten et al. 2010; Berazneva et al. 2019). (b) A later approach runs a process model offline for fixed practices or rates and carries the fitted response into the decision model (Antle et al. 2003; Nguyen, Nong, and Paustian 2019). (c) In this paper, a management system chosen at the start sets the rules, and the herd, the surrogate of the MEMS soil model, the animals, and the farm accounts exchange state every day for ten years.

3. The surrogate in the farm loop and the experiment

Earlier economic models represent the soil process in two ways, by a few equations or by a response fitted offline. In our model, the surrogate, the herd, and the accounts exchange state every day for ten years (Figure 1), and Table 1 compares this design with the earlier models.

3.1. Study site, weather, and start states

We apply the model to a single cow-calf site in Michigan, which isolates management effects from between-site variation. Its 16-layer soil profile comes from SSURGO (Soil

TABLE 1. **This paper relative to earlier models.** Each feature has a precedent in earlier work, but no earlier study combines them. "SOC as a decision state" records whether the decision problem carries SOC as a state, and "Daily feedback" records whether decisions and the simulated state exchange information each day. The row for this paper distinguishes the theory, in which SOC is a state, from the experiment, which compares a grid of fixed rules that do not respond to SOC.

Study	SOC as a decision state	How the soil process enters	Daily feedback from decisions	Animal condition responds to feed	Within-farm grazing pattern	Payment design
Antle and Capalbo (2001), Antle et al. (2003)	No. Century carbon rate per practice change	Century carbon rates by practice and zone, as a lookup table	No. Fall and spring crop choice	No	Across fields and zones only	Per hectare against per tonne, with measurement cost
Berazneva et al. (2019)	Yes, optimal control	Annual SOC equation with two parameters fitted to RothC, econometric yield	No, annual	No, maize cropland	No	No. Shadow value against carbon prices
Torell, Lyon, and Godfrey (1991), Ritten et al. (2010)	No. Forage state	Forage growth or production function, no soil	No. Annual stocking given the forage state	Partly. Steer gain falls with stocking or forage	No. Homogeneous range	No
Ritten, Bastian, and Rashford (2012)	No. Fixed regional credit rate	Regional offset rates, no process or forage model	No decision model	No livestock model	Between four regions	Offsets at fixed rates per hectare
Wang, Teague, and Park (2016); Wang et al. (2018)	No. Two-grass biomass state	No soil. Two-grass Lotka-Volterra forage model	Daily forage and grazing loop, fixed calendar rotation	2016 fixed cow weight. 2018 daily steer gain	30 paddocks, selective patch and species grazing	No
Nguyen, Nong, and Paustian (2019)	No. SOC after 30 years is a constraint in a static profit maximization	One-hidden-layer neural network trained on DayCent. It maps fixed annual rates to SOC after 30 years and mean corn grain and stover yields	No. Constant annual rates of nitrogen, irrigation, and residue removal	No livestock. Irrigated corn	No. Fields and a landscape in eastern Colorado	No. GHG is an objective of a regional planner, not priced
Liu et al. (2024)	No decision	Machine learning pretrained on *ecosys*, tuned to observations, cropland	No decisions, daily fluxes	No, cropland	No. Regional 250 m grid	No
Shang et al. (2024)	No soil carbon	Neural surrogate of the FarmDyn farm model, no soil	No. Static input-output map	No, arable farms	No	No
Zhang et al. (2026)	No decision. SOC per spatial unit	MEMS 3.0 process model, run directly	Daily. Herd moves by forage availability	No. Fixed 500 kg animal units	Habitat-suitability spatial units	No
This paper	**SOC per patch. A state in the theory, a grid of fixed rules in the experiment**	**Recurrent surrogate of MEMS, full daily state**	**Daily patch choice and hay, calendar rotation**	**Body weight, condition, conception**	**Selective grazing over 12 patches**	**Outcome, practice, premium**

Survey Staff 2026), and the surrogate trained on every MEMS simulation at this site, so the farm simulation is in-sample. The representative farm has 60 hectares, a scale that corresponds to the 50-to-99-head herd class reported for Michigan in the 2022 USDA Census of Agriculture (USDA-NASS 2024), and we divide it into twelve 5-hectare patches. We treat grazing within a patch as uniform because grazing is far less uneven in paddocks of a few hectares than in large pastures (Barnes et al. 2008).

The management decade 2026 to 2035 replays the daily gridMET weather of 1996 to 2005 at the site year by year (Abatzoglou 2013), at a constant atmospheric carbon dioxide concentration of 427 parts per million, the forcing of the surrogate training data. Every simulation receives this forcing, and the comparison is conditional on one common forcing path. Each simulation starts on 1 January 2026 with all twelve patches at the MEMS state of the site at the end of 2025 under one of five management histories (Section 3.3).

3.2. The surrogate

For one patch, the surrogate predicts the full MEMS state at the end of a day from the state at the end of the previous day, the fixed site characteristics, and the weather, calendar, and management of the day. The state holds 425 values, and its layered variables keep the 16 soil layers of MEMS. The surrogate does not predict the internal fluxes of MEMS.

A convolutional encoder over the 16 soil layers and an encoder for the other inputs feed a two-layer gated recurrent unit (GRU), a recurrent network layer that carries a learned memory of recent weather, vegetation, soil, and management from one day to the next. The two management inputs of a patch are its local grazing density, in demand-equivalent LSU per hectare, and the hay fed on it. The paddock count, the farm geometry, the calendar, and selective grazing stay in the farm model, outside the neural network. On each day of the grazing season, the farm model chooses the occupied patch and its hay and passes the two inputs to the surrogate, which advances all 12 patches. The realized forage and hay then determine the energy balance and the body weights of the next day. Appendix D specifies the surrogate, and Appendix C gives the herd, nutrition, movement, and accounting rules.

Training uses 1,667 thirty-year MEMS simulations (2026 to 2055) at 39 sites, 36 of them in Michigan, each starting after one of five management histories of its site. From 2026, each simulation follows one of 90 synthetic grazing schedules, with daily local densities from 0.5 to 36 LSU per hectare and hay of up to 54 grams of dry matter per square meter per day. The schedules are independent of the farm model of this paper, which lets the surrogate learn the response of MEMS to the two management inputs without a herd model. The training objective combines the one-day prediction error with a penalty for drift over long forecasts and a penalty for persistent level error in seven forage and SOC outputs (Appendix D.3).

Validation tests recursive prediction over the ten-year horizon of the management analysis. It holds out one start history at each of two sites, the low-pressure rotational history at test site 1 and the heavy continuous history at test site 2, and training sees these two sites only under their other histories. Each held-out history has nine MEMS simulations under different synthetic grazing schedules, two continuous and seven rotational, and we call these 18 simulations the test simulations. In each test simulation, the surrogate starts from the MEMS state at the end of 2025 and then advances from its own predictions for 3,652 days under the recorded weather and management. The same test simulations also guided the choice of the final training checkpoint, which makes the test accuracy slightly optimistic. Section 4 reports the comparison, and Appendix D.4 defines the measures.

3.3. Management experiment design

We simulate every management system of a finite grid. The planned stocking rate runs from 0 to 3.2 LSU per hectare in increments of 0.2. The single management system at SR = 0 keeps no herd: the surrogate advances the 12 patches with no grazing and no hay, and the accounts carry only the permanent fence. At each of the 16 stocking rates from 0.2 to 3.2 LSU per hectare, $N = 1$ represents continuous grazing, with all patches eligible throughout the grazing season. Calendar-based rotation uses $N \in \{2, 3, 4, 6, 12\}$, each with $\mathrm{gp} \in \{1, \ldots, 7\}$ days and with every gp from 10 to 75 days in steps of 5 for which the rotation cycle $N\,\mathrm{gp}$ fits the 151-day grazing season. The longest grazing period is 75 days with two paddocks, 50 with three, 35 with four, 25 with six, and 10 with twelve, and rest is the derived quantity $(N-1)\mathrm{gp}$. Within the active paddock, equation (C1) applies the same animal selective-grazing rule as under continuous grazing. Each stocking rate has one continuous-grazing design and $21+16+13+11+8 = 69$ calendar-based-rotation designs, for 70 paddock designs. Each paddock design runs at five levels of feed-gap supplementation, $\phi \in \{0, 0.25, 0.5, 0.75, 1\}$, and from five start states, the MEMS states of the site at the end of 2025 after the five management histories of the surrogate training data (ungrazed, low-pressure continuous grazing, low-pressure rotational grazing, heavy continuous grazing with supplement, and intensive rotational grazing with supplement). The experiment has $5 \times (1 + 16 \times 70 \times 5) = 28{,}005$ simulations. The grid stops at 3.2 LSU per hectare, where 9.7 to 14.9 percent of the simulations of each start history complete.

The farm and financial accounts cover the ten management years 2026 to 2035, and Appendix C.5 defines the feed, energy, and weight calculation. The profit optimum is the system with the highest NPV in this grid of fixed management systems. The speed of the surrogate permits a comparison of the full grid and provides a basis for later optimization over the mixed discrete and continuous decisions.

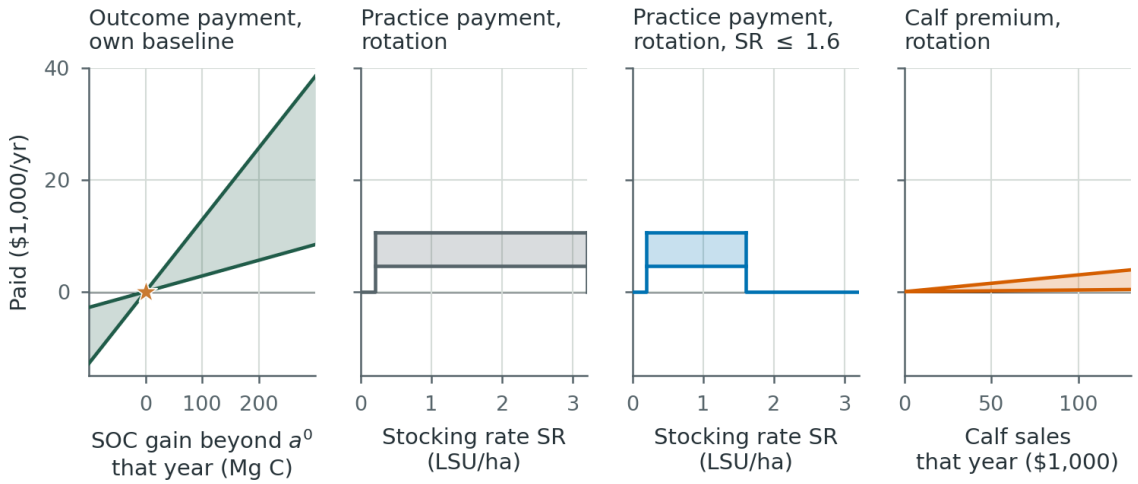


FIGURE 2. **What each payment design pays a farm in one year.** Each panel gives the payment to one 60-hectare farm in one year of the ten-year contract, shaded between the lowest and the highest observed level. Outcome payment against the farm's own baseline: \$28 to \$128 per Mg C of SOC gained in the year beyond the gain of the profit optimum a^0 (star), with a repayment at the same price per Mg C below it. Practice payments, plotted against the planned stocking rate for rotation alone and for rotation with a cap of 1.6 LSU per hectare: \$76.55 to \$175.84 per hectare per year, or \$4,593 to \$10,550 per farm, for a management system that rotates ($N \geq 2$). The management system without a herd does not rotate and receives no practice payment. Calf premium: 0.3 to 3.0 percent of the calf sales of the year for a management system that rotates, paid by the buyer of the calves.

3.4. Payment experiment

We add the payments of Section 2.4 to the complete management systems after the simulation. The practice payment and the premium use three qualifying rules, rotation ($N \geq 2$) alone and rotation with a planned stocking rate of at most 2.0 or 1.6 LSU per hectare, and the premium also uses the outcome-verified rule. The stocking cap represents the balance of stocking rate and forage supply that the NRCS Prescribed Grazing standard and grazing labels require (USDA-NRCS 2017; American Grassfed Association 2024; National Audubon Society 2023). The Michigan cost scenarios for prescribed grazing describe the situation before the practice as an inefficient use of forage, with stocking rates that are likely higher than the current level of production (USDA-NRCS 2024c). The premium, which the buyer of the calves pays, involves no public expense. Figure 2 shows what each design pays a farm in one year of the contract.

We evaluate each instrument over a band of observed payment levels. Outcome prices run from \$28 per Mg C, the 2024 average price of agricultural carbon credits (\$7.66 per t CO_2e, Procton 2025), to \$128 per Mg C, the highest price that U.S. farmer programs paid, which is \$35 "per metric ton of carbon sequestered" in Cargill's RegenConnect (Plastina, Wongpiyabovorn, and Jo 2024). Because three other programs in the same review state

their price per t CO_2e, we read this ton as one credit of 1 t CO_2e. For comparison, the social cost of carbon is $190 per t CO_2 for 2020 emissions at a 2 percent near-term discount rate, or $697 per Mg C (U.S. Environmental Protection Agency 2023). Practice rates run from $76.55 to $175.84 per hectare per year, the Michigan EQIP rates of fiscal year 2025 for Pasture Standard and Pasture Intensive prescribed grazing (USDA-NRCS 2024b). Premiums run from 0.3 to 3.0 percent of the calf price, a range that expresses the auction premiums of $0.73 to $7.92 per hundredweight in Appendix A as shares of the model calf price of $264 per hundredweight, and a premium of 10 percent serves as a stress test. The Michigan schedule does not state how many years prescribed grazing is paid. We assume a ten-year contract, and a three-year contract would need 2.84 times the rate for the same present value.

For each farm, design, and level, we report the chosen management system, the additional SOC, the present value of the payment, the payment per additional Mg C, and the windfall share, the part of the payment that the farm would receive at its profit optimum. For each farm, we also report the switch level of each design and, for the outcome payment, the lowest price at which the farm keeps no herd.

4. The surrogate: accuracy over ten years and speed in the loop

4.1. Accuracy over ten years

We score three outputs that carry forage and soil carbon. Plant carbon (C) and nitrogen (N), each the sum of the five MEMS plant pools (live leaf and stem, dead leaf and stem, and seed), are scored on the grazing-season days (day of year 120 to 270) of the ten test years, and total SOC is scored on the stock at the end of the tenth year. Table 2 reports the relative error and the mean absolute error (MAE) and compares the MAE with that of the site climatology, the mean MEMS value of each day of the year in the training simulations at the test site.

Over the ten years, the relative error is 7.95 percent for plant C and 8.84 percent for plant N, with MAEs of 36.5 kilograms of C and 1.58 kilograms of N per hectare against MEMS means of 459 and 17.9 kilograms per hectare. At the end of the tenth year, the total SOC MAE is 0.48 Mg C ha^{-1} against a mean stock of 119 Mg C ha^{-1}, a relative error of 0.40 percent. On all 18 test simulations and for all three outputs, the surrogate is more accurate than the site climatology and than persistence, which holds the MEMS value of 1 January 2026, with skill scores from 0.91 to 0.99 (Appendix Table D1). For SOC, the skill against persistence, 0.98, is more informative than the relative error, which divides by the whole stock, since the skill compares the SOC error with the SOC change in MEMS. The plant errors do not grow with the forecast horizon, and the SOC error stays at 1.3 to 2.3 percent of the site-climatology error in every year (Appendix D.4). Figure 3 shows one test

TABLE 2. **Surrogate accuracy over ten years and speed in the farm loop.** Panel A compares the surrogate with MEMS on the 18 test simulations. Each forecast starts from the MEMS state at the end of 2025 and advances from its own predictions for ten years. Plant C and N are scored on the grazing-season days of all ten years, and total SOC on the stock at the end of the tenth year. The relative error is the sum of the absolute errors divided by the sum of the absolute MEMS values, and MAE is the mean absolute error. The last column divides the surrogate MAE by the MAE of the site climatology, the mean MEMS value of each day of the year in the training simulations at the test site. For total SOC, the climatology adds the mean change of these simulations to the MEMS stock of the test simulation on 1 January 2026. Panel B gives the time of one ten-year farm simulation on one CPU core and the CPU time of all 28,005 simulations, each including the farm, herd, and animal model. The surrogate advances the paddock designs of one start history, planned stocking rate, and level of feed-gap supplementation together on one core, and the MEMS total is the time of one simulation multiplied by 28,005.

A. Ten-year recursive forecast, 18 test simulations

Output	Relative error (%)	Ten-year MAE	MAE divided by site-climatology MAE
Total SOC, end of year 10	0.40	0.48 Mg C ha^{-1}	0.013
Plant C	7.95	36.5 kg C ha^{-1}	0.062
Plant N	8.84	1.58 kg N ha^{-1}	0.070

B. Time in the same farm loop

	MEMS in the loop	Surrogate in the loop	Speed factor
One ten-year farm simulation, one CPU core	3.1 h	80 to 85 s	132 to 140
All 28,005 simulations (CPU-hours)	87,000 to 88,000	340	257 to 258

simulation at each test site.

4.2. Speed in the farm loop

In the same farm loop, one ten-year farm simulation takes 80 to 85 seconds on one CPU core with the surrogate and 3.1 hours with MEMS (Table 2), and both times include the farm, herd, and animal model. MEMS cannot save and resume its state, so a loop with MEMS must restart it for a patch from 1 January 2020, with all actions so far, on each day on which the patch is grazed or fed hay. One farm simulation restarts MEMS 1,511 times, at about 7 seconds per restart, and this restart coupling is the least costly coupling that MEMS permits. If the daily actions were known in advance, MEMS would need one simulation for each patch, about 83 seconds for the farm. The difference in run time comes from the restarts that a daily decision loop requires.

In the experiment, the surrogate advances the paddock designs of one start history, planned stocking rate, and level of feed-gap supplementation together on one CPU core, in batches of up to 70 simulations, which halves its time to an average of 43.7 seconds per simulation. All 28,005 simulations took 340 CPU-hours, in two runs of 1 hour 42 minutes and 2 hours 23 minutes on a computer cluster. With MEMS, they would take about 88,000 CPU-hours.

Over the 18,457 complete simulations with a herd, the local grazing density and the hay offer never exceed the training ranges of the surrogate. The crude protein of the plant forage eaten is clipped into its training band on 22.6 percent of the grazing days. The clip removes 2.0 times as much nitrogen as it adds and mostly lowers crude protein above the band. The simulated outcomes carry the process representation and the limitations of MEMS and have no independent field validation.

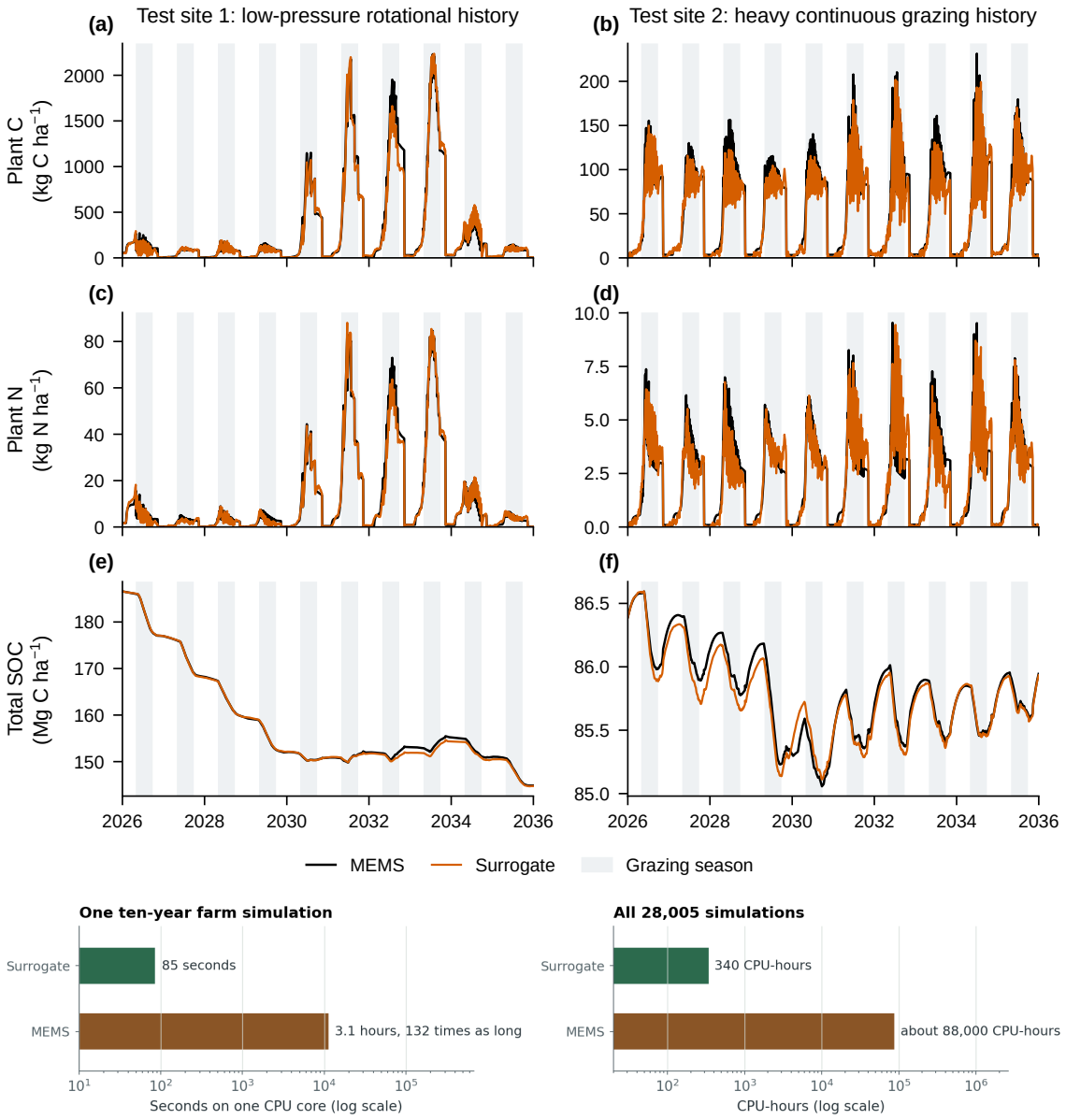


FIGURE 3. **Accuracy of the surrogate over ten years and its speed in the farm loop.** Top: plant C, plant N, and total SOC in MEMS and in the ten-year recursive forecast of the surrogate, for the test simulation (a MEMS simulation held out from training) with the median plant C error at each of the two test sites. Shaded bands mark the grazing season. Bottom left: run time of one ten-year farm simulation on one CPU core, with the surrogate and with MEMS in the same farm loop. Bottom right: CPU-hours of all 28,005 simulations.

5. Results

All results in this section come from the farm loop with the surrogate. Each start history has 5,600 simulations with a herd, and 41.7 percent of them complete the ten years in the heavy continuous history, 65.6 percent in the intensive rotational history, 73.2 and 74.4 percent in the low-pressure continuous and low-pressure rotational histories, and 74.7 percent in the ungrazed history (Appendix Table E2). The other simulations end early when animals reach a weight floor (Appendix C.5). Completion falls with the planned stocking rate. At 0.2 LSU per hectare, 80.0 percent of the heavy continuous simulations and all simulations of the other four histories complete, and at 3.2 LSU per hectare, 9.7 to 14.9 percent of the simulations of each history complete. The simulation without a herd completes in every history. A simulation that ends early has no ten-year NPV, so every outcome below, like the definitions of Section 2.1, refers to the complete simulations. Appendix E.1 maps which simulations complete (Figure E2), and Appendix E compares the simulated animal weights with a research herd (Figure E3).

5.1. Past grazing sets the SOC path, and management sets how far below it the farm ends

All five histories start from one MEMS spin-up near 197 Mg C per hectare in 1926 (Figure 4a). A century of different grazing then leaves them far apart at the end of 2025, from 84.1 Mg C per hectare after heavy continuous grazing through 158.5 after intensive rotational grazing, 196.5 after low-pressure continuous grazing, and 212.7 after low-pressure rotational grazing, to 218.9 without grazing.

Over the ten experiment years, the heavy continuous history gains SOC under every complete management system, by 4.3 to 52.6 Mg C per hectare (Figure 4b). The intensive rotational, low-pressure continuous, and low-pressure rotational histories have complete simulations on both sides of zero, with ten-year changes from −23.1 to 13.3, from −40.2 to 8.3, and from −47.0 to 3.2 Mg C per hectare, respectively. In the low-pressure continuous history, only systems at 1.0 LSU per hectare or less gain SOC, and in the low-pressure rotational history only systems at 0.4 LSU per hectare or less. The ungrazed history loses 0.2 to 50.6 Mg C per hectare under every complete management system with a herd and gains 1.3 only without a herd. Across the 18,457 complete simulations with a herd, the start history explains 73 percent of the variance of the ten-year SOC change and the planned stocking rate 23 percent (Appendix Figure E1).

The soil of each history is still adjusting to its past in 2026, and the management system without a herd, with no grazing and no hay from that year, traces this adjustment. Without a herd, SOC rises in every history, by 52.6, 13.3, 8.3, 3.2, and 1.3 Mg C per hectare in the heavy continuous, intensive rotational, low-pressure continuous, low-pressure rotational,

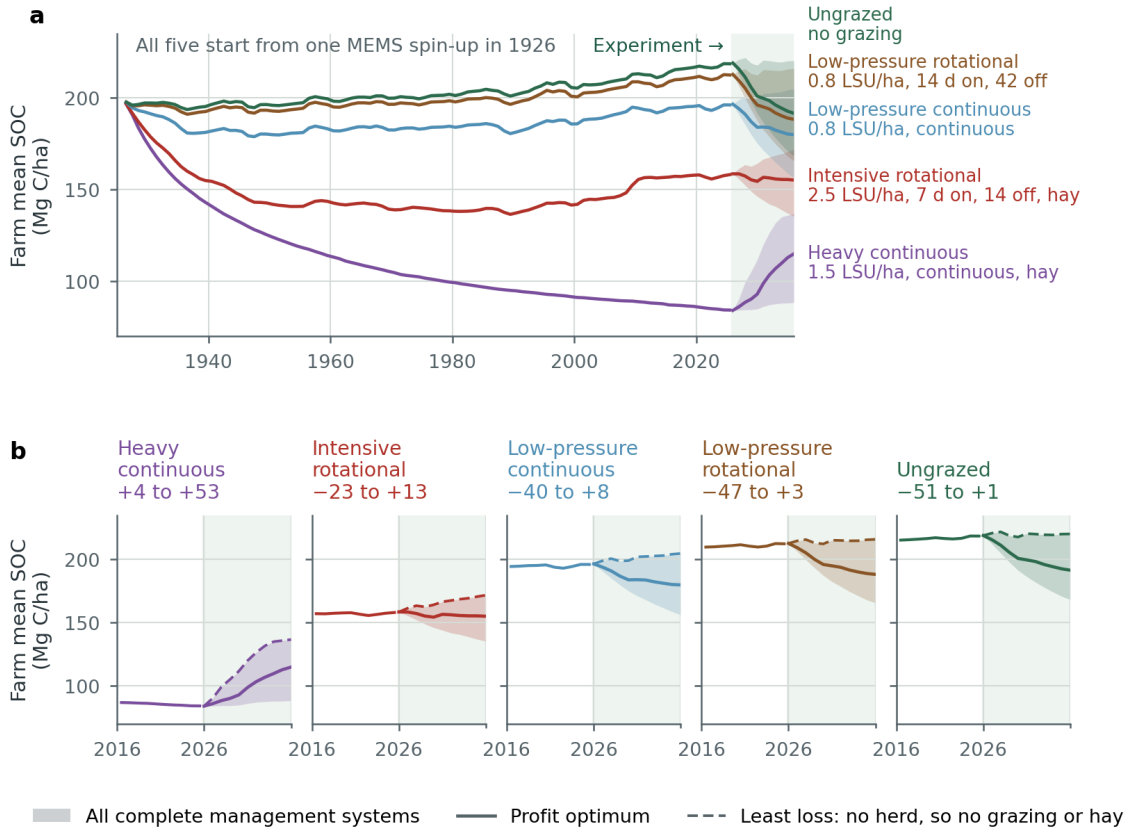


FIGURE 4. **Past grazing sets the SOC path, and management sets how far below it the farm ends.** The start histories are in the order of their start SOC. (a) Farm mean total SOC of the five start histories in the MEMS simulations that define them, from the end of the spin-up in 1926 to the end of 2025, followed over the ten experiment years by the range over all complete simulations (band) and the profit optimum (line). Each label gives the MEMS schedule that the history repeats every year from 1926 to 2025, and its rate is the grazing density on pasture rather than a planned stocking rate. (b) The years 2016 to 2035 for each history, on the same axes. The band spans all complete simulations, the solid line is the profit optimum, and the dashed line is the least-loss management system, which keeps no herd and has no grazing and no hay from 2026. Each panel title gives the range of the ten-year SOC change in Mg C per hectare.

and ungrazed histories, respectively. Every complete simulation with a herd ends below the no-herd path of its history.

Within a history, management moves the ten-year change over a wide range, whose width runs from 36.4 Mg C per hectare in the intensive rotational history to 51.9 in the ungrazed history. The histories start from different stocks, so we compare SOC only within a history, and for the same reason a payment that measures SOC against the start stock pays mainly for the history (Section 5.4).

5.2. The decisions interact

The value of each decision depends on the others, and the planned stocking rate, above all, shapes the best paddock design. Appendix Figure E1 splits the variance of the ten-year NPV across the complete simulations with a herd into Shapley shares. The Shapley share of a factor is its mean gain in R^2 when it joins a regression on indicator variables, over all orders of entry. The planned stocking rate explains 74 percent of the NPV variance, the paddock design 5 percent, the start history 3 percent, and the feed-gap supplementation 2 percent. Together these main effects explain 84 percent, and the pairwise interactions add 14 percent, a share that an additive model of the decisions leaves out.

The interaction appears in the best management system at each stocking rate (Appendix Table E1). Continuous grazing earns the most at low planned stocking rates and rotation at high ones. At 0.2 to 1.4 LSU per hectare, the best continuous grazing earns more than the best rotation in 30 of the 35 pairs of history and stocking rate. From 1.6 LSU per hectare, the best rotation earns more in every pair in which a continuous-grazing simulation completes, and from 2.6 LSU per hectare its lead is \$24,197 to \$124,760. The profit optimum of every history uses two paddocks and full feed-gap supplementation ($\phi = 1$) at 2.0 or 2.2 LSU per hectare.

Herd condition links these decisions to profit. The off-season ration, the same in every management system, returns the mature cows to BCS 5 before calving, but the first-calf cows carry the condition of the grazing season into calving, at a median BCS of 4.31 in the intensive rotational history to 4.53 in the low-pressure rotational history. The conception rate rises by 0.25 per score between BCS 4 and 5 and by 0.07 per score between BCS 5 and 6 (equation (B4)). Every open cow is culled, and thinner first-calf cows therefore give fewer calves and more cull sales in the next year.

The feed gap carries the stocking rate into herd condition. The planned stocking rate sets most of the size of the gap, the paddock design changes where the herd grazes, and the feed-gap supplementation ϕ fills part of it with hay. In the four histories that complete without hay, we match management systems that differ only in ϕ. At 0.8 and 1.0 LSU per hectare, there is little gap to fill, and the median NPV difference between $\phi = 1$ and $\phi = 0$ is at most \$239 in each history. At 2.2 LSU per hectare, in the two low-pressure histories

and the ungrazed history, $\phi = 1$ costs a median \$11,908 to \$15,337 of hay over ten years, but it raises the first-calf BCS at calving by 0.51 to 0.64 and the first-calf conception rate by 0.117 to 0.133 and earns \$18,385 to \$21,725 more NPV. The intensive rotational history has one matched pair at this stocking rate, and in it $\phi = 1$ earns \$19,329 more.

The paddock design acts on the feed gap through selective grazing. Within the eligible area, the herd occupies the patch with the highest grazing value, which rises with biomass and live-leaf share and favors the patches near water (equation (C1)). Under continuous grazing, the herd can stay on a preferred patch after grazing it down, and the feed gap opens while other patches still hold forage. A calendar rotation moves the herd at the end of each grazing period. In the ungrazed history at 1.0 LSU per hectare, this move barely matters, and continuous grazing leads by \$2,867 of NPV because the rotation pays more for fencing and labor. At 2.2 LSU per hectare, two paddocks with a four-day grazing period feed 3.0 tonnes of hay per year on pasture, against 8.1 tonnes under continuous grazing, and lead by \$7,100 of NPV after fencing and labor.

5.3. The least costly SOC gains change the whole management system

Profit and SOC move in opposite directions up to the most profitable stocking rate, and both fall beyond it. In every history, the highest NPV at each planned stocking rate rises up to 2.0 or 2.2 LSU per hectare and falls at higher rates, while the mean ten-year SOC change falls at every step from 0.2 to 2.8 LSU per hectare. A farm at its profit optimum can gain SOC only by giving up profit (Figure 5a).

Table 3 gives the profit optimum and the least-loss management system of each farm. The profit optimum uses 2.0 LSU per hectare in the heavy continuous and intensive rotational histories and 2.2 LSU per hectare in the other three. In every history, the least-loss management system keeps no herd. It has no sales and pays only the fence cost, an NPV of –\$10,344. From the profit optimum, the move to no herd gives up \$92,082 of NPV for 1,308 Mg C in the heavy continuous history, \$98,051 for 1,000 Mg C in the intensive rotational history, \$117,222 for 1,489 Mg C in the low-pressure continuous history, \$119,549 for 1,660 Mg C in the low-pressure rotational history, and \$119,250 for 1,718 Mg C in the ungrazed history. Among the systems with a herd, the least-loss one uses 0.2 LSU per hectare in every history, with twelve paddocks and a 1 to 4 day grazing period in four histories and two paddocks with a 75-day period in the intensive rotational history.

The profit and SOC frontier holds 110 management systems in the heavy continuous history, 73 in the intensive rotational history, 187 in the low-pressure continuous history, 186 in the low-pressure rotational history, and 194 in the ungrazed history (Figure 5a). In every history, the first step along the frontier keeps the stocking rate of the profit optimum and changes the paddock design. This step adds 8 Mg C for \$228 of NPV in the heavy continuous history, 33 Mg C for \$3,293 in the intensive rotational history, 13 Mg C

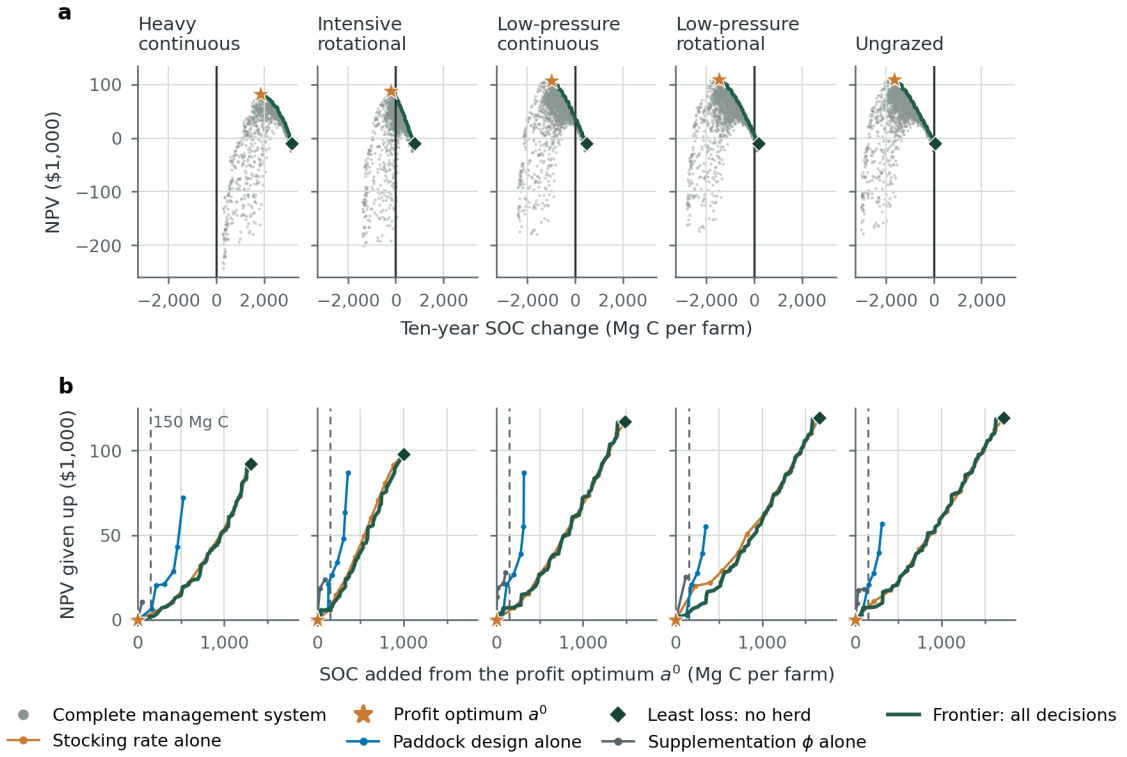


FIGURE 5. **From the profit optimum, more SOC always costs NPV, and moving all decisions adds it at the least cost.** One column for each start history, in the order of start SOC. (a) NPV against the ten-year SOC change per 60-hectare farm for every complete management system. The line is the profit and SOC frontier, the star is the profit optimum a^0, and the diamond is the least-loss management system, which keeps no herd. (b) NPV given up against SOC added from a^0. The frontier of all decisions is the frontier of panel (a). Each other line moves one decision and holds the others at a^0: the planned stocking rate (down to no herd), the paddock design (over the 70 designs), or the feed-gap supplementation ϕ. The dots on each line mark the tested values on the frontier of that decision. The dashed line marks 150 Mg C, the target of Table 4.

TABLE 3. **The profit optimum and the least-loss management system of each farm.** Each start history is one 60-hectare farm. The profit optimum a^0 is the complete management system with the highest ten-year NPV, and the least-loss system is the one with the highest ten-year SOC change. In every history, the least-loss system keeps no herd, so it has no paddocks, grazing period, or feed-gap supplementation, and its NPV is the fence cost. SR is the planned stocking rate, and ϕ is the share of the daily feed gap that hay on pasture fills. NPV is in $ thousand at a 5 percent discount rate, and the SOC change from 1 January 2026 to 31 December 2035 is in Mg C per farm.

	Heavy continuous		Intensive rotational		Low-pressure continuous		Low-pressure rotational		Ungrazed	
	a^0	Least loss	a^0	Least loss	a^0	Least loss	a^0	Least loss	a^0	Least loss
Decisions										
SR (LSU ha^{-1})	2.0	0	2.0	0	2.2	0	2.2	0	2.2	0
Paddocks N	2	none	2	none	2	none	2	none	2	none
Grazing period (d)	6	none	5	none	5	none	15	none	4	none
ϕ	1	none	1	none	1	none	1	none	1	none
Outcomes										
NPV ($ thousand)	81.7	−10.3	87.7	−10.3	106.9	−10.3	109.2	−10.3	108.9	−10.3
SOC change (Mg C)	+1,849	+3,157	−203	+797	−993	+496	−1,469	+191	−1,643	+76

for $881 in the low-pressure continuous history, 35 Mg C for $1,703 in the low-pressure rotational history, and 58 Mg C for $1,895 in the ungrazed history. The stocking rate then falls in steps to 0.2 LSU per hectare as the other decisions change with it, and the frontier ends at no herd. Beyond the profit optimum, 70 percent of the frontier systems with a herd use continuous grazing or a grazing period of seven days or less in the heavy continuous history, against only 23 percent in the intensive rotational history and 77, 79, and 77 percent in the low-pressure continuous, low-pressure rotational, and ungrazed histories, respectively.

SOC is cheapest when all four decisions move together. Figure 5b and Table 4 compare the frontier of all decisions with moves of one decision that hold the others at the profit optimum. The stocking rate alone reaches no herd and so adds as much SOC as all decisions together, but in most histories it gives up more NPV on the way. For the first 150 Mg C, moving all decisions together gives up $14 of NPV per Mg C in the heavy continuous history, $38 in the intensive rotational history, $48 in the low-pressure continuous history, $45 in the low-pressure rotational history, and $50 in the ungrazed history. In the low-pressure continuous history, the least costly first 150 Mg C change only the stocking rate, and lowering the stocking rate alone gives up the same $48 per Mg C. In the other four histories, the stocking rate alone gives up 1.4 (ungrazed) to 3.0 (low-pressure rotational) times as much as all decisions. The paddock design alone adds at most 310 to 527 Mg C per

TABLE 4. **SOC gains from the profit optimum when all decisions or one decision change.** Starting from the profit optimum a^0 of each farm (Table 3), each row varies the named decisions over all tested values and holds the other decisions at a^0. The paddock design combines the paddock count N and the grazing period. The stocking-rate row includes the system without a herd, for which the other decisions do not apply. Mg C added is the largest SOC gain over a^0 in the row, per 60-hectare farm. $ per Mg C is the NPV given up for the first 150 Mg C: the smallest NPV loss among the systems in the row that add at least 150 Mg C, divided by 150. "never" means that no system in the row adds 150 Mg C.

	Heavy continuous		Intensive rotational		Low-pressure continuous		Low-pressure rotational		Ungrazed	
	Mg C added	$ per Mg C	Mg C added	$ per Mg C	Mg C added	$ per Mg C	Mg C added	$ per Mg C	Mg C added	$ per Mg C
All decisions	1,308	$14	1,000	$38	1,489	$48	1,660	$45	1,718	$50
Stocking rate alone	1,308	$32	1,000	$99	1,489	$48	1,660	$133	1,718	$72
Paddock design alone	527	$44	355	$177	318	$179	346	$137	310	$139
Supplementation ϕ alone	50	never	89	never	104	never	117	never	109	never

farm and, for the first 150 Mg C, gives up 2.8 to 4.7 times as much as all decisions. Feed-gap supplementation alone adds at most 50 to 117 Mg C and never reaches 150 Mg C.

With the stocking rate and ϕ of the profit optimum, the paddock design alone moves SOC, but only at a high cost in NPV. The 70 paddock designs spread the ten-year SOC change over 14.8, 11.8, 11.1, 10.1, and 9.9 Mg C per hectare in the heavy continuous, intensive rotational, low-pressure continuous, low-pressure rotational, and ungrazed histories, respectively. In every history, twelve paddocks lose the least SOC, with a ten-day period in the heavy continuous, intensive rotational, and low-pressure continuous histories and a seven-day period in the other two. This design forces the herd onto every patch, and each paddock rests $N - 1$ grazing periods between visits (equation (4)), which gives 110 days of rest with a ten-day period and 77 days with a seven-day period, against 4 to 15 days at the profit optima.

At this stocking rate, the design with the least SOC loss earns an NPV of $9,519 against $81,738 at the profit optimum in the heavy continuous history, $863 against $87,707 in the intensive rotational history, $20,063 against $106,878 in the low-pressure continuous history, $54,144 against $109,205 in the low-pressure rotational history, and $52,104 against $108,906 in the ungrazed history. Each of its paddocks is one patch, so the calendar holds the whole herd on one 5-hectare patch for seven or ten days, and selective grazing cannot move the herd to more forage. Hay must then fill a larger feed gap, and the first-calf cows calve in poorer condition. At 0.8 LSU per hectare, where the feed gap is smaller, the same design earns $14,797 to $16,595 less than the best design at that stocking rate. The least costly path combines the decisions: it lowers the stocking rate, which shrinks the feed

gap, and adds paddocks and rest days once the herd no longer runs short of forage.

5.4. The qualifying rule decides what a payment buys

A practice payment or a calf premium buys SOC only when the profit optimum of the farm does not qualify for it. Each start history is one 60-hectare farm, which chooses the complete management system with the highest NPV plus the present value of the payment (Section 2.4). Figures 6 to 8 show the choice of each farm at each payment level, Table 5 gives the switch level of each design for each farm, and Figure 9 shows where each design moves each farm. Appendix Table E4 reports these choices at the observed levels.

An outcome payment against the farm's own baseline pays nothing at the profit optimum and pays only for change. On a grid with steps of $2, the switch prices of Proposition 2 are $14 per Mg C for the heavy continuous farm, $44 for the intensive rotational farm, $36 for the low-pressure continuous farm, $26 for the low-pressure rotational farm, and $38 for the ungrazed farm (Table 5). At $28 per Mg C, the lowest observed price, only the heavy continuous and low-pressure rotational farms move, and they add 184 and 120 Mg C. As the price rises, each farm lowers its stocking rate in steps (Figure 6). At $50 per Mg C, the heavy continuous farm uses 1.6 LSU per hectare and adds 405 Mg C, and the intensive rotational farm uses 1.8 LSU per hectare and adds 154 Mg C. The low-pressure continuous, low-pressure rotational, and ungrazed farms use 2.0 LSU per hectare with four paddocks and a three-day grazing period and add 282, 347, and 317 Mg C. At this price, each farm receives $43 to $45 per additional Mg C.

At high observed prices, the outcome payment ends beef production on the farm. The heavy continuous, intensive rotational, low-pressure continuous, low-pressure rotational, and ungrazed farms keep no herd from $160, $146, $120, $114, and $102 per Mg C, respectively, and the low-pressure and ungrazed farms reach this point within the observed range (Table 5). Just below these prices, the heavy continuous farm keeps 0.2 LSU per hectare, the intensive rotational farm 1.0, the low-pressure continuous farm 1.2, and the low-pressure rotational and ungrazed farms 0.6 LSU per hectare. At $128 per Mg C, the top of the observed range, the heavy continuous farm keeps 0.4 LSU per hectare and adds 1,181 Mg C for a payment of $139,735, or $118 per additional Mg C. The intensive rotational farm keeps 1.0 LSU per hectare and adds 567 Mg C for $61,909, or $109 per Mg C. The low-pressure continuous, low-pressure rotational, and ungrazed farms keep no herd and add 1,489, 1,660, and 1,718 Mg C for $164,737, $183,511, and $189,259, or $110 to $111 per Mg C. In the model, a farm without a herd earns nothing from its land, and no management system sells or buys the start herd. We expect a land rent or a herd sale to lower the prices at which the farms keep no herd. Other farms may raise the calves that such a farm no longer sells, and the experiment does not count those farms.

A practice payment for rotation alone buys no SOC (Figure 7). Every profit optimum

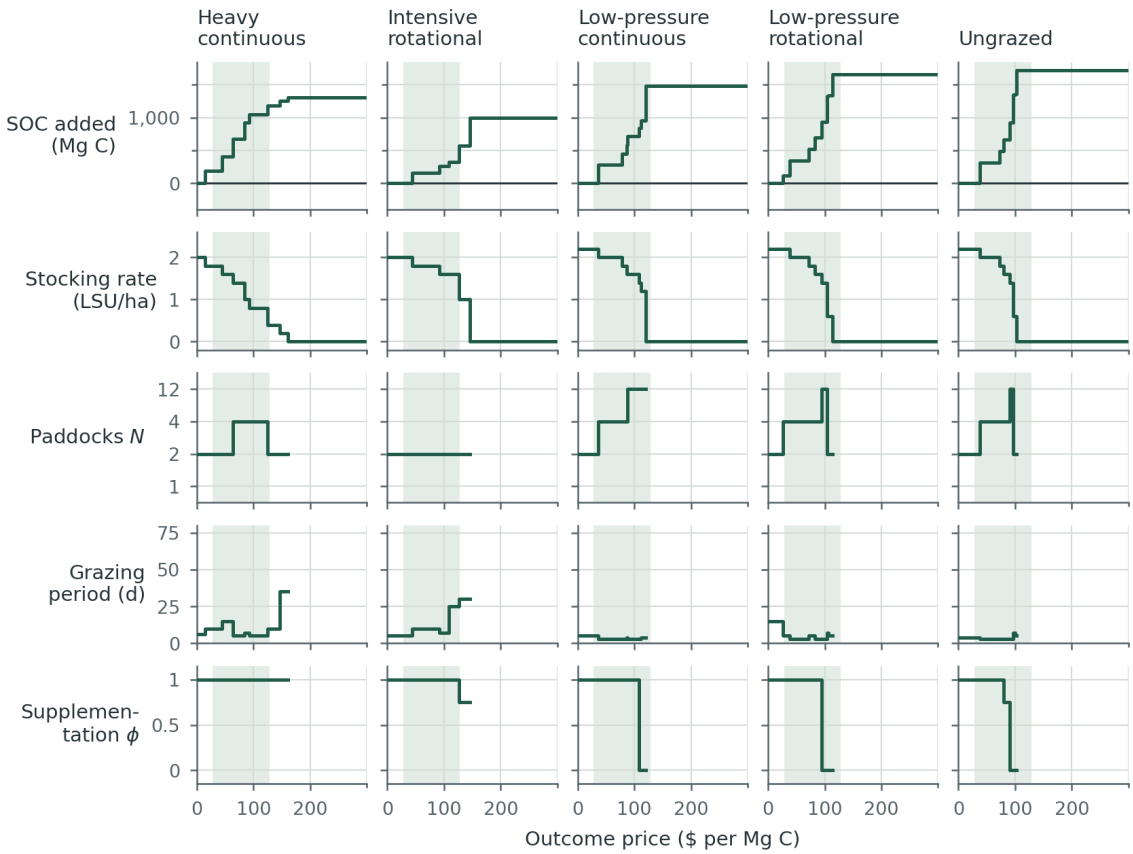


FIGURE 6. **The outcome payment adds more SOC as its price rises.** The management system that each farm chooses under the outcome payment against its own baseline, at prices from $0 to $300 per Mg C in steps of $2, with one column for each start history. The rows give the SOC added over the profit optimum a^0 per 60-hectare farm, the planned stocking rate, the paddock count N, the grazing period, and the feed-gap supplementation ϕ. The management system without a herd has no paddocks, grazing period, or ϕ, and these rows end where the farm keeps no herd. The shaded band marks the observed credit and program prices, $28 to $128 per Mg C. Each design runs a ten-year contract from 2026 to 2035.

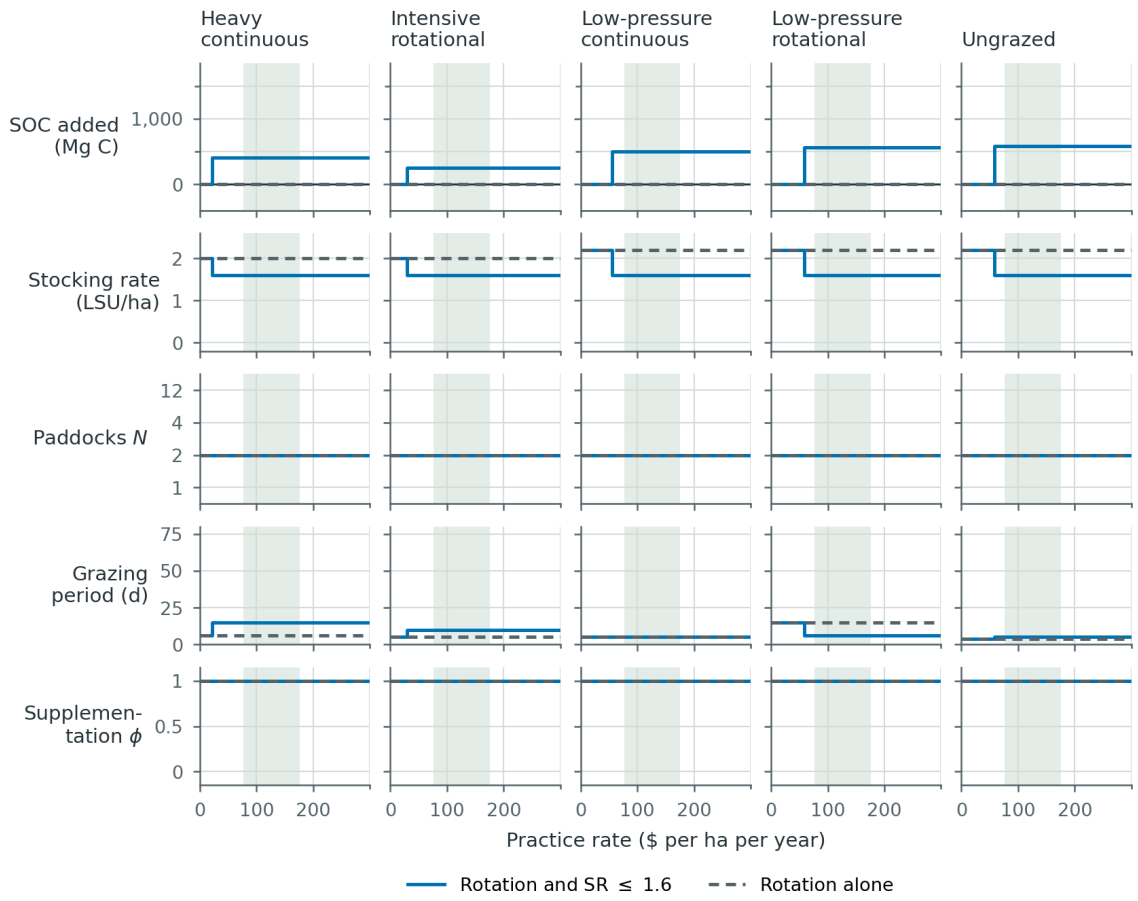


FIGURE 7. **The practice payment for rotation alone moves no farm, and the cap moves each farm once.** The panels are as in Figure 6, at practice rates from \$0 to \$300 per hectare per year in steps of \$2, for rotation alone (dashed) and for rotation with a cap of 1.6 LSU per hectare (solid). Every farm already rotates at a^0, so the payment for rotation alone leaves every farm at a^0. The shaded band marks the Michigan EQIP rates, \$76.55 to \$175.84 per hectare per year.

already uses two paddocks, so every farm qualifies without a change, and no rate up to \$300 per hectare per year moves a farm. At the Michigan EQIP rates of \$76.55 and \$175.84 per hectare per year (USDA-NRCS 2024b), the payment transfers \$37,240 and \$85,539 to each farm and adds no SOC. The whole transfer is windfall, as Proposition 1 states and as Chabé-Ferret and Subervie (2013) measured for cover crops.

This result depends on where the profit optimum lies. In the heavy continuous history, the best management system with continuous grazing earns only \$758 less than the profit optimum, and a cost of rotation higher by that amount would make continuous grazing the optimum. A payment for rotation alone would then move the farm, but it would buy little SOC. The move from the best system with continuous grazing to the best one with rotation, which is the profit optimum, adds 100 Mg C per farm in the heavy continuous

history. In the intensive rotational history, the same move lowers SOC by 119 Mg C because rotation also raises the most profitable stocking rate from 1.8 to 2.0 LSU per hectare. In the low-pressure continuous, low-pressure rotational, and ungrazed histories, it adds 124, 42, and 67 Mg C, respectively.

A practice payment for rotation with a stocking cap of 1.6 LSU per hectare excludes the profit optimum. It moves each farm once, to 1.6 LSU per hectare with two paddocks and $\phi = 1$, at $22 per hectare per year for the heavy continuous farm, $30 for the intensive rotational farm, $56 for the low-pressure continuous farm, and $58 for the low-pressure rotational and ungrazed farms, all below the observed EQIP rates (Table 5). The move adds 405 Mg C on the heavy continuous farm, 251 on the intensive rotational farm, 497 on the low-pressure continuous farm, 566 on the low-pressure rotational farm, and 580 on the ungrazed farm. A higher rate adds only transfer. At the EQIP Pasture Standard rate of $76.55 per hectare per year, each farm receives $37,240, and the payment costs $92 per additional Mg C on the heavy continuous farm, $148 on the intensive rotational farm, $75 on the low-pressure continuous farm, $66 on the low-pressure rotational farm, and $64 on the ungrazed farm. The Pasture Intensive rate pays $85,539 to each farm for the same SOC, from $148 per Mg C on the ungrazed farm to $341 on the intensive rotational farm. Unlike the outcome payment, the cap never moves a farm below 1.6 LSU per hectare, and every farm keeps its herd.

A calf premium for rotation alone lowers SOC (Figure 8). The profit optimum qualifies, and because the rule sets no limit on the stocking rate, a premium that raises the revenue of every calf rewards a larger herd. The heavy continuous and intensive rotational farms do not move at premiums up to 10 percent of the calf price. The low-pressure continuous, low-pressure rotational, and ungrazed farms move from 2.2 to 2.4 LSU per hectare at premiums from 1.95, 1.65, and 2.85 percent of the calf price and lose 317, 205, and 264 Mg C, respectively. Each of these moves raises calf revenue, as Proposition 3 requires, and all three rates lie inside the observed band of 0.3 to 3.0 percent of the calf price.

Table 5 collects the switch levels. Within the observed levels, the outcome payment and the practice payment with the cap of 1.6 LSU per hectare move every farm, the practice payment for rotation alone moves no farm, and the premium for rotation alone moves three farms to less SOC.

Every management system that the outcome payment chooses at any price from $0 to $300 per Mg C lies on the profit and SOC frontier of its history and buys its SOC at the least private cost (Figure 9). Besides the profit optimum, the outcome payment chooses 8 management systems in the heavy continuous history, 5 in the intensive rotational history, 7 in the low-pressure continuous history, 8 in the low-pressure rotational history, and 7 in the ungrazed history. The system that the cap of 1.6 LSU per hectare chooses also lies on the frontier of each history (Proposition 4). The private cost of this move is

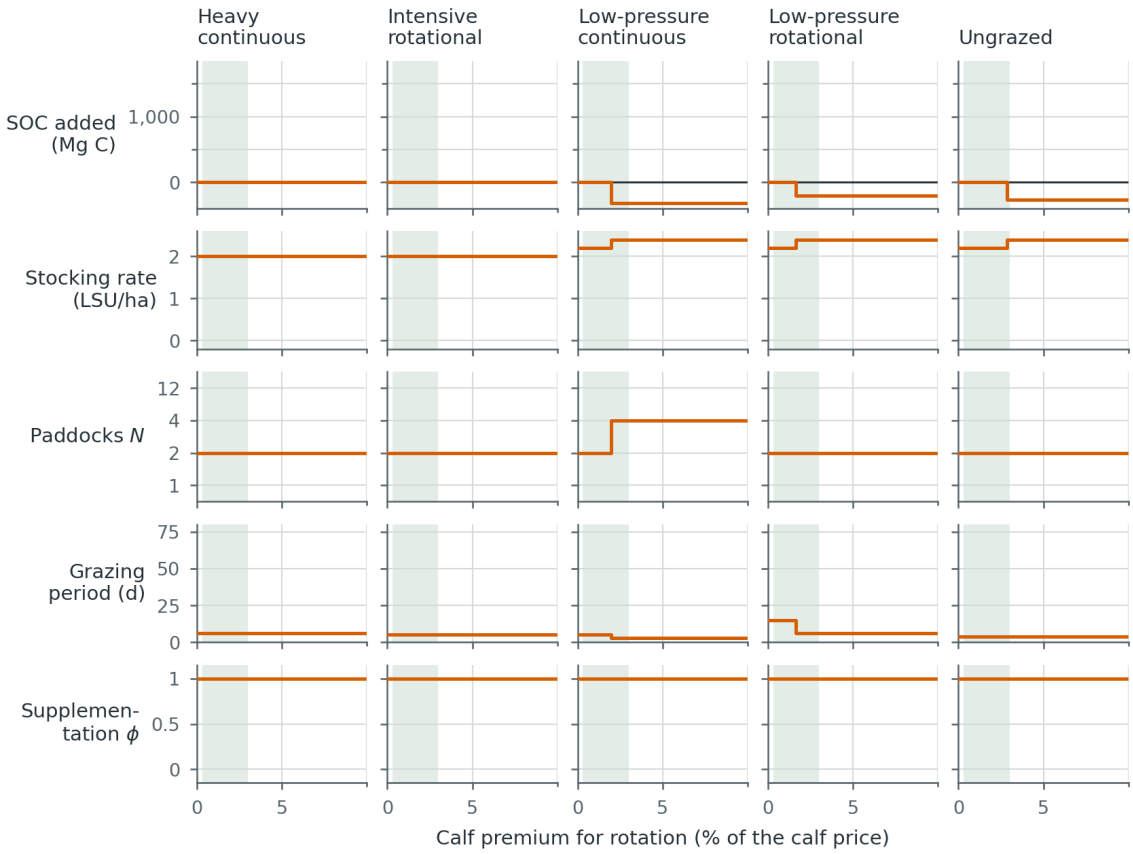


FIGURE 8. **The calf premium for rotation raises the stocking rate of three farms.** The panels are as in Figure 6, at calf premiums for rotation from 0 to 10 percent of the calf price in steps of 0.05 percentage points. The shaded band marks the observed calf auction premiums, 0.3 to 3.0 percent.

TABLE 5. **The lowest payment level that moves each farm.** Each start history is one 60-hectare farm that chooses the complete management system with the highest NPV plus the present value of the payment over the ten-year contract. The switch level is the lowest tested level at which the farm leaves its profit optimum a^0. "none" means no move up to \$300 per hectare per year or 10 percent of the calf price. "No herd" gives the lowest outcome price at which the farm keeps no herd. The outcome price runs from \$0 to \$400 per Mg C in steps of \$2, the practice rate from \$0 to \$300 per hectare per year in steps of \$2, and the premium from 0 to 10 percent of the calf price in steps of 0.05 percentage points. The last row gives the observed levels of Section 3.4.

	Outcome payment, own baseline (\$ per Mg C)		Practice payment (\$ ha^{-1} yr^{-1})		Calf premium (% of calf price)
Start history	Switch	No herd	Rotation	Rotation, SR ≤ 1.6	Rotation
Heavy continuous	\$14	\$160	none	\$22	none
Intensive rotational	\$44	\$146	none	\$30	none
Low-pressure continuous	\$36	\$120	none	\$56	1.95%
Low-pressure rotational	\$26	\$114	none	\$58	1.65%
Ungrazed	\$38	\$102	none	\$58	2.85%
Observed levels	\$28 to \$128		\$76.55 to \$175.84		0.3 to 3.0%

\$10,393 in the heavy continuous history, \$13,767 in the intensive rotational history, \$26,761 in the low-pressure continuous history, \$28,149 in the low-pressure rotational history, and \$28,156 in the ungrazed history. The premium moves three farms off the frontier, to systems with less SOC and less NPV before the premium than the profit optimum.

A baseline at the start stock pays mainly for history. Against this baseline, the heavy continuous farm is paid for SOC that it gains under every management system. At \$28 per Mg C, both start-stock designs pay this farm \$46,256 for a move from 2.0 to 1.8 LSU per hectare that adds 184 Mg C, and 90 percent of the payment is windfall. At \$128 per Mg C, they pay it \$329,432 for 1,181 Mg C, with a windfall share of 58 percent (Appendix E.7).

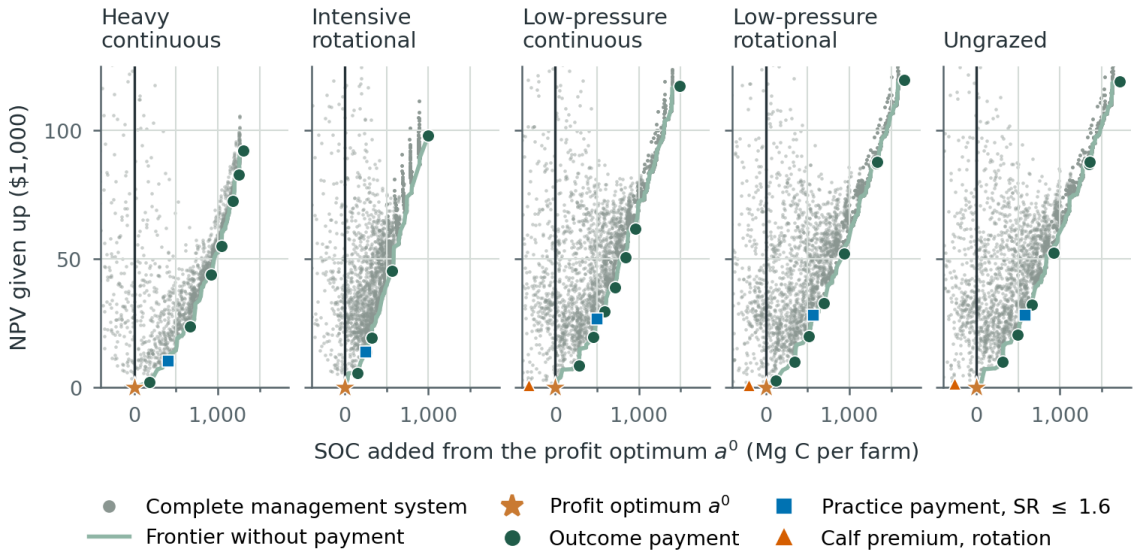


FIGURE 9. **The outcome payment and the cap move each farm along its frontier.** NPV given up against SOC added from the profit optimum a^0 per 60-hectare farm, one panel for each start history, with every complete management system and the profit and SOC frontier without payment. Circles mark every management system that the outcome payment chooses from $0 to $300 per Mg C, squares the choice of the practice payment with a cap of 1.6 LSU per hectare, and triangles the choice of the calf premium for rotation up to 10 percent of the calf price. The star marks a^0, where the practice payment for rotation alone leaves every farm. The heavy continuous and intensive rotational farms do not move under the premium.

6. Conclusion

We develop SoFAME, an integrated model of a cow-calf farm, and place a learned surrogate of MEMS, a biogeochemical process-based model, inside its daily management loop. The surrogate advances the soil and forage of 12 patches from each day's local grazing density and hay, while herd movement, animal nutrition, and the accounts remain in the farm model, where the researcher can change them. On 18 test simulations, each for a combination of site and start history held out from training, the mean absolute error of its total SOC forecasts after ten years is 0.40 percent of the MEMS stock. In the same farm loop, the surrogate runs a ten-year farm simulation 130 to 140 times faster than MEMS. At this cost, a study can evaluate every system of a management grid: the 5,601 systems under each of five start histories take 340 CPU-hours, against about 88,000 CPU-hours with MEMS.

Across these systems, the soil that 100 years of past grazing leave behind sets the path that SOC follows over the next decade without a herd, and management determines how far below that path the farm ends. The decisions interact: although the planned stocking rate explains 74 percent of the variance of NPV, the best paddock design shifts with it, and the pairwise interactions of the decisions explain another 14 percent. The least costly route to more SOC therefore changes the whole management system. On a 60-hectare farm, it gives up $14 to $50 of NPV per Mg C over the first 150 Mg C, depending on the start history, while a lower stocking rate alone gives up as much in the low-pressure continuous history and 1.4 to 3.0 times as much in the other four histories.

A payment for a practice that the profit optimum already uses buys no SOC. On this farm, the profit optimum of every start history rotates, so a practice payment for rotation alone is a pure windfall. A calf premium for rotation alone, within the observed range of 0.3 to 3.0 percent of the calf price, raises the stocking rate of the ungrazed farm and the two low-pressure farms and adds to their SOC loss. If continuous grazing were the profit optimum, a payment for rotation alone would move the farm, but the move from the best management system with continuous grazing to the best one with rotation adds at most 124 Mg C per farm and lowers SOC in the intensive rotational history. The SOC goal must therefore enter the qualifying rule, because no payment rate moves a farm whose profit optimum already qualifies.

A stocking cap turns a practice payment into a contract that binds. With a cap of 1.6 LSU per hectare, a practice payment moves each farm at $22 to $58 per hectare per year, below the observed EQIP rates, and adds 251 to 580 Mg C, depending on the farm. An outcome payment against the farm's own baseline, which the experiment assumes the program knows, moves each farm at $14 to $44 per Mg C. Both designs move every farm onto its profit and SOC frontier and so buy their SOC at the least private cost. They differ at high payment levels: the cap moves each farm once and the farm keeps its herd, while

the outcome payment moves the farm further down its frontier as the price rises and, from $102 to $160 per Mg C, pays it to keep no herd. Three of the five farms reach this point within the observed range of $28 to $128 per Mg C. At these prices, a program buys part of its SOC by ending beef production on the farm. If other farms then raise the calves, they may lose SOC or emit carbon elsewhere, a leakage that a model of one farm does not count. In the model, the farm without a herd earns nothing from its land, and no management system sells or buys the start herd. We expect a land rent or a herd sale to lower the price at which a farm keeps no herd.

Setting a cap requires the program to know where the profit optimum lies. For an outcome payment, the choice of baseline determines who receives the money. Against the start stock, most of the money goes to the heavy continuous farm, which gains SOC under every management system. This outcome is the self-selection that the additionality literature predicts (Ferraro and Pattanayak 2006; Chabé-Ferret and Subervie 2013). The ten-year payments of the experiment are stewardship payments and do not meet a permanence rule for offsets (Climate Action Reserve 2022; Integrity Council for the Voluntary Carbon Market 2025).

These results come from one Michigan site, one weather path, five start histories, and ten years. The cow-calf producer chooses among fixed rules that do not change as SOC changes, so the profit optimum is the best rule in the grid. The rankings count private value and exclude the social value of carbon, and the experiment has no transaction, measurement, or verification costs and no SOC loss after a contract ends. The surrogate inherits the structure of MEMS and overstates farm SOC at high stocking rates. When farm simulations of an earlier specification of the experiment are replayed in MEMS with the same daily actions, the surrogate's farm SOC exceeds the MEMS value by a median of 3.9 Mg C per hectare at 2.0 to 2.2 LSU per hectare, and the difference grows with the stocking rate. The surrogate thus understates the SOC gain of a move to a lower stocking rate, and we expect this bias to make the switch levels too high. The replay does not cover the low stocking rates and the farm without a herd that the outcome payment chooses at high prices. Appendix E.9 reports the replay in full and lists the limits of the animal, grazing, and payment models.

At less than 90 seconds per ten-year simulation on one CPU core, the model can address questions that a fitted response cannot. An evaluation of the management systems across sites, soils, and climate scenarios would show where the rankings hold and how much the response to a payment varies among farms, which a program needs to know to target payments. Monitoring networks such as the 3M project, with four experimental hubs and 59 producer-managed farms and ranches (Stanley et al. 2026), provide field data to calibrate and test the model at such sites. Rules that respond to the farm state can replace the fixed rules, for example a rule that moves the herd when the occupied patch

has no grazeable forage or one that sets the stocking rate each year from the forage. Within the training conditions of the surrogate, the model can also support the dynamic optimization of grazing with SOC as a state. The same model can test payment designs under the frictions of a real program, including a baseline estimated with error, the costs of measurement and verification, and contracts long enough to meet a permanence rule. The structure is not specific to grazing: a farm decision model that needs the daily state of a soil process model, for cover crops, nitrogen, or irrigation, can use it once a surrogate is trained for that model and those decisions.

References

Abatzoglou, John T. 2013. "Development of Gridded Surface Meteorological Data for Ecological Applications and Modelling." *International Journal of Climatology* 33 (1): 121–131. 10.1002/joc.3413.

Adler, Peter B., D. A. Raff, and William K. Lauenroth. 2001. "The Effect of Grazing on the Spatial Heterogeneity of Vegetation." *Oecologia* 128 (4): 465–479. 10.1007/s004420100737.

Aglasan, Serkan, Roderick M. Rejesus, Stephen Hagen, and William Salas. 2024. "Cover Crops, Crop Insurance Losses, and Resilience to Extreme Weather Events." *American Journal of Agricultural Economics* 106 (4): 1410–1434. 10.1111/ajae.12431.

American Grassfed Association. 2024. "Grassfed Ruminant Standards, Version 9.0." American Grassfed Association, Denver, CO.

Antle, John M. and Susan M. Capalbo. 2001. "Econometric-Process Models for Integrated Assessment of Agricultural Production Systems." *American Journal of Agricultural Economics* 83 (2): 389–401. 10.1111/0002-9092.00164.

Antle, John M., Susan M. Capalbo, Siân Mooney, Edward T. Elliott, and Keith H. Paustian. 2003. "Spatial Heterogeneity, Contract Design, and the Efficiency of Carbon Sequestration Policies for Agriculture." *Journal of Environmental Economics and Management* 46 (2): 231–250. 10.1016/S0095-0696(02)00038-4.

Antle, John M., Susan M. Capalbo, Keith Paustian, and Md Kamar Ali. 2007. "Estimating the Economic Potential for Agricultural Soil Carbon Sequestration in the Central United States Using an Aggregate Econometric-Process Simulation Model." *Climatic Change* 80: 145–171. 10.1007/s10584-006-9176-5.

Barnes, M. K., B. E. Norton, M. Maeno, and J. C. Malechek. 2008. "Paddock Size and Stocking Density Affect Spatial Heterogeneity of Grazing." *Rangeland Ecology & Management* 61 (4): 380–388. 10.2111/06-155.1.

Berazneva, Julia, Jon M. Conrad, David T. Güereña, Johannes Lehmann, and Dominic Woolf. 2019. "Agricultural Productivity and Soil Carbon Dynamics: A Bioeconomic Model." *American Journal of Agricultural Economics* 101 (4): 1021–1046. 10.1093/ajae/aaz014.

Blanc, Élodie. 2017. "Statistical Emulators of Maize, Rice, Soybean and Wheat Yields from Global Gridded Crop Models." *Agricultural and Forest Meteorology* 236: 145–161. 10.1016/j.agrformet.2016.12.022.

Bosch, Darrell J., Kurt Stephenson, Gordon Groover, and Blair Hutchins. 2008. "Farm Returns to Carbon Credit Creation with Intensive Rotational Grazing." *Journal of Soil and Water Conservation* 63 (2): 91–98. 10.2489/jswc.63.2.91.

Briske, David D., Steven R. Archer, Emily Burchfield, William Burnidge, Justin D. Derner, Hannah Gosnell, Jerry Hatfield, Clare E. Kazanski, Mona Khalil, Tyler J. Lark, et al. 2023. "Supplying Ecosystem Services on U.S. Rangelands." *Nature Sustainability* 6 (12): 1524–1532. 10.1038/s41893-023-01194-6.

Briske, David D., Justin D. Derner, Joel R. Brown, Samuel D. Fuhlendorf, W. Richard Teague, Kris M. Havstad, Robert L. Gillen, Andrew J. Ash, and Walter D. Willms. 2008. "Rotational Grazing on Rangelands: Reconciliation of Perception and Experimental Evidence." *Rangeland Ecology and Management* 61 (1): 3–17. 10.2111/06-159R.1.

Burt, Oscar R. 1981. "Farm Level Economics of Soil Conservation in the Palouse Area of the

Northwest." *American Journal of Agricultural Economics* 63 (1): 83–92. 10.2307/1239814.

Chabé-Ferret, Sylvain and Julie Subervie. 2013. "How Much Green for the Buck? Estimating Additional and Windfall Effects of French Agro-Environmental Schemes by DID-Matching." *Journal of Environmental Economics and Management* 65 (1): 12–27. 10.1016/j.jeem.2012.09.003.

Chen, Le, Roderick M. Rejesus, Serkan Aglasan, Stephen Hagen, and William Salas. 2023. "The Impact of No-Till on Agricultural Land Values in the United States Midwest." *American Journal of Agricultural Economics* 105 (3): 760–783. 10.1111/ajae.12338.

Chen, Le, Roderick M. Rejesus, Zachary S. Brown, Christopher N. Boyer, and James A. Larson. 2025. "Dynamically Optimal Cover Crop Adoption." *European Review of Agricultural Economics* 52 (3): 301–333. 10.1093/erae/jbaf018.

Chen, Le, Roderick M. Rejesus, Hongqiang Yan, and Serkan Aglasan. 2026. "Lost Ground, Lost Value: Investigating the Relationship Between Soil Erosion and Agricultural Land Value." *American Journal of Agricultural Economics* pp. 1–24. 10.1002/ajae.70057.

Clark, Colin W. 1976. *Mathematical Bioeconomics: The Optimal Management of Renewable Resources*. New York: Wiley.

Climate Action Reserve. 2022. "Soil Enrichment Protocol, Version 1.1." Climate Action Reserve, Los Angeles, CA. 31 May 2022, with the errata of 21 October 2025.

Commonwealth of Australia. 2021. "Carbon Credits (Carbon Farming Initiative—Estimation of Soil Organic Carbon Sequestration using Measurement and Models) Methodology Determination 2021." Federal Register of Legislation, F2021L01696.

Conant, Richard T., Keith Paustian, and Edward T. Elliott. 2001. "Grassland Management and Conversion into Grassland: Effects on Soil Carbon." *Ecological Applications* 11 (2): 343–355. 10.1890/1051-0761(2001)011[0343:GMACIG]2.0.CO;2.

Derner, Justin D., David J. Augustine, David D. Briske, Hailey Wilmer, Lauren M. Porensky, María E. Fernández-Giménez, Dannele E. Peck, and John P. Ritten. 2021. "Can Collaborative Adaptive Management Improve Cattle Production in Multipaddock Grazing Systems?" *Rangeland Ecology and Management* 75: 1–8. 10.1016/j.rama.2020.11.002.

Drewnoski, Mary. 2020. "Producer Questions about Feeding Corn, Corn Stalks, or Corn Byproduct." Nebraska Extension BeefWatch, University of Nebraska-Lincoln, February 2, 2020. Available at https://beef.unl.edu/feedingcorn.

Engel, Stefanie, Stefano Pagiola, and Sven Wunder. 2008. "Designing Payments for Environmental Services in Theory and Practice: An Overview of the Issues." *Ecological Economics* 65 (4): 663–674. 10.1016/j.ecolecon.2008.03.011.

Ferraro, Paul J. and Subhrendu K. Pattanayak. 2006. "Money for Nothing? A Call for Empirical Evaluation of Biodiversity Conservation Investments." *PLoS Biology* 4 (4): e105. 10.1371/journal.pbio.0040105.

Fox, D. G., T. P. Tylutki, L. O. Tedeschi, M. E. Van Amburgh, L. E. Chase, A. N. Pell, T. R. Overton, and J. B. Russell. 2003. "The Net Carbohydrate and Protein System for Evaluating Herd Nutrition and Nutrient Excretion: CNCPS Version 5.0, Model Documentation." Animal Science Mimeo 213, Cornell University, Department of Animal Science, Ithaca, NY.

Franke, James A., Christoph Müller, Joshua Elliott, Alex C. Ruane, Jonas Jägermeyr, Abigail Snyder, Marie Dury, Pete D. Falloon, Christian Folberth, Louis François, et al. 2020. "The GGCMI Phase

2 Emulators: Global Gridded Crop Model Responses to Changes in CO_2, Temperature, Water, and Nitrogen (version 1.0)." *Geoscientific Model Development* 13 (9): 3995–4018. 10.5194/gmd-13-3995-2020.

Gassman, Philip W., Jimmy R. Williams, Xiuying Wang, Ali Saleh, Edward Osei, Larry M. Hauck, R. Cesar Izaurralde, and Joan D. Flowers. 2010. "The Agricultural Policy/Environmental eXtender (APEX) Model: An Emerging Tool for Landscape and Watershed Environmental Analyses." *Transactions of the ASABE* 53 (3): 711–740. 10.13031/2013.30078.

Gersie, Samuel P., David J. Augustine, and Justin D. Derner. 2019. "Cattle Grazing Distribution in Shortgrass Steppe: Influences of Topography and Saline Soils." *Rangeland Ecology & Management* 72 (4): 602–614. 10.1016/j.rama.2019.01.009.

Gold Standard. 2020. "Soil Organic Carbon Framework Methodology, Version 1.0." Gold Standard Foundation, Geneva.

Gordon, H. Scott. 1954. "The Economic Theory of a Common-Property Resource: The Fishery." *Journal of Political Economy* 62 (2): 124–142. 10.1086/257497.

Homans, Frances R. and James E. Wilen. 1997. "A Model of Regulated Open Access Resource Use." *Journal of Environmental Economics and Management* 32: 1–21. 10.1006/jeem.1996.0947.

Huffaker, Ray G. and Kevin Cooper. 1995. "Plant Succession as a Natural Range Restoration Factor in Private Livestock Enterprises." *American Journal of Agricultural Economics* 77: 901–913. 10.2307/1243813.

Huffaker, Ray G. and James E. Wilen. 1991. "Animal Stocking Under Conditions of Declining Forage Nutrients." *American Journal of Agricultural Economics* 73 (4): 1213–1223. 10.2307/1242449.

Integrity Council for the Voluntary Carbon Market. 2025. "Decision on the Approval of a Carbon-Credit Category: U.S. Soil Enrichment Protocol Version 1.1." Decision M45_CAR_US_SEP_v1.1_2025, Integrity Council for the Voluntary Carbon Market. Governing Board decision of 23 October 2025.

Kunkle, William E., Robert S. Sand, and D. Owen Rae. 1994. "Effects of Body Condition on Productivity in Beef Cattle." Extension Special Publication SP-144, Florida Cooperative Extension Service, Institute of Food and Agricultural Sciences, University of Florida.

Lal, Rattan. 2004. "Soil Carbon Sequestration Impacts on Global Climate Change and Food Security." *Science* 304 (5677): 1623–1627. 10.1126/science.1097396.

Liu, Licheng, Wang Zhou, Kaiyu Guan, Bin Peng, Shaoming Xu, Jinyun Tang, Qing Zhu, Jessica Till, Xiaowei Jia, Chongya Jiang, et al. 2024. "Knowledge-Guided Machine Learning Can Improve Carbon Cycle Quantification in Agroecosystems." *Nature Communications* 15: 357. 10.1038/s41467-023-43860-5.

Lu, Dan and Daniel M. Ricciuto. 2019. "Efficient Surrogate Modeling Methods for Large-Scale Earth System Models Based on Machine-Learning Techniques." *Geoscientific Model Development* 12: 1791–1807.

Mason, Charles F. and Andrew J. Plantinga. 2013. "The Additionality Problem with Offsets: Optimal Contracts for Carbon Sequestration in Forests." *Journal of Environmental Economics and Management* 66 (1): 1–14. 10.1016/j.jeem.2013.02.003.

May, Robert M. 1977. "Thresholds and Breakpoints in Ecosystems with a Multiplicity of Stable States." *Nature* 269: 471–477.

McConnell, Kenneth E. 1983. "An Economic Model of Soil Conservation." *American Journal of*

Agricultural Economics 65 (1): 83–89.

Melkonyan, Tigran and Michael H. Taylor. 2013. "Regulatory Policy Design for Agroecosystem Management on Public Rangelands." *American Journal of Agricultural Economics* 95 (3): 606–627. 10.1093/ajae/aas170.

Moore, Andrew D., John R. Donnelly, and Michael Freer. 1997. "GRAZPLAN: Decision Support Systems for Australian Grazing Enterprises. III. Pasture Growth and Soil Moisture Submodels, and the GrassGro DSS." *Agricultural Systems* 55 (4): 535–582. 10.1016/S0308-521X(97)00023-1.

National Academies of Sciences, Engineering, and Medicine. 2016. *Nutrient Requirements of Beef Cattle: Eighth Revised Edition*. Washington, DC: The National Academies Press. 10.17226/19014.

National Academies of Sciences, Engineering, and Medicine. 2021. *Nutrient Requirements of Dairy Cattle: Eighth Revised Edition*. Washington, DC: The National Academies Press. 10.17226/25806.

National Audubon Society. 2023. "Audubon Conservation Ranching Program Protocols." National Audubon Society, New York, NY.

National Research Council. 2000. *Nutrient Requirements of Beef Cattle: Seventh Revised Edition: Update 2000*. Washington, DC: The National Academies Press. 10.17226/9791.

Nguyen, Trung H., Duy Nong, and Keith Paustian. 2019. "Surrogate-Based Multi-Objective Optimization of Management Options for Agricultural Landscapes Using Artificial Neural Networks." *Ecological Modelling* 400: 1–13. 10.1016/j.ecolmodel.2019.02.018.

Noy-Meir, Imanuel. 1975. "Stability of Grazing Systems: An Application of Predator-Prey Graphs." *Journal of Ecology* 63 (2): 459–481.

Odde, K. G., M. E. King, E. D. McCabe, M. J. Smith, K. L. Hill, G. M. Rogers, and K. E. Fike. 2019. "Trends in "Natural" Value-Added Calf Programs at Superior Livestock Video Auction." *Kansas Agricultural Experiment Station Research Reports* 5 (1). 10.4148/2378-5977.7718.

Pates, Nicholas J. and Nathan P. Hendricks. 2020. "Additionality from Payments for Environmental Services with Technology Diffusion." *American Journal of Agricultural Economics* 102 (1): 281–299. 10.1093/ajae/aaz028.

Plastina, Alejandro, Oranuch Wongpiyabovorn, and H. Jo. 2024. "How to Grow and Sell Carbon Credits in US Agriculture." Ag Decision Maker A1-76, Iowa State University Extension and Outreach.

Procton, Alex. 2025. "State of the Voluntary Carbon Market 2025: Meeting the Moment: Renewing Trust in Carbon Finance." Forest Trends' Ecosystem Marketplace, Washington, DC.

Reichstein, Markus, Gustau Camps-Valls, Bjorn Stevens, Martin Jung, Joachim Denzler, Nuno Carvalhais, and Prabhat. 2019. "Deep Learning and Process Understanding for Data-Driven Earth System Science." *Nature* 566: 195–204.

Ritten, John P., Christopher T. Bastian, and Benjamin S. Rashford. 2012. "Profitability of Carbon Sequestration in Western Rangelands of the United States." *Rangeland Ecology and Management* 65 (4): 340–350. 10.2111/REM-D-10-00191.1.

Ritten, John P., W. Marshall Frasier, Christopher T. Bastian, and Stephen T. Gray. 2010. "Optimal Rangeland Stocking Decisions Under Stochastic and Climate-Impacted Weather." *American Journal of Agricultural Economics* 92 (4): 1242–1255. 10.1093/ajae/aaq052.

Rouhi Rad, Mani, Jordan F. Suter, Dale T. Manning, and Christopher Goemans. 2026. "Abatement Subsidies for Groundwater Conservation." *American Journal of Agricultural Economics* 108 (5): 1437–1465. 10.1111/ajae.70021.

Saitone, T. L. 2018. "Value-Added Programs that Really Add Value to Your Cattle?" Siskiyou Stockman, University of California Cooperative Extension, June 2018.

Sanderman, Jonathan and Jeffrey A. Baldock. 2010. "Accounting for Soil Carbon Sequestration in National Inventories: A Soil Scientist's Perspective." *Environmental Research Letters* 5 (3): 034003. 10.1088/1748-9326/5/3/034003.

Santos, Rafael S., Emma K. Hamilton, Paige L. Stanley, Keith Paustian, M. Francesca Cotrufo, and Yao Zhang. 2024. "Simulating Adaptive Grazing Management on Soil Organic Carbon in the Southeast USA Using MEMS 2." *Journal of Environmental Management* 365: 121657. 10.1016/j.jenvman.2024.121657.

Schaefer, Milner B. 1954. "Some Aspects of the Dynamics of Populations Important to the Management of the Commercial Marine Fisheries." *Bulletin of the Inter-American Tropical Tuna Commission* 1 (2): 27–56.

Shang, Linmei, Jifeng Wang, David Schäfer, Thomas Heckelei, Juergen Gall, Franziska Appel, and Hugo Storm. 2024. "Surrogate Modelling of a Detailed Farm-Level Model Using Deep Learning." *Journal of Agricultural Economics* 75 (1): 235–260. 10.1111/1477-9552.12543.

Smith, Vernon L. 1969. "On Models of Commercial Fishing." *Journal of Political Economy* 77 (2): 181–198. 10.1086/259507.

Soil Survey Staff. 2026. "Soil Survey Geographic (SSURGO) Database." Natural Resources Conservation Service, U.S. Department of Agriculture. Available online at https://sdmdataaccess.sc.egov.usda.gov, accessed via USDA Soil Data Access.

Stanley, Paige L., Alexandria S. Kuhl, Isabella C. F. Maciel, M. Francesca Cotrufo, Douglas J. Goodwin, Jennifer Hodbod, et al. 2026. "Merging Rigor and Relevance in Grazingland Research: A Comprehensive Social-Ecological Monitoring Approach." *Carbon Balance and Management* 21: 67. 10.1186/s13021-026-00431-7.

Stanley, Paige L., Chris Wilson, Erica Patterson, Megan B. Machmuller, and M. Francesca Cotrufo. 2024. "Ruminating on Soil Carbon: Applying Current Understanding to Inform Grazing Management." *Global Change Biology* 30 (3): e17223. 10.1111/gcb.17223.

Teague, W. Richard, Steven Apfelbaum, Rattan Lal, Urs P. Kreuter, Jason Rowntree, C. A. Davies, Russ Conser, Mark Rasmussen, Jerry Hatfield, Tong Wang, et al. 2016. "The Role of Ruminants in Reducing Agriculture's Carbon Footprint in North America." *Journal of Soil and Water Conservation* 71 (2): 156–164. 10.2489/jswc.71.2.156.

Torell, L. Allen, Kenneth S. Lyon, and E. Bruce Godfrey. 1991. "Long-Run Versus Short-Run Planning Horizons and the Rangeland Stocking Rate Decision." *American Journal of Agricultural Economics* 73: 795–807.

U.S. Environmental Protection Agency. 2023. "Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances." Docket EPA-HQ-OAR-2021-0317, U.S. Environmental Protection Agency, Washington, DC.

USDA-NASS. 2024. "2022 Census of Agriculture." United States Department of Agriculture, National Agricultural Statistics Service, Washington, DC. Volume 1, Geographic Area Series. Available at https://www.nass.usda.gov/AgCensus/.

USDA-NASS. 2026. “Crop Values Annual Summary.” United States Department of Agriculture, National Agricultural Statistics Service. Editions of 2023 to 2026, Michigan corn price received, marketing years 2020 to 2024.

USDA-NRCS. 2017. “Conservation Practice Standard: Prescribed Grazing, Code 528.” United States Department of Agriculture, Natural Resources Conservation Service, Wisconsin.

USDA-NRCS. 2024a. “Fiscal Year 2025 Michigan CSP Payment Schedule.” United States Department of Agriculture, Natural Resources Conservation Service, East Lansing, MI.

USDA-NRCS. 2024b. “Fiscal Year 2025 Michigan EQIP Payment Schedule.” United States Department of Agriculture, Natural Resources Conservation Service, East Lansing, MI.

USDA-NRCS. 2024c. “Michigan Practice Scenarios, Fiscal Year 2025.” United States Department of Agriculture, Natural Resources Conservation Service, East Lansing, MI.

Verra. 2025. “VM0042 Improved Agricultural Land Management, Version 2.2.” Verra, Washington, DC. Verified Carbon Standard methodology, 21 October 2025.

Wang, Tong, W. Richard Teague, and Seong C. Park. 2016. “Evaluation of Continuous and Multipaddock Grazing on Vegetation and Livestock Performance—A Modeling Approach.” *Rangeland Ecology and Management* 69 (6): 457–464. 10.1016/j.rama.2016.07.003.

Wang, Tong, W. Richard Teague, Seong C. Park, and Stan Bevers. 2018. “Evaluating Long-Term Economic and Ecological Consequences of Continuous and Multi-Paddock Grazing – A Modeling Approach.” *Agricultural Systems* 165: 197–207. 10.1016/j.agsy.2018.06.012.

Wang, Yangchuan, Olga Isengildina-Massa, and Shamar Stewart. 2023. “U.S. Grass-Fed Beef Premiums.” *Agribusiness* 39 (3): 664–690. 10.1002/agr.21790.

Wongpiyabovorn, Oranuch, Alejandro Plastina, and John M. Crespi. 2023. “Challenges to Voluntary Ag Carbon Markets.” *Applied Economic Perspectives and Policy* 45 (2): 1154–1167. 10.1002/aepp.13254.

Zhang, Yao, A. E. King, Emma K. Hamilton, and M. Francesca Cotrufo. 2024. “Representing Cropping Systems with the MEMS 2 Ecosystem Model.” *Agronomy Journal* 116 (5): 2328–2345. 10.1002/agj2.21611.

Zhang, Yao, Rafael S. Santos, Emma K. Hamilton, Paige L. Stanley, Hao Yang, Keith Paustian, Erica L. Patterson, Stephen M. Ogle, Isabella C. F. Maciel, Guilhermo F. S. Congio, et al. 2026. “Modeling the Effects of Livestock Rotation Frequency on Forage Production and Soil Carbon Using a Spatially Explicit Grazing Distribution.” *Preprints (preprints.org)*. 10.20944/preprints202602.0923.v1.

Zimmerman, L. C., T. C. Schroeder, K. C. Dhuyvetter, K. C. Olson, G. L. Stokka, J. T. Seeger, and D. M. Grotelueschen. 2012. “The Effect of Value-Added Management on Calf Prices at Superior Livestock Auction Video Markets.” *Journal of Agricultural and Resource Economics* 37 (1): 128–143. 10.22004/ag.econ.122317.

Appendix A. Related literature and program details

This appendix reviews the literature and the program details behind Section 1.

SOC is the slow-moving stock at the center of the grazing system. It changes little from one year to the next, yet over a decade it governs both the forage a pasture can grow (Lal 2004; Stanley et al. 2024) and the climate benefit a grazing operation can claim (Conant, Paustian, and Elliott 2001; Teague et al. 2016). Grazing differs from the cropland settings of the first soil-as-capital models in the mechanism that links carbon to net value. In cropping, SOC raises crop yield and revenue through an agronomic response. In grazing, the link runs through the effect of SOC on forage quantity and quality and, in our model, on the hay that fills the feed gap on pasture and on the body condition of the animals, which sets the calf crop. With hay as an economic substitute for pasture-supplied energy, the value of SOC depends on its ability to produce more forage within a season, reduce the annual feed bill, and keep the herd in condition.

Treating soil as a depletable capital stock whose value should enter farm-level decisions began with Burt (1981) and McConnell (1983), and the latter derived the shadow value of soil depth. In discrete time, with SOC as the stock, the steady-state shadow value λ satisfies $\lambda^{ss} = (\partial\pi/\partial\mathrm{SOC})/(1 - \delta\,\partial\mathrm{SOC}_{t+1}/\partial\mathrm{SOC}_t)$. We follow Berazneva et al. (2019), who paired an annual SOC equation with an econometric maize-yield function for smallholder farms in Kenya. The two parameters of their SOC equation were fitted to equilibria of the Rothamsted Carbon Model, and the simultaneous solution of the steady-state optimality conditions gave shadow values of $95 to $168 per Mg C. We use the same state variable with a different solution strategy: we compare fixed management systems over ten years instead of solving the steady-state conditions. The production systems also differ. Their link from carbon to revenue is the direct response of crop yield to SOC, and ours runs indirectly through forage quantity and quality and through the annual stored-feed bill, which more abundant forage reduces.

Pairing a calibrated biophysical model with an estimated economic one, as in Berazneva et al. (2019), is an instance of the Econometric-Process Model (EPM) strategy of Antle and Capalbo (2001), in which a biophysical model maps management to yields and stocks and an economic model chooses management in response. Antle et al. (2003) used this strategy to design carbon-sequestration contracts, and Antle et al. (2007) applied it to estimate agricultural carbon supply curves for the central United States. Our coupling of a MEMS surrogate with a comparison of stocking and rotation rules adapts the EPM strategy to a process-based grassland model.

Soil carbon on grazing land, or grassland, accumulates differently from that on cropland. It enters through root turnover and the manure that the herd returns to the pasture, and perennial, untilled pasture avoids the tillage that accelerates carbon loss on cropland (Stanley et al. 2024; Teague et al. 2016). That improved management can build the SOC

stock is well documented. Conant, Paustian, and Elliott (2001) synthesized 115 studies and found a mean gain of 0.54 Mg C per hectare per year, which we use as empirical context for the single-site model. Lal (2004) listed improved grazing among the strategies that increase the soil carbon pool. On cropland, soil-health practices were found to capitalize into farmland prices (Chen et al. 2023, 2026) and to build resilience to climate stress (Aglasan et al. 2024). Chen et al. (2025) solved a cover-crop adoption problem with soil fertility as the state variable, the closest cropland analog to ours, and we draw on their method.

A parallel grazing-bioeconomics literature supplies the dynamic structure. Grazing systems can exhibit nonmonotonic responses. Noy-Meir (1975), generalized by May (1977), showed that moderate stocking admits both a productive and a degraded stable state, separated by a threshold across which a small change in stocking can tip the system into collapse. Huffaker and Cooper (1995) separated rangeland into a fast forage process and a slow successional one and formalized the split as a slow-manifold decomposition, a structure we keep with SOC as the slow stock. Huffaker and Wilen (1991) cast the within-season decline of forage as a fishery-style control problem. Whether the slow stock matters determines whether long-horizon and short-horizon analyses differ. Torell, Lyon, and Godfrey (1991) found the two nearly identical because the falling weight gain of the steers stopped the stocking before the range was damaged, and Ritten et al. (2010) solved the annual stocking problem by stochastic dynamic programming, with logistic forage growth whose rate depends on precipitation. Ritten, Bastian, and Rashford (2012) found carbon payments at prevailing prices well below the management costs they were meant to cover. We therefore track production and SOC within one dynamic system and exclude standalone carbon revenue from the private objective. The value of rotation is contested. Wang et al. (2018) found that multipaddock rotation raises ranch NPV, and Bosch et al. (2008) reported ranch-scale gains. Briske et al. (2008) found no advantage in controlled plot-scale experiments, and Derner et al. (2021) found that adaptive management recovered part of the steer gain lost at high stocking density in a multipaddock rotation, but not enough to cover its cost. These experiments are too short to resolve the decade-scale SOC feedbacks that our simulation captures. Section 5.2 and Appendix E.5 return to the debate.

Grazing differs from cropping in a second respect: cattle use space unevenly within a farm. Using GPS tracking, Gersie, Augustine, and Derner (2019) found that 4 to 16 percent of a pasture went ungrazed while a small fraction received more than four times the average use, and the pattern sharpened as forage quality differentiated through the season. Adler, Raff, and Lauenroth (2001) showed that such preferential use reshapes vegetation into a mosaic of heavily used lawns and lightly used roughs that carries through to the soil. Zhang et al. (2026), the closest modeling predecessor of our patch-based representation, embedded explicit livestock movement in MEMS to capture this within-pasture variation.

A representative-paddock model cannot express this spatial pattern, so we represent the farm as patches grazed preferentially and let SOC diverge across them under management. This focus on within-farm heterogeneity complements the between-site heterogeneity that Antle et al. (2003) used for contract design.

The optimization template comes from renewable-resource theory. Gordon (1954) showed that open access drives a stock below the social optimum, and Schaefer (1954) supplied the logistic growth function. Clark (1976) synthesized fishery economics into the modified golden rule, the direct precursor of our shadow-value formula, and Smith (1969) added the effort dynamics that parallel herd expansion and contraction. Homans and Wilen (1997) showed that a season-length catch limit lets fishing capacity expand until rents dissipate, a result that carries over to practice-based grazing payments that may invite overstocking within a rotation. In our within-season problem, the producer chooses the grazing period conditional on an installed paddock design, and the resulting rest interval is nested inside the across-year Bellman equation for SOC. The daily dynamics of this problem resemble the drawdown of fish stocks in Homans and Wilen (1997), with the producer as the resource manager and with multiple spatially distinct drawdown periods per year.

Process-based models are slow to run, which motivates a surrogate trained on their output for scaling this single-site analysis to many sites and for repeated evaluation of management. Reichstein et al. (2019) argued that hybrid mechanistic-and-learning models suit Earth-system problems, and Liu et al. (2024) applied the idea to cropland carbon by training networks on the *ecosys* model to $R^2 > 0.9$ against its output. Such surrogates can be trained from process simulations (Lu and Ricciuto 2019) and can link mechanistic models to economic integrated assessment (Blanc 2017; Franke et al. 2020). On the economic side, Shang et al. (2024) built a deep-learning surrogate of a farm optimization model. Our structured recurrent surrogate emulates the complete daily MEMS state and takes two management inputs directly: the local LSU per hectare and the hay fed on the patch. With this design, the same farm-management and animal modules evaluate thousands of management systems without explicitly constructing the internal action fluxes of MEMS.

We use the framework to compare three payments for SOC: an outcome payment per Mg C, a practice payment per hectare, and a calf premium for qualifying management. This application relates to the literature on payment design for ecosystem services. In the dryland Northern Great Plains, Antle et al. (2003) found per-hectare (practice-based) carbon contracts to be as much as five times more costly than per-tonne (outcome-based) contracts. Spatial heterogeneity drove the gap, both in soil carbon rates across subregions and in the opportunity cost of changing practices across fields. Rouhi Rad et al. (2026) compared an incremental abatement subsidy, which pays for water use below a threshold, with land retirement for groundwater conservation. The subsidy had the lower budgetary

cost for small reductions in water use and the higher cost for large reductions, because a high uniform threshold also paid users who already used less water. A threshold set for each well from its own past use tended to give the lowest cost to the policymaker. Engel, Pagiola, and Wunder (2008) laid out the three design dimensions of payments for environmental services: conditionality on outcomes, additionality of conservation, and targeting of payments to high-impact sites. Melkonyan and Taylor (2013) compared input regulation, cost-sharing, and performance regulation for public rangelands and found performance regulation theoretically efficient but operationally costly. Ferraro and Pattanayak (2006) gave the additionality problem its sharpest formulation ("money for nothing"), and Pates and Hendricks (2020) extended it to dynamic settings in which technology diffusion erodes additionality over time. Wongpiyabovorn, Plastina, and Crespi (2023) documented the practical barriers to outcome-based payments (fixed certification costs of $5,000 or more and ongoing verification expenses), which may exclude small cow-calf operations regardless of their theoretical efficiency advantage. Wang, Isengildina-Massa, and Stewart (2023) found that spending on food away from home was the main driver of U.S. retail premiums for grass-fed beef and that media information on climate change had no significant effect, a pattern that complicates the use of market premiums as payments for ecosystem services. Retail premiums also differ from the premiums that a cow-calf producer receives for weaned calves. In the years with a significant premium at video auctions, calves eligible for natural programs earned $0.73 to $1.09 per hundredweight more, age-and-source-verified calves $0.99 to $3.23, and calves of the non-hormone-treated cattle program $1.02 to $4.04. Organic calves earned $7.92 per hundredweight more in 2017 (Zimmerman et al. 2012; Odde et al. 2019; Saitone 2018). We found no auction estimate of a calf premium for a grazing standard.

The baseline is the central design choice of an outcome payment, and offset protocols for agricultural soils set it in two ways. Verra VM0042 and the Soil Enrichment Protocol of the Climate Action Reserve model the business-as-usual counterfactual with the weather of the project and credit avoided loss (Verra 2025; Climate Action Reserve 2022). The Australian soil carbon method of 2021 and the Gold Standard methodology credit only a gain above the stock at the start of the project (Commonwealth of Australia 2021; Gold Standard 2020). The difference matters for land that loses SOC, because a relative SOC gain often comes from a reduction in losses rather than an increase in stocks (Sanderman and Baldock 2010). Mason and Plantinga (2013) wrote an offset contract as a per-unit payment plus a lump-sum transfer, which separates the incentive from the payment to landowners who would supply carbon without it. Payments to such landowners are large in practice: a French subsidy for cover crops added 10 hectares of cover crops on the average recipient farm and paid for almost 7 hectares that would have been covered without it (Chabé-Ferret and Subervie 2013).

These literatures have developed largely in parallel. Bioeconomic SOC models focus on cropland, grazing models generally omit detailed SOC dynamics, grazing-land biogeochemistry remains largely absent from existing surrogate applications to crop-process and economic models, and studies of carbon-payment design concentrate on cropland. This paper integrates these strands in a coupled private-profit management framework that treats SOC as productive natural capital and reports it as a separate environmental outcome.

Appendix B. Model details

This appendix states the full farm model behind Section 2. Appendix B.1 writes the dynamic problem of a forward-looking cow-calf producer, and Appendix B.9 derives the shadow value of SOC that this problem implies. The experiment compares fixed rules and does not solve this problem or compute the shadow value.

The model links the whole farm to the local patch in both directions. At the whole-farm level, the producer selects a planned annual herd target, a calendar grazing rule conditional on installed infrastructure, and a feed-gap supplementation. A herd controller converts the herd target into realized cow, heifer, and calf counts, and an animal demand schedule converts these counts and the current body weights into a daily feed target and a demand-equivalent herd input. The calendar restricts the herd to a management paddock, animal selectivity determines the patch actually used, and the resulting local grazing pressure and hay drive the daily forage and soil transitions. The realized forage and hay set the energy balance of each animal class and thereby its body weight on the next day. At weaning and at the year boundary, the herd-composition transition promotes animals between classes, records sales and replacements, and sets the next year's calf crop from cow body condition at calving. Aggregation of the daily transitions and herd outcomes yields the annual SOC, feed cost, and profit of the producer's dynamic problem.

B.1. Whole-farm management and dynamic decision problem

Years are indexed by t and calendar days within a year by d. Let $\mathcal{C} = \{\text{cow}, \text{heifer}, \text{calf}\}$ index animal classes. Let $\mathbf{X}_t$ collect the complete farm biophysical state at the start of year t, including the spatial distributions of vegetation, soil water and temperature, nutrients, surface organic matter, and SOC. Here $X_{k,t,d}$ denotes the state vector of patch k at the end of day d. Let $\mathbf{n}_t = (n_{c,t}, w_{c,t})_{c\in\mathcal{C}}$ collect the opening herd state: the animal count $n_{c,t}$ and the representative body weight $w_{c,t}$ of each class. Finally, let $W_t = (w_{t,d})_d$ denote the year's exogenous daily forcing path, including weather, atmospheric carbon dioxide, and calendar-indexed inputs. Farm area A, the grazing-season start and end dates d_{on} and d_{off}, and the herd-demographic parameters are fixed features of the production setting. The producer's annual operating controls are the planned whole-farm herd target H_t, measured in 500-kilogram reference LSU, the feed-gap supplementation $\phi_t \in [0, 1]$, and, for rotational designs, the calendar grazing-period length gp_t, measured in days spent in each management paddock before advancing. With area fixed, the planned stocking rate is $\text{SR}_t = H_t/A$. Like the herd target, the feed-gap supplementation is fixed for the year, and Appendix B.6 defines the daily hay that it implies. When $N = 1$, the timing control is inactive, and the maximization below runs over H_t and ϕ_t only.

The number of management paddocks N is a discrete, long-lived design choice from the

feasible set $\mathcal{N}$. A change in N requires fencing, which rules out costless annual adjustment. The operating problem is therefore solved conditional on installed N, fence investment is annualized in the farm account, and designs are compared separately. The exogenous forcing affects the forage and soil transitions and, through the stored-feed requirement, current profit. Let $\mathcal{W}_{t:\infty} = (W_t, W_{t+1}, \dots)$ be a specified forcing sequence and let $\delta \in (0,1)$ be the annual discount factor. Conditional on that sequence, the producer's time-dependent operating value is

$$
\begin{aligned}
V_{N,t}(\mathbf{X}_t, \mathbf{n}_t; \mathcal{W}_{t:\infty}) = \max_{H_t,\, \mathrm{gp}_t,\, \phi_t} \Bigg\{ & \\
& \pi_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t) \\
& + \delta V_{N,t+1}\Big(G_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t), \\
& \qquad \mathcal{H}(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t); \mathcal{W}_{t+1:\infty} \Big) \Bigg\}.
\end{aligned} \tag{B1}
$$

The preferred installed design is $N^* \in \arg\max_{N \in \mathcal{N}} V_{N,0}(\mathbf{X}_0, \mathbf{n}_0; \mathcal{W}_{0:\infty})$, with annualized fencing costs included in π_N. The function G_N advances the biophysical state, and $\mathcal{H}$ advances the herd state. The herd transition depends on the pasture and on management, because forage and hay set body weights and cow condition sets the next calf crop. The formulation evaluates management conditional on a deterministic forcing scenario, as does the numerical replay below. Under uncertain forcing, the same equation carries an expectation conditional on the producer's information when H_t, gp_t, and ϕ_t are chosen. The following subsections specify the mechanisms that G_N, $\mathcal{H}$, and π_N summarize.

B.2. Herd composition and transition model

The planned herd target is expressed through class shares, which set the target class counts. Observed opening proportions may differ from these shares, and the demographic controller does not reimpose them each day or year. Let α_c be the planned share of class $c \in \mathcal{C}$, let ℓ_c be its representative LSU coefficient, and let $\bar{\ell} = \sum_{c \in \mathcal{C}} \alpha_c \ell_c$. The planned class target associated with H_t is

$$
\widetilde{n}_{c,t}(H_t) = \alpha_c \frac{H_t}{\bar{\ell}}. \tag{B2}
$$

We use $\boldsymbol{\alpha} = (0.50, 0.10, 0.40)$ and $\boldsymbol{\ell} = (1.00, 0.70, 0.20)$, ordered as cows, replacement heifers, and the annual calf cohort. The vector $\widetilde{\mathbf{n}}_t$ gives the planned management target, and the realized herd composition evolves through the demographic controller. At numerical initialization, the planned cow target is translated into attrition-consistent opening counts of cows and heifers, and the calf cohort remains pending until the calving date. Within

year t, $n_{c,t,d}$ denotes the realized count of class c on day d after any scheduled demographic event.

The general demographic balance of each class is

$$n_{c,t+1} = (1 - \mu_c) n_{c,t}^{\text{end}} + b_{c,t} + p_{c,t}^{\text{buy}} - v_{c,t}^{\text{sale}} - q_{c,t}^{\text{cull}} + \sum_{c' \neq c} \tau_{c' \to c,t} - \sum_{c' \neq c} \tau_{c \to c',t}. \tag{B3}$$

Here $n_{c,t}^{\text{end}}$ is the class count immediately before year-boundary flows, μ_c is mortality, b denotes births, p^{buy} purchases, v^{sale} sales, q^{cull} culls, and τ promotions between classes. In the implemented three-class rule, pending calves enter the live class at calving. At weaning, the surviving calves are split by sex. Male calves and the female calves not needed for replacement are sold, existing heifers are promoted to cows, and the retained female calves become the new heifer cohort. Replacement heifers are purchased only if retained females cannot fill the cohort needed to return the herd toward its target. The cow class holds two cohorts, each with its own count and body weight: the promoted heifers form the first-calf cohort until their first calf is weaned and then join the mature cohort. On the calving date, the cows of cohort g conceive at the rate

$$\rho_{g,t} = \rho\big(\text{BCS}_{g,t}\big), \tag{B4}$$

where $\text{BCS}_{g,t}$ is the body condition score of the cohort on the calving date of year t, a linear function of its body weight, and ρ is an increasing function. At the year boundary, mortality applies and every open cow is culled. When fewer than 12 percent of the cows are open, pregnant cows are also culled up to 12 percent. The cows that remain are all pregnant and form the next pending calf cohort. A herd that loses condition before calving therefore produces fewer calves in the next year and sells more cull cows. These rules define $\mathcal{H}$, supply the sales and replacement flows to the annual accounts, and pass $\mathbf{n}_{t+1}$ to the next annual cycle. The same realized class counts and body weights drive animal demand and farm accounting, and all management systems use the same demographic rates and event dates. Equation (B1) permits annual retargeting of H_t, but the reported numerical comparison holds $H_t \equiv H$ within each management system.

B.3. Calendar movement rule

Let the farm contain K biophysical patches indexed by k, with areas A_k satisfying $\sum_k A_k = A$. The installed design partitions those patches into N non-overlapping management paddocks,

$$\mathcal{G}(N) = \{\mathcal{P}_1, \ldots, \mathcal{P}_N\}, \qquad \bigcup_{j=1}^{N} \mathcal{P}_j = \{1, \ldots, K\}.$$

Biophysical patches are the units at which local state is simulated, and management paddocks are the groups of patches that the farm calendar uses.

The fixed grazing season has length $L_g = d_{\text{off}} - d_{\text{on}} + 1$. Under equal-duration sequential rotation, the full calendar cycle and the scheduled rest interval of a management paddock are

$$c_t = N\,\text{gp}_t, \qquad r_t = (N-1)\,\text{gp}_t. \tag{B5}$$

The producer's primitive timing decision is the grazing-period length, and once N and gp_t are set, the calendar determines rest. Farm area and paddock count alone do not determine a duration.

Let $j_{t,0}$ denote the first management paddock used in year t. During the grazing season, the scheduler holds the herd in one management paddock for gp_t days and then advances it to the next paddock in sequence. The first paddock may rotate across years to distribute incomplete end-of-season cycles. Residual forage is an outcome and never triggers a move to the next management paddock. When $N = 1$, the calendar places no within-farm restriction on the herd. Appendix B.14.1 gives the exact day-index rule and the boundary bookkeeping.

B.4. Animal demand schedule

The animal demand schedule converts the realized class counts and the current body weights into a daily feed target. Let $w_{c,t,d}$ be the representative body weight of class c at the start of day d, and let $D_{c,t,d}$ be the dry matter per head that the class can eat on that day. For cows and heifers, $D_{c,t,d}$ is the dry-matter intake capacity at the current weight, physiological stage, and air temperature. For calves, it is the allowance of solid feed at the current age and weight, which the calf eats in addition to milk. The herd feed target is

$$D_{t,d} = \sum_{c \in \mathcal{C}} n_{c,t,d} D_{c,t,d}.$$

The MEMS grazing interface measures grazing pressure in demand-equivalent LSU. Under the conversion inside MEMS, one demand-equivalent LSU removes $\kappa_{\text{MEMS}} = 15$ kilograms of forage dry matter per day, and the daily herd input is

$$L_{t,d} = \frac{D_{t,d}}{\kappa_{\text{MEMS}}}. \tag{B6}$$

The annual target H_t is the intended whole-farm herd size, and $L_{t,d}$ is the demand-equivalent herd that drives the daily biophysical model. Since $D_{t,d}$ depends on the current weights, a herd that loses weight on day d eats less and places a smaller demand on the

pasture on day $d+1$ (Appendix B.8).

B.5. Animal selective-grazing model

Within the active management paddock, cattle may use patches selectively. Let $\mathcal{E}_{t,d} = \mathcal{P}_{j_{t,d}}$ be the eligible set and let $\Omega(\cdot)$ be the grazing value of a patch as a function of its preceding state, the herd's preceding location, and fixed spatial attributes. The occupied patch is

$$k_{t,d} = \arg\max_{k \in \mathcal{E}_{t,d}} \Omega\Big(X_{k,t,d-1}, k_{t,d-1}, k\Big). \tag{B7}$$

The theory needs only this abstract selection rule, and Appendix C.4 specifies the numerical score. Under continuous grazing, which the model represents by $N = 1$, all K patches remain eligible throughout the grazing season, and animal selective grazing determines the occupied patch through equation (B7). Spatial use can therefore remain uneven. Under calendar-based rotation, $N > 1$, and the active management paddock restricts the eligible set. When $N = K$, every management paddock contains one patch, and the calendar fully determines location.

B.6. Local grazing-pressure mapping

The annual management choices produce daily local grazing pressure

$$u_{k,t,d} = \mathbf{1}\{d_{\text{on}} \le d \le d_{\text{off}}\}\mathbf{1}\{k = k_{t,d}\}\frac{L_{t,d}}{A_k}, \tag{B8}$$

measured in LSU per hectare. The occupied patch receives the whole herd, and every other patch receives zero.

The feed-gap supplementation sets the hay fed on pasture. Let $F_{k,t,d-1}$ be the grazeable forage on patch k at the end of day $d-1$, the forage dry matter above a retained residual, in kilograms. On a grazing day, the feed gap of the herd is $\max\{0, D_{t,d} - F_{k_{t,d},t,d-1}\}$, and the producer fills the fixed share ϕ_t of it with hay on the occupied patch:

$$h_{t,d} = \phi_t \max\big\{0,\ D_{t,d} - F_{k_{t,d},t,d-1}\big\}, \qquad s_{k,t,d} = \mathbf{1}\{k = k_{t,d}\}\frac{h_{t,d}}{10A_k}. \tag{B9}$$

Here $h_{t,d}$ is in kilograms of dry matter, and the local hay input $s_{k,t,d}$ is in grams per square meter per day when A_k is in hectares. The rule sets no minimum amount. With $\phi_t = 0$, the herd eats forage only, and with $\phi_t = 1$, forage and hay together meet the feed target.

B.7. Biophysical transition model

Hay fed on pasture enters the patch transition through the local input $s_{k,t,d}$ of equation (B9), and hay fed off pasture in winter enters no patch. Local grazing pressure enters the transition as a management input and carries no direct carbon or nutrient flux. Let

$$\mathbf{a}_{k,t,d} = \mathcal{A}\left(X_{k,t,d-1}, u_{k,t,d}, s_{k,t,d}\right) \tag{B10}$$

denote the causal action map from the preceding patch state and current management inputs to realized forage removal, trampling, fecal carbon and nitrogen, urine carbon and nitrogen, and hay intake. The animals eat forage first, up to the feed target, and then hay, so realized pasture intake is $I^{\text{past}}_{t,d} = \min\{D_{t,d}, F_{k_{t,d},t,d-1}\}$, and the hay of equation (B9) is eaten in full. The forage available at the start of the day caps realized removal, so all action components are known before the patch state advances. Hay changes the patch state through the excreta of the animals that eat it.

Let z_k denote fixed site characteristics and $w_{t,d}$ the day's exogenous forcing and calendar context. The generic local transition is

$$X_{k,t,d} = F\left(X_{k,t,d-1}, z_k, w_{t,d}, \mathbf{a}_{k,t,d}\right), \tag{B11}$$

where all arguments are predetermined at the transition boundary. The representation leaves the transition technology unspecified, and Appendix C.3 describes its process-model and surrogate implementations.

B.8. Annual farm accounting

Feed account and body weight. The feed account sums the feed that the producer buys. On a grazing day, this feed is the hay $h_{t,d}$ of equation (B9). Off pasture, the herd eats a fixed winter ration of hay $h^{\text{off}}_{t,d}$ and corn grain $m_{t,d}$ under the same rule in every management system. The ration depends on the body weights, so its quantity responds to management through the condition in which the herd leaves the pasture. The annual quantities are

$$S^{g}_{t} = \sum_{d=d_{\text{on}}}^{d_{\text{off}}} h_{t,d}, \qquad S^{\text{off}}_{t} = \sum_{d \notin [d_{\text{on}}, d_{\text{off}}]} h^{\text{off}}_{t,d}, \qquad M_t = \sum_{d \notin [d_{\text{on}}, d_{\text{off}}]} m_{t,d}, \tag{B12}$$

and the annual hay quantity priced as purchased feed is

$$S^{\text{store}}_{t} = S^{\text{off}}_{t} + S^{g}_{t}. \tag{B13}$$

The realized forage and feed are divided among the animal classes, and the diet of each class sets its energy balance. Let $E_{c,t,d}$ be the net energy that the daily diet of one

animal of class c supplies minus its fixed obligations for maintenance, for pregnancy in a pregnant cow, and for lactation in a nursing cow, in megacalories per day. When $E_{c,t,d} < 0$, the animal mobilizes body reserves and loses weight:

$$w_{c,t,d+1} = w_{c,t,d} + \frac{E_{c,t,d}}{r_c}, \tag{B14}$$

where r_c is the net energy that one kilogram of body weight supplies. When $E_{c,t,d} \geq 0$, the animal gains the weight that the surplus energy supports. The new weight sets the next day's feed target through the demand schedule of Appendix B.4. A feed-gap supplementation below one therefore lowers the hay bill, but in a forage shortage it also lowers body weight, the next day's grazing pressure, and, through equation (B4), the next calf crop. Appendix C.5 gives the allocation among classes, the energy terms, and the weight limits.

Farm state and profit accounts. Let $\mathbf{X}_t = (X_{1,t}, \ldots, X_{K,t})$ collect the patch states at the start of year t, and let $\mathrm{SOC}(X_{k,t})$ extract total SOC from patch k. Farm-average SOC is

$$\mathrm{SOC}_t = \sum_{k=1}^{K} \frac{A_k}{A} \, \mathrm{SOC}(X_{k,t}). \tag{B15}$$

Composing equation (B11) over all days and patches under the calendar, selection, feed, and local-pressure rules defines the annual farm transition

$$\mathbf{X}_{t+1} = G_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t), \qquad \mathrm{SOC}_{t+1} = g_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t), \tag{B16}$$

with $\mathbf{n}_{t+1} = \mathcal{H}(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t)$, where W_t is the annual exogenous forcing path. Besides farm-average SOC, the complete transition depends on the spatial distribution of state and on the herd counts and weights. The scalar function g_N used below is therefore a reduced-form SOC projection of the full daily system, conditional on its other state components, the herd state, and forcing.

Annual profit conditional on installed design N is

$$\begin{aligned} \pi_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t) = {} & R_{\mathrm{calf}} + R_{\mathrm{cull}} - C_{\mathrm{feed}} - C_{\mathrm{herd}} - C_{\mathrm{min}} \\ & - C_{\mathrm{labor}}(N, \mathrm{gp}_t) - C_{\mathrm{fence}}(N) - C_{\mathrm{repl}}, \end{aligned} \tag{B17}$$

with every term evaluated for year t under forcing path W_t. Calf revenue is the number of calves sold times their realized weaning weight, and cull revenue is the number of culled cows times their realized weight. Feed cost C_{feed} prices the hay quantity in equation (B13) and the corn M_t of equation (B12). Herd nonfeed cost and mineral cost are recorded separately as C_{herd} and C_{min}. Rotation affects labor cost through the scheduled moves

between management paddocks and fencing cost through installed N, with the durable fence cost converted to an annual equivalent. Higher SOC can raise private profit by increasing forage quantity and quality, which lowers C_{feed} when $\phi_t > 0$ and raises animal weights and the calf crop when the feed gap is not filled. Terminal SOC and the carbon benchmark are excluded from π_N and reported as separate environmental criteria.

The model distinguishes forage, SOC, myopic-profit, and forward-looking management benchmarks, which Appendix B.14.2 defines formally. A longer calendar grazing period simultaneously lengthens scheduled rest, changes the movement frequency, and changes how animal selective grazing distributes use across patches. The resulting ranking of these benchmarks is an empirical property of G_N and π_N, which the calendar identity alone does not determine.

B.9. Dynamic optimization and SOC valuation

Equation (B1) gives the private-profit operating problem conditional on installed infrastructure, and the outer comparison over N reflects the discreteness and durability of paddock design.

For a transparent scalar expression of the shadow value, we condition on the non-SOC components of $\mathbf{X}_t$, the herd state $\mathbf{n}_t$, and a stationary reference forcing environment $\overline{W}$ and evaluate both profit and the transition under that environment. Suppressing these conditioned arguments gives the autonomous reduced operating problem

$$\widetilde{V}_N(\mathrm{SOC}_t) = \max_{H_t,\, \mathrm{gp}_t,\, \phi_t} \left\{ \pi_N(\mathrm{SOC}_t, H_t, \mathrm{gp}_t, \phi_t) + \delta \widetilde{V}_N(g_N(\mathrm{SOC}_t, H_t, \mathrm{gp}_t, \phi_t)) \right\}.$$

Let $\lambda_{t+1} = \widetilde{V}'_N(\mathrm{SOC}_{t+1})$ denote the private marginal continuation value of farm-average SOC. For a rotational design $N \geq 2$, an interior solution satisfies

$$\frac{\partial \pi_N}{\partial H_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial H_t} = 0, \tag{B18}$$

$$\frac{\partial \pi_N}{\partial \mathrm{gp}_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial \mathrm{gp}_t} = 0, \tag{B19}$$

$$\frac{\partial \pi_N}{\partial \phi_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial \phi_t} = 0. \tag{B20}$$

Each condition balances a current margin against a discounted SOC effect. The current margin is the marginal return to another planned LSU in the herd condition and the feed, labor, and movement consequences of a change in the grazing period in the calendar condition. The supplementation condition weighs the current hay cost against the weight and calf crop that the hay preserves and against its discounted SOC effect. Hay changes SOC through the excreta that it adds to the occupied patch and through the herd weight,

which sets later grazing pressure. With $\phi_t \in [0,1]$, the supplementation condition holds with equality only at an interior level: at $\phi_t = 0$ its left side is not positive, and at $\phi_t = 1$ it is not negative. With the planned herd target and installed N held fixed, let $\mathrm{gp}^m(H;N)$ denote a unique interior myopic timing optimum, at which $\partial\pi_N/\partial\mathrm{gp} = 0$. If $\lambda_{t+1} > 0$, $\partial g_N/\partial\mathrm{gp} > 0$ there, and the dynamic objective is strictly concave in gp over the feasible interval, the forward-looking timing choice satisfies $\mathrm{gp}^d(H;N) > \mathrm{gp}^m(H;N)$, and by equation (4) the scheduled rest is longer as well. This comparative result applies only to timing conditional on fixed H, ϕ, and N. Joint adjustment of the herd target, timing, and feed-gap supplementation requires the full system of equations (B18) to (B20), and the forage, profit, and SOC optima may follow different rankings.

Applying the envelope theorem and imposing a stationary scalar equilibrium yields the private steady-state shadow value of SOC:

$$\lambda_N^{ss} = \frac{\partial\pi_N/\partial\mathrm{SOC}}{1 - \delta\,\partial g_N/\partial\mathrm{SOC}}. \tag{B21}$$

It rises with the marginal profit of SOC, with the persistence of SOC, and with the producer's patience. The scalar expression is exact for the autonomous reduced problem under $\overline{W}$, and direct computation of the steady-state shadow value requires this stationary reference environment. The numerical experiment instead replays ten years of historical weather, a nonstationary finite forcing path, and the resulting paths support a finite-horizon comparison of management. The numerical model retains the complete patch-state vector and can evaluate how the reduction depends on the distribution of state across the farm.

For each installed N and a fixed feed-gap supplementation, Appendix B.14.4 gives four stationary equations for the planned herd target, the grazing period, SOC, and the shadow value of SOC, and a free feed-gap supplementation adds equation (B20) to the system. Eliminating λ_N^{ss} from the herd and calendar conditions gives the operating-management isoquant

$$\frac{\partial\pi_N/\partial H}{\partial\pi_N/\partial\mathrm{gp}} = \frac{\partial g_N/\partial H}{\partial g_N/\partial\mathrm{gp}}. \tag{B22}$$

The isoquant compares the two annual operating margins conditional on N. Because the design choice is discrete, installed paddock designs are compared directly by their values.

B.10. Payments for SOC

A payment adds a transfer P_t to annual profit, so the producer maximizes the discounted sum of $\pi_N + P_t$ in equation (B1). We consider three instruments. Let Ω be a set of qualifying management choices, for example rotation ($N \geq 2$) with a planned stocking rate at or

below a cap $\overline{\mathrm{SR}}$, and let $\mathbb{1}_{\mathcal{Q}}$ equal one when the management of year t belongs to $\mathcal{Q}$. The outcome payment at rate p per Mg C, the practice payment at rate r per hectare, and the calf premium at share q of the calf price are

$$P_t^{\mathrm{out}} = p\,A\left(\mathrm{SOC}_{t+1} - \mathrm{SOC}_t - b_t\right), \qquad P_t^{\mathrm{pr}} = r\,A\,\mathbb{1}_{\mathcal{Q}}, \qquad P_t^{\mathrm{prem}} = q\,R_{\mathrm{calf}}\,\mathbb{1}_{\mathcal{Q}}, \tag{B23}$$

where b_t is the baseline SOC change of year t.

The outcome payment is linear in the SOC transition g_N, so it adds pA to the discounted shadow value in each first-order condition. For the herd target, equation (B18) becomes

$$\frac{\partial \pi_N}{\partial H_t} + \left(\delta \lambda_{t+1} + pA\right)\frac{\partial g_N}{\partial H_t} = 0, \tag{B24}$$

and the grazing-period and supplementation conditions change in the same way. The rate therefore sets the response. If the producer cannot affect b_t, the baseline drops out of these conditions. It changes the size of the transfer and, in a voluntary program, the decision to join, but not the management of a participant. The contracts of Mason and Plantinga (2013) used the same split into a per-unit payment and a lump-sum transfer. A producer joins if the highest NPV with the payment exceeds the highest NPV without it. Two baselines bracket the offset protocols of Appendix A. Let the no-payment management be the optimum of the farm without a payment, with SOC path SOC_t^B. With the no-payment change as the baseline, $b_t = \mathrm{SOC}_{t+1}^B - \mathrm{SOC}_t^B$, the no-payment management earns nothing, every payment is for additional SOC, and the producer never loses by joining. With the start stock as the baseline, $b_t = 0$, a farm whose SOC falls under its no-payment management would owe money at that management. It therefore joins only if the payment under some other management exceeds the NPV that the change gives up. A one-sided rule that pays $pA \max(0, \mathrm{SOC}_{t+1} - \mathrm{SOC}_t)$ removes the debt, but it pays nothing for a smaller loss, and the marginal value pA applies only in years with a gain.

The practice payment is constant on $\mathcal{Q}$, so it leaves equations (B18) to (B20) unchanged within the qualifying set. It acts only through a discrete comparison: the producer moves to the best qualifying management if the present value of the payment exceeds the NPV given up. If the no-payment management already qualifies, the payment changes nothing and is a pure transfer. A cap $\overline{\mathrm{SR}}$ below the stocking rate of the no-payment management makes the payment bind. The SOC that the payment then buys is the difference in SOC between the no-payment management and the best management at the cap. The calf premium scales calf revenue on $\mathcal{Q}$: within the qualifying set, it multiplies the calf-revenue term $\partial R_{\mathrm{calf}}/\partial H_t$ of equation (B18) by $1 + q$. Where calf revenue rises with the herd target, the premium therefore raises the private return to a higher stocking rate, which works against the SOC goal. A stocking-rate cap in $\mathcal{Q}$ limits this effect, and the premium then acts like a practice payment whose size is proportional to calf revenue.

B.11. Spatial aggregation

The farm-management layer separates management paddocks from biophysical patches. Cattle concentrate on preferred locations and leave others lightly used (Gersie, Augustine, and Derner 2019), and this preferential use reshapes vegetation and, through it, soil (Adler, Raff, and Lauenroth 2001). Equation (B7) therefore lets patch states diverge even when all patches start with the same soil and weather. Farm-average SOC in equation (B15) aggregates these heterogeneous stocks, and with a nonlinear daily transition, its shadow value may differ from the area-weighted average of the patch-specific values. The reduced scalar problem provides the transparent shadow-value expression, and the full transition G_N preserves the spatial distribution needed to evaluate it.

B.12. Theory-to-numerical mapping

The steady-state system (SS-1) to (SS-4) and the operating-management isoquant (B22) are reduced-form implications conditional on an installed N and stationary reference forcing. The theory establishes the decision structure and one conditional comparative result. With the planned herd target fixed, a forward-looking producer chooses a longer calendar grazing period than a myopic producer when the marginal SOC effect of gp remains positive at the myopic optimum. The ranking of the forage, profit, and SOC optima and the magnitude of λ_N^{ss} are empirical questions, since both depend on the daily transition and profit functions under the chosen reference environment.

We give the framework empirical content by specifying each component of equations (B10) and (B11). The numerical animal model supplies the feed target $D_{t,d}$, the demand-equivalent daily herd input $L_{t,d}$, the class energy balances, and the weight changes, and the biophysical transition supplies the grazeable forage, the realized pasture intake, and its quality. Equation (B9) sets the hay on pasture, and a fixed winter ration sets the feed off pasture. Calf weaning weights, cull weights, and the calf crop therefore respond to management through the body weights.

The analytical and numerical models use the same controls. Both condition on an installed paddock count N, treat the planned herd target H, the grazing-period length gp, and the feed-gap supplementation ϕ as the operating margins, hold the grazing-season dates fixed, and derive the scheduled rest of a management paddock as $(N-1)\text{gp}$. Both apply selective patch use within the active paddock and map the resulting location and the demand-equivalent herd $L_{t,d}$ into daily local LSU per hectare. The numerical experiment restricts $H_t \equiv H$ and $\phi_t \equiv \phi$, evaluates a discrete grid of fixed management systems with the surrogate, and conditions on one replayed weather path. The Bellman problem provides the theoretical benchmark for future dynamic optimization. The numerical private objective is ten-year discounted profit, and terminal SOC is reported separately.

The purchased-feed account follows equations (B12) and (B13), and the hay on pasture enters the biophysical transition through equation (B9).

B.13. Outcome payments against the start stock and above a threshold

Two contrast cases of the outcome payment use the start stock as the baseline (Commonwealth of Australia 2021; Gold Standard 2020). One pays only the annual gains, with present value $pA \sum_t \delta^t \max\{0, \Delta_t(a)\}$. The other pays the net change, $pA \sum_t \delta^t \Delta_t(a)$, and the farm joins only if the payment raises its NPV above $\mathrm{NPV}(a^0)$. This payment differs from equation (5) by a lump sum, so a farm that joins chooses the same management system under both (Mason and Plantinga 2013). Most of the transfer of the two start-stock designs goes to the heavy continuous farm, which gains SOC under every management system. A variant after Rouhi Rad et al. (2026) pays $\delta^9 pA \max\{0, \Delta\mathrm{SOC}(a) - \theta\}$ at the end of 2035, with either one threshold θ for all farms or one at each farm's own no-payment change plus a margin. At \$100 per Mg C, the threshold at each farm's own no-payment change buys 2,136 Mg C over the five farms for \$137,707 (Appendix E.7).

B.14. Theory derivations

This subsection gives the bookkeeping and envelope steps behind Section 2.3 and Appendix B.9.

B.14.1. Calendar identity and local grazing input

Under the equal-duration calendar, a complete cycle contains N consecutive grazing periods of gp_t days, so its length is $c_t = N\mathrm{gp}_t$. A management paddock is active for one such period and is inactive while the other $N - 1$ paddocks are active. Its scheduled rest is therefore

$$r_t = c_t - \mathrm{gp}_t = (N-1)\mathrm{gp}_t, \tag{B25}$$

which gives equation (4). The identity is calendar bookkeeping: biomass and residual forage respond to the schedule, but the herd advances by the calendar alone.

Let $j_{t,0}$ be the first management paddock used in year t. During the grazing season, the exact scheduler is

$$j_{t,d} = 1 + \mathrm{mod}\left(j_{t,0} - 1 + \left\lfloor \frac{d - d_{\mathrm{on}}}{\mathrm{gp}_t} \right\rfloor, N\right). \tag{B26}$$

If the grazing season ends partway through a cycle, the final realized interval can be truncated, and rotating the starting paddock across years distributes this boundary effect while preserving the planned identity.

The rest of a management paddock differs from the biological rest of an individual patch. Equation (B26) makes every patch in the active management paddock eligible, but under equation (B7) the herd occupies only the patch with the highest grazing value. An unselected patch can therefore remain ungrazed while its management paddock is active, and all patches outside the active paddock receive zero local LSU by construction.

The local-input mapping concentrates the whole-farm herd spatially without changing its size. With exactly one patch occupied during the grazing season, equation (B8) implies

$$\sum_{k=1}^{K} \frac{A_k}{A} u_{k,t,d} = \frac{L_{t,d}}{A}. \tag{B27}$$

The right-hand side is the demand-equivalent whole-farm LSU per hectare, and the occupied patch receives $L_{t,d}/A_k$. The grazing regime determines the eligible set and its timing, animal selective grazing chooses the occupied patch, and the farm-management layer passes the resulting daily local LSU per hectare to MEMS or the surrogate, together with the hay input of equation (B9), which is colocated with the herd.

B.14.2. Management benchmarks

Let $Y_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t)$ be the pasture forage supplied to the herd during the year, and let $\Delta\mathrm{SOC}_N = g_N - \mathrm{SOC}_t$ be the annual SOC change. Conditional on the opening state, H_t, ϕ_t, and N, the forage and SOC benchmarks are

$$\mathrm{gp}_F^* \in \arg\max_{\mathrm{gp}} Y_N, \qquad \mathrm{gp}_{\mathrm{SOC}}^* \in \arg\max_{\mathrm{gp}} \Delta\mathrm{SOC}_N.$$

The myopic producer chooses

$$(H_\pi^*, \mathrm{gp}_\pi^*, \phi_\pi^*) \in \arg\max_{H,\mathrm{gp},\phi} \pi_N(\mathbf{X}_t, \mathbf{n}_t, H, \mathrm{gp}, \phi),$$

and the forward-looking producer solves equation (B1). These definitions separate biological and current-profit benchmarks from the dynamic choice and leave their ordering unrestricted.

B.14.3. Envelope derivation of the shadow value

The shadow value (B21) follows from the envelope theorem applied to the reduced scalar, stationary-reference-forcing version of equation (B1) defined in Appendix B.9. Write S for SOC, and let $H^*(S)$, $\mathrm{gp}^*(S)$, and $\phi^*(S)$ denote the maximizing controls conditional on installed N. Subscripts on π_N and g_N below denote partial derivatives. Differentiating

$\widetilde{V}_N(S) = \pi_N(S, H^*, \mathrm{gp}^*, \phi^*) + \delta \widetilde{V}_N(g_N(S, H^*, \mathrm{gp}^*, \phi^*))$ gives

$$\begin{aligned}\widetilde{V}'_N(S) = \pi_{N,S} &+ \delta \widetilde{V}'_N(g_N) g_{N,S} \\ &+ \left[\pi_{N,H} + \delta \widetilde{V}'_N(g_N) g_{N,H}\right] \frac{dH^*}{dS} + \left[\pi_{N,\mathrm{gp}} + \delta \widetilde{V}'_N(g_N) g_{N,\mathrm{gp}}\right] \frac{d\mathrm{gp}^*}{dS} \\ &+ \left[\pi_{N,\phi} + \delta \widetilde{V}'_N(g_N) g_{N,\phi}\right] \frac{d\phi^*}{dS}. \end{aligned} \tag{B28}$$

The three bracketed terms are the operating first-order conditions (B18) to (B20), and they vanish at an interior optimum. At a corner feed-gap supplementation, ϕ^* stays at the corner for small changes in S, which gives $d\phi^*/dS = 0$, and the third term vanishes as well. The envelope condition is therefore

$$\widetilde{V}'_N(S) = \pi_{N,S} + \delta \widetilde{V}'_N(g_N) g_{N,S}. \tag{B29}$$

Writing $\lambda_t = \widetilde{V}'_N(\mathrm{SOC}_t)$ and imposing a stationary scalar equilibrium gives $\lambda_N^{ss} = \pi_{N,S} + \delta \lambda_N^{ss} g_{N,S}$. Solving this equation yields equation (B21). The value is positive and finite when $\pi_{N,S} > 0$ and $\delta g_{N,S} < 1$, that is, when SOC raises current profit, for example through avoided hay purchases or heavier animals, and its discounted persistence is less than one.

B.14.4. Stationary operating system

Conditional on installed N and a fixed feed-gap supplementation ϕ, the stationary system has four unknowns, $(H^*, \mathrm{gp}^*, \mathrm{SOC}^{ss}, \lambda_N^{ss})$, and four equations:

$$\frac{\partial \pi_N}{\partial H} + \delta \lambda_N^{ss} \frac{\partial g_N}{\partial H} = 0, \tag{SS-1}$$

$$\frac{\partial \pi_N}{\partial \mathrm{gp}} + \delta \lambda_N^{ss} \frac{\partial g_N}{\partial \mathrm{gp}} = 0, \tag{SS-2}$$

$$g_N(\mathrm{SOC}^{ss}, H^*, \mathrm{gp}^*) = \mathrm{SOC}^{ss}, \tag{SS-3}$$

$$\lambda_N^{ss} = \frac{\partial \pi_N / \partial \mathrm{SOC}}{1 - \delta\, \partial g_N / \partial \mathrm{SOC}}. \tag{SS-4}$$

Here H^* is the stationary planned herd target. The first two equations determine the operating margins, the third is the SOC fixed point, and the fourth is the envelope-based shadow value. A free feed-gap supplementation adds the unknown ϕ^* and the stationary form of equation (B20), with the corner conditions of Appendix B.9. The discrete installed-paddock designs are compared directly by value and have no additional first-order condition.

B.14.5. Conditional dynamic extension of the grazing period

Fix the current state, installed design N, planned herd target H, and feed-gap supplementation ϕ, and suppress ϕ in the notation. Let $\mathrm{gp}_m = \mathrm{gp}^m(H;N)$ be an interior maximizer of current profit, so $\pi_{N,\mathrm{gp}}(S,H,\mathrm{gp}_m) = 0$. The derivative of the dynamic objective with respect to the grazing period, evaluated at that myopic choice, is

$$\frac{\partial}{\partial \mathrm{gp}}\left[\pi_N(S,H,\mathrm{gp}) + \delta \widetilde{V}_N(g_N(S,H,\mathrm{gp}))\right]\Big|_{\mathrm{gp}=\mathrm{gp}_m} = \delta \lambda_{t+1} g_{N,\mathrm{gp}}(S,H,\mathrm{gp}_m). \tag{B30}$$

If $\lambda_{t+1} > 0$ and $g_{N,\mathrm{gp}} > 0$ at gp_m, this derivative is positive. If, in addition, the dynamic objective is strictly concave in gp over the relevant interval, its conditional maximizer lies to the right: $\mathrm{gp}^d(H;N) > \mathrm{gp}^m(H;N)$. Equation (B25) then gives $r^d(H;N) > r^m(H;N)$ for $N > 1$. The result holds conditional on H and N. When the herd target, timing, and feed-gap supplementation change together, the full system of equations (B18) to (B20) governs their interaction and leaves the coordinate-by-coordinate ordering unrestricted.

Appendix C. Numerical implementation and calibration

This appendix specifies the herd, movement, nutrition, coupling, and accounting rules of the numerical model in Section 3, together with its assumptions and parameter values.

C.1. Herd composition and grazing calendar

The planned targets for mature cows, replacement heifers, and annual calves use class shares of 0.50, 0.10, and 0.40 and representative coefficients of 1.00, 0.70, and 0.20 LSU per head relative to a 500-kilogram reference animal. These values convert the planned annual herd target H to a cow target. The realized opening class proportions emerge from the demographic controller, which initializes attrition-consistent cow and heifer counts around that target, holds the calves as a pending cohort until calving, and then supplies the same daily class counts to the animal-demand and accounting modules. Cows calve on day 75, and calves are weaned on day 270 at their simulated weight after a pre-weaning mortality of 3.0 percent. At weaning, the existing heifer cohort is promoted to cows, the surviving calves are split equally by sex, and the retention controller keeps enough female calves to form the next replacement-heifer cohort. Male and surplus female calves are sold, and any replacement shortfall is purchased. The promoted heifers calve for the first time the following spring as the first-calf cohort. At the year boundary, cow and heifer mortality is 1.5 percent, every open cow is culled, and at least 12 percent of the cows are culled. The resulting counts and the next pending calf cohort from equation (B4) initialize the next annual cycle. Following the Michigan grazing calendar, cattle graze on pasture from day 120 to 270 and eat stored feed off pasture from late September through April.

C.2. Integrated computational architecture

Figure C1 arranges the model components by time scale and shows how they nest. The annual setup consists of the whole-farm management system and the realized opening herd. On each day, the calendar and the animal demand schedule provide parallel inputs, animal selective grazing chooses the occupied patch, and the local LSU per hectare and the hay of equation (B9) drive the management-responsive biophysical transition. MEMS constructs the corresponding removal, trampling, and animal-return fluxes internally. The realized forage and hay set the body weights for the next day, and the resulting patch state becomes the input for that day. At calving, weaning, and the year boundary, the herd-composition controller updates the class counts and sets the pending calf cohort of the next year. A single annual farm account combines the two hay streams, the winter corn, the herd outcomes, SOC, and all revenue and cost items. The ten-year management comparison sits outside the one-year box and aggregates the repeated annual cycles.

Inside the farm loop, the management-responsive surrogate of MEMS (Appendix C.3) provides this transition from the preceding patch state, the site and weather context, the daily local LSU per hectare, and the hay input. Every simulation runs on the surrogate. Hay fed on pasture enters both the surrogate and the economic account, and winter hay and corn enter only the economic account.

C.3. MEMS process model and patch representation

MEMS is a process-based model of carbon and nitrogen cycling among soil layers, vegetation, and the atmosphere at daily resolution (Zhang et al. 2024; Santos et al. 2024). It resolves SOC into measurable pools whose turnover responds to weather, plant inputs, and grazing offtake. We take this pool structure as given. The outputs relevant to this paper are the daily aboveground forage state and the annual farm-level SOC stock.

We represent the 60-hectare farm as K biophysically identical patches, which the surrogate advances with one shared soil profile, weather sequence, and forage species mix. The patches differ only in their dynamic state, the biomass and leaf-to-stem composition that grazing history produces. With a patch area of 5 hectares, the farm contains $K = 12$ patches. This resolution, at which grazing within a patch is treated as approximately uniform, is held fixed across all management comparisons.

C.4. Calendar movement and selective patch use

The numerical implementation specifies the abstract grazing value in equation (B7) as

$$\Omega_{k,t,d} = P_k\big[1 - \mathcal{T}(k_{t,d-1} \to k)\big]\mathcal{I}(B_{k,t,d-1})Q^{\gamma}_{k,t,d-1}, \tag{C1}$$

where P_k is a fixed spatial-preference weight for non-forage attractors such as water, $\mathcal{T}$ is the travel-cost term, $\mathcal{I}(B)$ is a saturating (Type II) intake response to biomass B, Q is the live-leaf share, and γ sets the strength of quality selection. The score is read from the state at the end of the preceding day, so selection is causal.

The two grazing regimes share the patches, the planned herd target, and the season length and differ only in how the calendar restricts the eligible area of the herd. Under continuous grazing, represented by $N = 1$, all K patches remain eligible throughout the grazing season. Under calendar-based rotation, equation (B26) activates one of $N > 1$ management paddocks for gp days. In both regimes, equations (C1) and (B7) implement animal selective grazing by choosing the patch with the highest grazing value within the eligible area. At a fixed planned herd target, the contrast between the regimes isolates the within-farm redistribution channel of Section B.11. Figure C2 shows the patch layout and the two regimes.

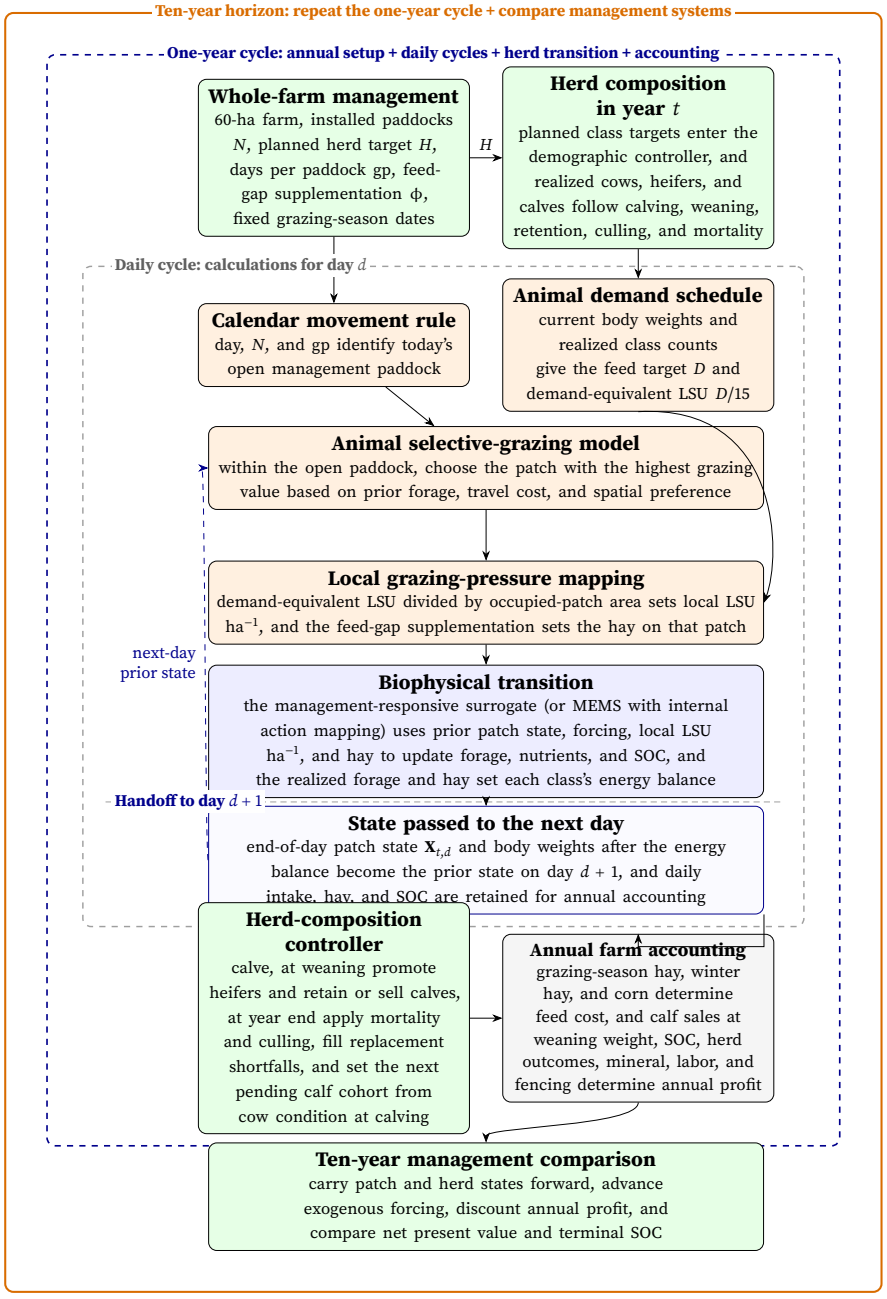


FIGURE C1. Temporal nesting and information flow. The annual setup fixes the whole-farm management system and the realized herd composition. On day d, the calendar and animal demand determine the eligible area and the demand-equivalent herd. Animal selective grazing, the local-pressure and hay mapping, and the biophysical transition then produce the patch state and the body weights used on day $d+1$. Once the daily cycle has run over the year, a one-way handoff carries the accumulated daily outcomes into the annual farm account, which remains downstream of the daily transition. The herd controller and the farm account close the year. The outer loop carries the patch and herd states and the exogenous forcing across ten annual cycles and compares discounted private profit and terminal SOC.

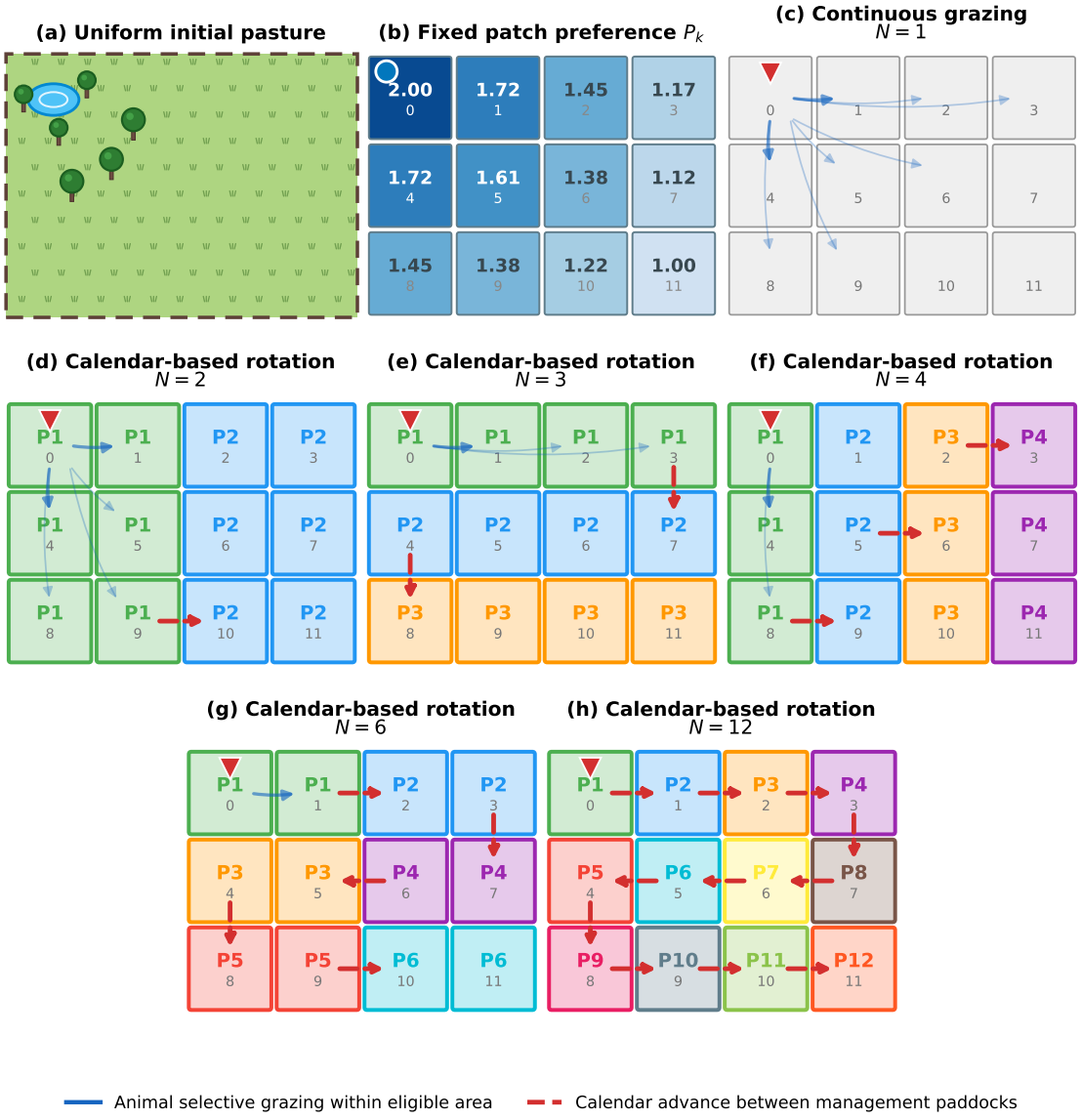


FIGURE C2. **Continuous grazing, calendar-based rotation, and animal selective grazing.** The 60-hectare farm contains $K = 12$ equal 5-hectare biophysical patches. Panel (a) shows their common initial pasture conditions and the fixed corner water-and-shade attractor, and panel (b) shows the resulting patch-preference gradient P_k. Under continuous grazing (c), represented by $N = 1$, all patches remain eligible throughout the grazing season. Under calendar-based rotation (d) to (h), equation (B26) activates one of $N > 1$ management paddocks for gp days. In both regimes, equations (C1) and (B7) select the highest-value eligible biophysical patch. When $N = 12$, each management paddock contains one patch, and the calendar fully determines location. Colors identify management-paddock membership, blue arrows show animal selective grazing within the eligible area, and red dashed arrows show calendar advances between management paddocks.

C.5. Herd nutrition, body weight, and feed accounting

The animal model follows National Research Council relationships (National Academies of Sciences, Engineering, and Medicine 2016). The herd has three classes (cows, replacement heifers, and calves) with the planned shares and reference LSU coefficients of Appendix C.1, and the demographic controller determines realized class counts that can differ from the planned shares. Let $n_{c,t,d}$ be the number of animals in class c on day d and $w_{c,t,d}$ their representative shrunk body weight, which changes from day to day, and the cow class carries one weight for each of its two cohorts. The herd starts on 1 January 2026 with pregnant mature cows of 527 kilograms at BCS 5 on the nine-point scale and with heifers of 255 kilograms, the median 1 January weight of the later heifer cohorts in a preliminary simulation of the model.

The feed target per head $D_{c,t,d}$ is the intake capacity of the class. For cows and heifers, it is the dry-matter intake capacity as a fraction of body weight, 2.4 percent for a lactating cow, 2.1 percent for a dry cow, and 2.5 percent for a heifer, adjusted for air temperature (National Academies of Sciences, Engineering, and Medicine 2016). For nursing calves, for which the nutrient-requirement models have no intake equation, the model sets a solid-feed allowance that depends on calf age and weight, and the calves also drink milk.

On a grazing day, let C_j and N_j be the carbon and nitrogen of forage pool j of the occupied patch at the end of the previous day, in grams per square meter. The five plant pools are the live and dead leaf, live and dead stem, and seed of the MEMS crop output, and the three standing dead pools are the leaf, stem, and seed of the MEMS standing dead biomass. The dry matter of a pool is C_j/f_j, with $f_j = 0.48$ for the plant seed and $f_j = 0.43$ for every other pool. With P and S the plant and standing dead dry matter, the grazeable forage of the 5-hectare patch, in kilograms, is

$$F_{k,t,d-1} = 10A_k \max\{0,\ P + S - 20\} \tag{C2}$$

and the pasture intake $I^{\text{past}}_{t,d} = \min\{D_{t,d}, F_{k_{t,d},t,d-1}\}$ is split between plant and standing dead forage in proportion to the grazeable dry matter of each group. The nitrogen of each group follows its carbon in proportion. The crude protein of the plant group is 6.25 times its nitrogen-to-dry-matter ratio. When this value lies outside the 1st to 99th percentile band of the MEMS training simulations at the study site in the same month (0.102 to 0.157 in May and 0.082 to 0.094 in September), it is set to the nearest bound, and the day is counted. Standing dead forage has a crude protein of 0.0838, the median MEMS value at the site. The total digestible nutrients of each forage group are set so that its net energy for maintenance is 0.95 + 3.30 CP megacalories per kilogram of dry matter, bounded to the range 0.8 to 2.2. The hay of equation (B9) has 45 percent carbon and 1.5 percent nitrogen in its dry matter, equivalent to 9.4 percent crude protein, and 53 percent total digestible

nutrients, which give 1.07 megacalories of net energy for maintenance per kilogram. The corn grain has 88 percent total digestible nutrients, 2.18 megacalories of net energy for maintenance and 1.50 for gain per kilogram, and 9.8 percent crude protein (feed 410 of National Research Council 2000).

Winter feeding lies outside the management comparison, and every management system feeds the same winter program off pasture. Heifers and first-calf cows eat hay, together with their corn, to their intake capacity, and before turnout the calves eat hay up to their solid-feed allowance. Heifers receive 1.5 kilograms of dry corn grain per head per day on every off-pasture day, and cows receive the same amount from day 15 to day 119, which spans the last 60 days of gestation and the period from calving to turnout. This amount is the upper end of Extension guidance for high-forage cow diets (Drewnoski 2020), and the corn replaces hay of the same weight. For mature cows, let $w^5 = 527$ kilograms be the BCS-5 weight, let $\tau_{t,d}$ be the number of days to the next calving, and let m be the corn per head. A mature cow at or above w^5 receives the hay

$$h^{\text{cow}}_{t,d} = \min\Big\{D_{\text{cow},t,d} - m,\ \max\Big[0,\ \frac{O_{t,d} + r^{+}\,(w^5 - w_{\text{cow},t,d})/\tau_{t,d} - m\,\text{NE}^{\text{corn}}_{m}}{\text{NE}^{\text{hay}}_{m}}\Big]\Big\}, \tag{C3}$$

where $O_{t,d}$ is the maintenance, pregnancy, and lactation energy and r^{+} is the energy stored per kilogram of gain in equation (C4). Between calving and turnout, the target weight change is zero. A mature cow below w^5 eats hay to capacity. The ration reaches the target exactly when the capacity does not bind.

The realized feed is allocated among the classes in two steps. The first step searches for a split of each feed source among the classes that meets the fixed energy obligations of every class, namely maintenance for all classes, pregnancy for pregnant cows, and lactation for nursing cows. Milk yield on lactation day ℓ follows the curve $y(\ell) = \alpha_y\,\ell^{0.15}e^{-0.003\ell}$, with $\alpha_y = 5.169$ so that the Angus peak of 8.0 kilograms per day falls on day 50 (National Research Council 2000), and milk enters the calf diet before solid feed. First-calf cows give 0.74 $y(\ell)$, the factor for two-year-old dams, every calf receives the herd mean milk, and each cow cohort bears the energy cost of its own milk. Milk is an obligation of the cow, so a cow that is short of energy loses weight but does not give less milk. The second step distributes the remaining feed in proportion to $w^{0.75}_{c,t,d}$ times the remaining intake capacity of each class. On pasture, cow cohort g receives the share $n_g w_g / \sum_{g'} n_{g'} w_{g'}$ of the cow feed, which equals its share of the cow intake capacity because both cohorts use the intake rate of the cow class. Off pasture, each class and cohort eats its own ration. If the feed cannot meet every obligation, the allocation keeps all feed in the diets, and the deficit appears in the energy balance.

For an adult class, let $E_{c,t,d}$ be the net energy for maintenance that the diet supplies minus the maintenance, pregnancy, and lactation requirement, in megacalories per

head per day. For a pregnant cow on gestation day t_g, the pregnancy requirement is $0.576\,\mathrm{CBW}\,(0.4504 - 0.000766\,t_g)\exp[(0.03233 - 0.0000275\,t_g)\,t_g]/1000$ megacalories per day, with calf birth weight CBW = 38 kilograms (National Research Council 2000, Equations 4-4 and 4-7), multiplied by the pregnant share of the cohort. The weight change is

$$w_{c,t,d+1} - w_{c,t,d} = \begin{cases} E_{c,t,d}/r_c^-, & E_{c,t,d} < 0, \\ g^E_{c,t,d}, & E_{c,t,d} \geq 0, \end{cases} \tag{C4}$$

where g^E is the gain that the surplus energy supports. For a cow, the tissue energy per kilogram rises with condition, $\mathrm{NE_{DLW}} = 0.5381\,\mathrm{BCS} + 3.2855$ megacalories per kilogram (Fox et al. 2003, p. 2.15). A loss supplies $r^- = 0.80\,\mathrm{NE_{DLW}}$ of dietary net energy per kilogram, and a gain requires $r^+ = \mathrm{NE_{DLW}}$, so $g^E = E/r^+$ (National Research Council 2000, p. 118). Cow weight does not rise above 605 kilograms, 1.10 times the 550-kilogram mature weight of the animal model. A heifer loses $E/2.75$ kilograms per day. Its gain follows the growth equation of the nutrient-requirement model, $g^E = 13.97\,\mathrm{RE}^{0.9116}\,\mathrm{EQSBW}^{-0.6837}$, which is the exact inverse of the retained-energy equation of the animal model (National Research Council 2000 prints the coefficient as 13.91). Here RE is the surplus net energy for maintenance times the ratio of the net energies for gain and maintenance of the diet, and $\mathrm{EQSBW} = 478\,w/527$ is the equivalent shrunk body weight. The model computes the gain that metabolizable protein supports only to count the days on which protein would limit gain.

For a nursing calf, the energy balance E^{ME} is in metabolizable energy. A calf with a deficit changes weight by $k_m E^{\mathrm{ME}}/(1.20 + 0.006\,w)$, with $k_m = 0.719$ on milk only and $k_m = 0.69$ on milk and solid feed. A calf with a surplus retains $\mathrm{RE} = k_g E^{\mathrm{ME}}$, where k_g is the mean of 0.55 for milk (National Academies of Sciences, Engineering, and Medicine 2021) and the ratio of net energy for gain to metabolizable energy of the solid diet (National Research Council 2000), weighted by the shares of milk and solid feed in the metabolizable energy supply of the calf. Its empty-body gain EBG solves the young-calf equation $\mathrm{RE} = \mathrm{EBG}^{1.1}\,\mathrm{EBW}^{0.205}$ (National Academies of Sciences, Engineering, and Medicine 2021), where the empty body weight EBW is 0.94 times the live weight on milk only and 0.93 times the live weight on milk and solid feed. The shrunk weight gain is $0.96\,\mathrm{EBG}/0.94$ on milk only and $0.96\,\mathrm{EBG}/0.91$ on milk and solid feed. Calves above 225 kilograms of live weight do not gain.

The body condition score of a cow cohort on the nine-point scale is linear in its weight,

$$\mathrm{BCS} = 5 + \frac{w/w_g^5 - 1}{0.0685}, \tag{C5}$$

so each score step is 6.85 percent of the BCS-5 weight (Fox et al. 2003, p. 2.15). For mature

cows, w^5 = 527 kilograms, and BCS 3, 4, 5, and 6 correspond to 455, 491, 527, and 563 kilograms. The BCS-5 weight of 527 kilograms comes from the mean October weight of 448 cows of the Michigan State University research herd, 568 kilograms of live weight or 545 kilograms of shrunk weight, which we take as BCS 5.5. For first-calf cows, w^5 = 0.85 × 527 = 447.95 kilograms, which is above the 80 percent of mature weight at first calving that National Research Council (2000, p. 32) assumes. The conception rate ρ of equation (B4) is 0.43, 0.61, 0.86, and 0.93 at BCS 3, 4, 5, and 6 (Kunkle, Sand, and Rae 1994), linear between these points and constant outside them, and it applies to the score of each cohort on the calving date, day 75. A simulation ends as infeasible, and its management system has no NPV, on the first day on which the mature cows reach 385 kilograms (BCS 1.1), the first-calf cows reach 330 kilograms, or a heifer cohort reaches 0.70 times its weight at entry into the herd. The first-calf floor of 330 kilograms is the lower edge of the adult animal model, 0.60 times its mature weight.

C.6. Daily coupled simulation

The farm loop follows the daily ordering of Figure C1 on every calendar day from 1 January 2026 to 31 December 2035, a total of 3,652 days, and treats leap days as off-pasture days. On each day d, the loop runs the following steps:

1. The herd model applies the calendar events of the day, calving and conception on day 75 and weaning on day 270, derives the class counts of the day from the cohort sizes, and computes $D_{c,t,d}$ and $D_{t,d}$ from the current weights and the air temperature.
2. On a grazing day, the calendar of equation (B26) and the selective-grazing rule of equation (B7), applied to the patch states at the end of day $d - 1$, give the occupied patch. The loop computes $L_{t,d}$, the local density $u_{k,t,d}$, the grazeable forage F, and the hay $h_{t,d}$ and $s_{k,t,d}$. Off pasture, it computes the winter ration of each class and cohort.
3. The surrogate advances all twelve patches by one day. The occupied patch receives $(u_{k,t,d}, s_{k,t,d})$, and every other patch receives zero inputs. Off pasture, every patch receives zero inputs.
4. The loop allocates the realized feed among the classes, computes the energy balances, and updates the weights by equation (C4). If a weight reaches its floor, the simulation ends.
5. On 31 December, the herd model applies mortality and culling and sells the culled cows at their cohort weight, and the annual account computes profit.

The animal demand, patch selection, and hay of a day thus enter the same daily transition, and the demand of the next day depends on its outcome. At weaning, the first-calf cohort joins the mature cohort at the head-weighted mean weight, and the promoted heifers form

the new first-calf cohort at their own weight, so the transition conserves live weight. The calves retained for replacement keep their weaning weight. To save computation time, the loop advances the paddock designs of one start history, level of feed-gap supplementation, and stocking rate together in one surrogate batch of up to 70 designs, or 840 patches.

C.7. Economic accounting

Profit is computed downstream of the biophysical and herd simulation, so the accounting result does not affect grazing, SOC, or herd dynamics within a model run. On a cash basis, annual profit is revenue minus cost:

$$\pi_N(y) = \underbrace{R_{\text{calf}}(y) + R_{\text{cull}}(y)}_{\text{revenue}} - \underbrace{\left[C_{\text{feed}}(y) + C_{\text{herd}}(y) + C_{\text{min}}(y) + C_{\text{labor}}(y) + C_{\text{fence}}(y) + C_{\text{repl}}(y)\right]}_{\text{cost}}. \tag{C6}$$

Each term is evaluated for simulation year y. Revenue comprises calf sales R_{calf} at the simulated weaning weight and cull-cow sales R_{cull} at the simulated cohort weight on 31 December, both priced per hundredweight. Feed cost C_{feed} prices the winter hay and the grazing-season hay of equation (B13) at a common price and the corn of equation (B12) at its own price. The account excludes the broad purchased-feed, homegrown-feed, and grazed-feed aggregates of the USDA Economic Research Service (ERS) to avoid double counting. The remaining costs are herd nonfeed operations C_{herd}, mineral C_{min}, labor C_{labor} (a per-cow base plus scheduled-move labor), the annual equivalent of perimeter and interior fencing C_{fence}, and replacement-heifer purchases C_{repl}. For a T-year simulation that begins in year y_0, the reported net present value is

$$\text{NPV}_N^{(T)} = \sum_{\tau=0}^{T-1} \delta^{\tau} \pi_N(y_0 + \tau) + \delta^{T-1} p_{\text{cull}} \sum_g n_g \left(w_g - w_g^5\right), \qquad \delta = (1 + 0.05)^{-1}. \tag{C7}$$

We set $T = 10$ and $y_0 = 2026$, so the first management year has a discount factor of one. The second term values the condition of the cows that remain at the end, where n_g and w_g are the count and weight of cow cohort g on 31 December 2035, w_g^5 is its BCS-5 weight, and p_{cull} is the cull price per kilogram. A herd that ends below BCS 5 has a negative terminal value, so a management system cannot gain NPV by letting its cows lose condition in the last years. The private-profit objective carries no carbon revenue, consistent with the cash basis, and Section 3.4 adds the payments after the run. Terminal SOC is reported as a separate environmental outcome, and the theoretical shadow value λ of Section B.9 measures the private continuation value of SOC through forage and feed-cost savings.

C.8. Numerical assumptions

The numerical implementation rests on the following assumptions. The fixed management grid, the feed-driven body weights, the intake capacities, and the common forcing and winter program matter most for interpreting the results.

1. *Fixed management grid.* Each management system holds its planned herd target, installed paddock count, grazing period, and feed-gap supplementation constant across all years, and the grazing-season dates are fixed. The analysis compares these fixed systems and leaves the dynamic optimization of equation (B1) to future work.
2. *Planned and daily LSU are distinct.* The planned target H is the intended annual herd size, and the design label H/A uses the reference class coefficients of equation (B2). The realized class counts and body weights set the feed target and the demand-equivalent $L_{t,d}$ of equation (B6), which sets the daily local pressure.
3. *Consistent demographic accounting.* Pending calves before calving and live calves afterward belong to the same cohort, and the realized cow, heifer, and calf counts of the demographic controller drive both daily demand and annual accounting.
4. *Identical start.* The twelve patches share one soil, weather sequence, species mix, and start state, and their within-farm heterogeneity emerges from divergent grazing and hay histories.
5. *Deterministic movement.* The herd follows equation (C1) deterministically, so each simulation yields a single trajectory.
6. *Feed-driven body weight.* Body weights change with the energy balance of the realized diet through equation (C4). A protein shortage does not limit gain, and the days on which it would limit gain are counted. Milk follows a fixed curve and does not respond to cow condition. The herd lifecycle is not validated, and Appendix E.2 compares the simulated weights with the October weights of a Michigan State University research herd, to which the model is not calibrated.
7. *Intake capacities.* The intake capacities that set the feed target, and the conversion of 15 kilograms of dry matter per demand-equivalent LSU, are not verified against official nutrient-requirement cases for the three classes.
8. *One representative weight per class and cohort.* All animals in a class, or in a cow cohort, share one weight, and the model has no within-group weight distribution.
9. *Hay rule.* On pasture, the producer feeds the fixed share ϕ of the feed gap, with no minimum. Off pasture, every management system uses the same winter program of hay and corn. All hay has one price and no waste.
10. *Conception from cow condition.* The conception rate of a cow cohort depends only on

its body condition at calving. All open cows are culled at the year boundary.

11. *Infeasible management.* A simulation whose cows or heifers reach their weight floor ends on that day and is excluded from the NPV ranking. The results report how often this happens.

12. *Surrogate forage quality.* When the surrogate state implies a plant crude protein outside the monthly band of the MEMS training simulations at the site, the crude protein is clipped into the band. Standing dead forage has a fixed crude protein.

13. *Exogenous forcing.* Every simulation receives the same deterministic forcing path described in Section 3.1: ten years of historical weather at a constant carbon dioxide concentration.

14. *No economic feedback within a simulation.* Profit is computed after the simulation and never alters the biophysical or herd dynamics. The cash basis excludes land rent, the opportunity cost of grazed forage, and carbon payments.

15. *Calendar management-paddock advancement.* The calendar alone determines when the herd advances to the next management paddock. Within the active paddock, the biomass and forage quality of the previous day affect the daily grazing value and can change which biophysical patch equation (C1) selects. A forage-triggered advancement rule is left to future analysis.

C.9. Parameterization and calibration

Table C1 reports the parameters needed to interpret and reproduce the management comparison, and Appendix D gives the surrogate architecture and validation settings. The biological and technical parameters fix the 60-hectare, twelve-patch Michigan farm, the grazing season from day of year 120 to 270, the three-class herd with its nutrition, weight, and demographic rules, and the calendar and animal selective-grazing rules described above. These quantities are the same in every management system, and only the planned stocking rate, installed paddock count, grazing-period length, feed-gap supplementation, and start history vary.

The study site is one of the Michigan sites of the surrogate training data. The 60-hectare farm is a representative-farm design assumption of the numerical experiment, the soil profile comes from SSURGO, and the five start states come from the MEMS simulations of the site under its five management histories.

The planned class shares and the per-head reference LSU coefficients, defined relative to a 500-kilogram reference animal, come from the USDA Census class structure together with herd-survey composition. They convert H to a planned cow target, from which the demographic controller constructs attrition-consistent opening counts and event-driven

daily counts. For example, $H = 60$ gives the planned target vector $(46.15, 9.23, 36.92)$ for cows, heifers, and calves, but the first opening state contains 40.01 cows, 6.24 heifers, no live calves, and 40.01 pending calves that enter on the calving date. Calves are born at 38 kilograms on day 75, and from then on every weight follows equation (C4). The calving and weaning dates and the demographic rates of Table C1 are the same in all management systems. The demand-equivalent conversion of equation (B6) uses the MEMS value of 15 kilograms of dry matter per LSU per day.

The half-saturation biomass, handling time, and quality exponent of the movement rule (C1) are set from the functional-response and bite-mechanics literature on grazing intake and from the quality-preference literature. The travel-cost coefficients and the nine-hour daily grazing time budget come from grazing-behavior studies and movement-rule calibration, and the preference map encodes a corner-water gradient as the non-forage attractor.

Calves and cull cows are valued at $264 and $115 per hundredweight of their simulated weights, respectively. Winter hay costs $220 per short ton of consumed dry matter, a rounded hay-equivalent value derived from Michigan hay prices with 85 percent dry matter and 90 percent recovery, and hay fed on pasture has the same price. Corn costs $233.8 per metric ton of dry matter, the mean Michigan price received for the marketing years 2020 to 2024 (USDA-NASS 2026), which is 3.6 percent below the hay price per ton of dry matter. The annual nonfeed herd cost is $280.84 per cow, and the mineral cost is $54.75 per cow. Labor combines a $27.99 per-cow base, a $19.70 hourly wage, and a move time of 30 minutes plus 0.10 minute per cow over a 24-week labor season. Permanent fence costs $2.51 per foot with a 20-year life, temporary fence costs $1.05 per foot with a 10-year life, and replacement heifers cost $1,200 per head. The discount rate is 5 percent. The calf and cull prices are fixed calibration values, and the replacement-heifer price is a model assumption.

TABLE C1. Selected parameter values and model settings for the numerical experiment of 28,005 simulations.

Quantity	Value	Unit	Source / note
Site and farm			
Latitude; elevation	43.57; 208	°N; m	surrogate training site
Farm area	60	ha	representative-farm design assumption
Patch area, count	5, 12	ha, —	within-patch uniformity scale
Soil texture (sand, clay)	40 to 48, 21 to 24	%	SSURGO, 16-layer profile
Soil pH	7.5 to 8.0	—	SSURGO

continued on next page

Quantity	Value	Unit	Source / note
Grazing season	120 to 270	day of year	Michigan off-pasture October to April
Herd composition and weight			
Planned class shares (cow / heifer / calf)	0.50 / 0.10 / 0.40	—	management-target conversion; USDA Census + herd survey
LSU coefficients	1.00 / 0.70 / 0.20	—	500 kg reference weight
Planned weighted LSU per head $\bar{\ell}$	0.65	—	derived from target shares
Demand capacity κ_{MEMS}	15	kg DM LSU^{-1} d^{-1}	MEMS conversion, equation (B6)
Calving, weaning	75, 270	day of year	fixed herd calendar
Initial weight (cow / heifer)	527 / 255	kg head^{-1}	herd start on 1 January 2026, cows at BCS 5; heifers at the median of later cohorts in a preliminary simulation
Calf birth weight	38	kg head^{-1}	animal model default
BCS-5 weight (mature / first-calf)	527 / 447.95	kg head^{-1}	MSU October cows as BCS 5.5; 0.85 of mature
BCS step	6.85	% of BCS-5 weight	Fox et al. (2003), equation (C5)
Intake capacity (lactating / dry cow / heifer)	2.4 / 2.1 / 2.5	% of body weight	times a cold factor
Peak milk; first-calf factor	8.0; 0.74	kg d^{-1}; —	Angus, National Research Council (2000)
Cow tissue energy $\mathrm{NE}_{\mathrm{DLW}}$	0.5381 BCS + 3.2855	Mcal kg^{-1}	Fox et al. (2003); loss 0.80, gain 1.00 times, National Research Council (2000)
Heifer weight change	loss $E/2.75$; gain NRC growth	kg d^{-1}	National Research Council (2000)
Calf weight gain	young-calf RE = $\mathrm{EBG}^{1.1}\mathrm{EBW}^{0.205}$; milk k_g 0.55	kg d^{-1}	National Academies of Sciences, Engineering, and Medicine (2021); solid k_g from National Research Council (2000)
Pregnancy energy	NRC Eq. 4-4, 4-7	Mcal d^{-1}	National Research Council (2000), CBW 38 kg
Cow weight cap	605	kg head^{-1}	1.10 times 550 kg mature weight

continued on next page

Quantity	Value	Unit	Source / note
Weight floor (mature / first-calf / heifer)	385 / 330 / 0.70 entry	kg head^{-1}	reaching it ends the simulation
Conception at BCS 3 / 4 / 5 / 6	0.43 / 0.61 / 0.86 / 0.93	fraction	Kunkle, Sand, and Rae (1994); linear between points
Mortality (cow / heifer / calf)	0.015 / 0.015 / 0.030	fraction yr^{-1}	demographic controller
Cow culling	all open cows, at least 0.12	fraction yr^{-1}	open cows first
Female-calf ratio; heifer retention rate	0.50; 0.15	fraction	demographic controller
Movement rule (equation C1)			
Half-saturation biomass $B_{1/2}$	1,200	kg DM ha^{-1}	intake-response calibration
Handling time h	1.5	s g^{-1} DM	bite-mechanics literature
Quality exponent γ	2	—	quality-preference literature
Travel cost (α_0, α)	0.25, 0.02	—, per 100 m	movement-rule calibration
Grazing time budget	9	h d^{-1}	grazing-behavior literature
Preference map P_k	corner-water gradient	—	non-forage attractor (water)
Feed and economics (cash basis, equation C6)			
Calf price	264	\$ cwt^{-1}	fixed calibration value; realized weaning weight
Cull-cow price	115	\$ cwt^{-1}	fixed calibration value
Cull-cow sale weight	cohort weight on 31 December	kg head^{-1}	simulated
Hay price (pasture and winter)	220	\$ short ton^{-1} consumed DM	Michigan hay price, 85% DM, 90% recovery; rounded
Hay composition (C, N; TDN)	0.45, 0.015; 0.53	fraction of DM	9.4% crude protein
Corn (cows day 15 to 119; heifers off pasture)	1.5	kg DM head^{-1} d^{-1}	Drewnoski (2020), upper end
Corn price; TDN	233.8; 0.88	\$ t^{-1} DM; fraction	USDA-NASS (2026); National Research Council (2000) feed 410
Retained forage residual	20	g DM m^{-2}	grazeable forage, equation (C2)
Plant crude protein band	monthly p1 to p99	fraction of DM	MEMS at the study site; surrogate values outside are clipped
Standing dead crude protein	0.0838	fraction of DM	MEMS median at the study site

continued on next page

Quantity	Value	Unit	Source / note
Forage net energy for maintenance	$0.95 + 3.30\,\mathrm{CP}$	Mcal kg^{-1} DM	bounded to 0.8 to 2.2
Herd nonfeed cost	280.84	\$ cow^{-1} yr^{-1}	broad ERS feed aggregates excluded
Mineral cost	54.75	\$ cow^{-1} yr^{-1}	separate annual account
Labor base, wage	27.99, 19.70	\$ cow^{-1}, \$ h^{-1}	farm labor parameterization
Move time	$30 + 0.10 n_{cow}$	min $move^{-1}$	24-week labor season
Fence (permanent, temporary)	2.51 / 20, 1.05 / 10	\$ ft^{-1} / yr life	Missouri Extension costs; NRCS/product-life assumptions
Replacement heifer	1,200	\$ $head^{-1}$	model assumption
Discount rate	5	% yr^{-1}	analysis convention
Design grid			
Planned stocking rate H/A	0 to 3.2 by 0.2	LSU ha^{-1}	12 to 192 planned LSU on 60 ha, no herd at 0
Paddock count N	1, 2, 3, 4, 6, 12	—	continuous grazing ($N = 1$); calendar-based rotation ($N > 1$)
Grazing period, $N = 2$	1 to 7, 10 to 75 by 5	d	21 levels
Grazing period, $N = 3$	1 to 7, 10 to 50 by 5	d	16 levels
Grazing period, $N = 4$	1 to 7, 10 to 35 by 5	d	13 levels
Grazing period, $N = 6$	1 to 7, 10 to 25 by 5	d	11 levels
Grazing period, $N = 12$	1 to 7, 10	d	8 levels
Scheduled management-paddock rest	$(N - 1)\mathrm{gp}$	d	calendar identity
Rotation start	advances by year	management paddock	fixed rotation; balances incomplete seasonal cycles
Feed-gap supplementation ϕ	0, 0.25, 0.5, 0.75, 1	—	share of the feed gap, equation (B9)
Management history	5	—	ungrazed, low-pressure continuous, low-pressure rotational, heavy continuous with supplement, intensive rotational with supplement
Start state	MEMS state at the end of 2025	per history	all twelve patches equal
Forcing path	gridMET 1996 to 2005	weather year	replayed for 2026 to 2035; CO_2 427 ppm

continued on next page

Quantity	Value	Unit	Source / note
Management years; SOC endpoints	2026 to 2035	calendar yr	SOC on 1 January 2026 and 31 December 2035
Biophysical transition engine	surrogate of MEMS	—	advances all twelve patches one day at a time
Surrogate management inputs	local grazing density; hay	LSU ha^{-1}; g DM m^{-2} d^{-1}	occupied patch only
Feed allocation	maintenance first, then $w^{0.75}\times$ remaining capacity	—	Appendix C.5
Simulations	28,005	—	16 SR × 70 paddock designs × 5 levels of feed-gap supplementation × 5 histories, and one no-herd simulation for each history

Appendix D. Surrogate architecture and validation

This appendix gives the input and output specification, the architecture, the training objective and data, and the validation details of the surrogate of Section 3.2.

D.1. Recurrence and management inputs

The model has 7.89 million parameters and combines a convolutional encoder over the 16 soil layers with an encoder for the non-layered state, site, weather, calendar, and management inputs. Its management vector $\mathbf{a}_{k,t,d} = (u_{k,t,d}, s_{k,t,d})$ holds the local grazing density, in demand-equivalent LSU per hectare, and the hay fed on the patch, in grams of dry matter per square meter per day, where one demand-equivalent LSU has a daily dry-matter demand of 15 kilograms in MEMS. The fused representation enters a two-layer gated recurrent unit (GRU) with 768 hidden features.

For patch k on day d, the recursion is

$$\begin{aligned} e_{k,t,d} &= \Phi_\theta(\widehat{X}_{k,t,d-1}, z_k, w_{t,d}, \mathbf{a}_{k,t,d}), \\ h_{k,t,d} &= \mathrm{GRU}_\theta(e_{k,t,d}, h_{k,t,d-1}), \\ \widehat{X}_{k,t,d} &= \mathcal{R}_{\mathrm{phys}}\Big(\widehat{X}_{k,t,d-1}, \Psi_\theta(h_{k,t,d})\Big). \end{aligned} \tag{D1}$$

Here θ collects the learned parameters, Φ_θ encodes the state and daily inputs into the feature vector $e_{k,t,d}$, $h_{k,t,d}$ is the GRU memory carried from day to day, and Ψ_θ maps that memory to the output heads. The farm-management layer constructs $u_{k,t,d}$ causally before the transition, and $\mathcal{R}_{\mathrm{phys}}$ reconstructs the next physical state and applies the output links. The heads of 409 state values predict their daily change, and the heads of the 16 soil-temperature layers predict their level. In addition to the explicit preceding state, the GRU gives the recursive emulator a learned memory of recent weather, vegetation, soil, and management history.

In a farm simulation, the calendar layer and the animal selective-grazing layer determine $u_{k,t,d}$ before the transition, and the management-responsive approximation is

$$\begin{aligned} &F_{\mathrm{MEMS}}\Big(X_{k,t,d-1}, z_k, w_{t,d}, \mathcal{A}(X_{k,t,d-1}, u_{k,t,d}, s_{k,t,d})\Big) \\ &\qquad \simeq \widehat{F}_\theta\Big(X_{k,t,d-1}, z_k, w_{t,d}, u_{k,t,d}, s_{k,t,d}\Big), \end{aligned} \tag{D2}$$

where $\mathcal{A}$ is the internal action construction of MEMS. Paddock count, farm geometry, calendar scheduling, and selective movement stay in the farm-management layer, outside the neural model. The hay input s follows the feed-gap supplementation of each management system (Section 3.3), and Appendix D.2 gives the full input and output specification.

D.2. Input and output specification

The recurrent physical state holds 425 values from nine MEMS output families: soil organic matter, crop, soil nutrients, soil temperature, soil water content, surface organic matter, surface feces, standing dead biomass, and water-storage states. It consists of 23 layered variables over the 16-layer soil profile and 57 non-layered values. Each daily transition also receives 177 fixed site values (11 soil profiles over 16 layers, and the elevation) and 14 weather and calendar values for the target day: latitude, maximum and minimum air temperature, precipitation, solar radiation, relative humidity, wind speed, mean air temperature, daily growing degree days, the seven-day precipitation sum, the atmospheric carbon dioxide concentration, the sine and cosine of the day of year, and the photoperiod. The neural model predicts the full next state, and the forage available for grazing is computed after the prediction as the dry matter of the plant and standing dead pools above a retained residual of 20 grams per square meter.

The surrogate takes the two management inputs of the target day, the local grazing density and the hay fed on the patch, and learns the direct mapping on the right side of equation (D2), while MEMS constructs the corresponding removal, trampling, fecal, and urinary fluxes internally on the left side. In MEMS the animals eat forage before hay, so hay changes the patch state through the excreta of the animals that eat it.

D.3. Architecture, training objective, and training data

The structured recurrent model has 7,886,698 parameters. A one-dimensional convolution with 64 channels runs over the 16 soil layers and reads the 23 layered state variables, the 11 site profiles, and an eight-value depth embedding. Three residual depth blocks and a projection to 256 profile features follow. A two-layer multilayer perceptron encodes the 57 non-layered state values, the elevation, the 14 weather and calendar values, and the two management inputs into 256 features, and a fusion layer combines the two representations into 512 features before the two-layer, 768-feature GRU of equation (D1). An output head with 512 hidden features maps the GRU state to the 425 daily outputs. The heads of 409 state values predict the daily change, which is added to the preceding state, and the heads of the 16 soil-temperature layers predict the level. The output links clamp at zero the 411 state values whose training minimum is not negative, and they clamp every state value to the range of the training data.

Let $\mathcal{D}$ be the set of days in a training window, and let $x_{j,d}$ and $\widehat{x}_{j,d}$ be the target and prediction for state value j on day d in the native representation of its head, a daily change or a level. Let $\rho_1(v)$ denote the smooth-L1 function with unit threshold, let m_j mark the state values that carry a loss, let σ_j be the daily-change scale of a change head or the level scale of a level head, and let w_j be the channel weight. The principal state-transition loss

is

$$\mathcal{L}_{\text{trans}} = \frac{1}{\sum_j w_j m_j} \sum_j w_j m_j \left[\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \rho_1 \left(\frac{\widehat{x}_{j,d} - x_{j,d}}{\sigma_j} \right) \right]. \tag{D3}$$

The channel weights come from a fixed recursive rollout on training simulations, repeated every 300 updates: $w_j = \min\{10, \max[0.1, (r_j/0.2912)^{0.5}]\}$, where r_j is the rollout MAE of state value j divided by its persistence MAE, the error of holding the first state of each window constant. A value that drifts more than its own change over the horizon receives more weight. For the eight forage carbon pools, each training sample also receives the level weight $[(L_s + 0.1R)/R]^{-1}$, normalized to a batch mean of one, where L_s is the mean true forage carbon of the sample window and R is the forage carbon of the normalizer reference state. This weight brings the forage loss close to a relative error, so that simulations with little forage still carry weight in the loss.

Two auxiliary terms address failure modes that the one-day transition loss misses. The drift term $\mathcal{L}_{\text{drift}}$ penalizes the mean error of the 409 change heads over a training window, divided by the daily-change scale, so that small daily biases do not accumulate in the recursion. For the seven decision outputs $\mathcal{J}_7$ (total forage, live leaf, live stem, and dead plus seed dry matter, total SOC, mineral-associated organic matter, and particulate organic matter), the level term is

$$\mathcal{L}_{\text{level}} = \frac{1}{7|\mathcal{D}|} \sum_{j \in \mathcal{J}_7} \sum_{d \in \mathcal{D}} \rho_1 \left(\frac{\widehat{y}_{j,d} - y_{j,d}}{|y_{j,d}| + 0.15\, \bar{y}_j} \right), \tag{D4}$$

where $y_{j,d}$ and $\widehat{y}_{j,d}$ are the true and predicted physical levels and $\bar{y}_j$ is the value of output j at the normalizer reference state. This term makes persistent errors visible in low-biomass or nearly stationary periods. The objectives are

$$\mathcal{L}^{\text{teacher}} = \mathcal{L}_{\text{trans}} + 0.30\, \mathcal{L}_{\text{drift}}, \qquad \mathcal{L}^{\text{push}} = \mathcal{L}^{\text{teacher}} + 0.20\, \mathcal{L}_{\text{level}}. \tag{D5}$$

Teacher-forced windows train against the true preceding states. Pushed windows feed the surrogate's own preceding prediction back through equation (D1), each pushed day carries a loss with probability 0.25, and the gradient passes through 60 days of the physical state. The horizon curriculum lengthens the windows from one day to 1,095 days while the pushed-window probability rises to 0.90.

Training runs as a chain of five parts, each starting from the weights of the one before, for 59,826 optimizer steps on one graphics processor (NVIDIA A100), about 52 hours in all, with one random seed. The final part runs 5,000 steps with batches of 4,096 pushed windows of 1,095 days and uses AdamW with a peak learning rate of 1.2×10^{-4} after a 500-step warmup, a cosine decay to 1.2×10^{-5} at step 2,700 and a constant rate thereafter,

weight decay 10^{-4}, gradient-norm clipping at 1.0, and bfloat16 mixed precision.

The training split of the MEMS simulation set contains 1,667 simulations with 18,252,901 daily transitions. Each simulation starts on 1 January 2026 and runs for 30 years with the gridMET weather of 1996 to 2025 (Abatzoglou 2013), repeated for 2026 to 2055, and a constant carbon dioxide concentration of 427 parts per million. The simulations cover 39 sites, 36 in Michigan, 2 in Ohio, and 1 in Wisconsin, and each site has five management histories before 2026: ungrazed, low-pressure continuous, low-pressure rotational, heavy continuous with supplement, and intensive rotational with supplement. Each history yields one MEMS state at the end of 2025, from which nine grazing schedules continue. The schedules come from 90 templates, 72 rotational and 18 continuous, with daily local densities of 0.5 to 36 LSU per hectare between day of year 120 and 270. On 56 percent of the grazing days the schedule also feeds hay, at a median of 2.45 and a maximum of 54 grams of dry matter per square meter per day. Two simulations in which the MEMS forage collapses remain in the training data.

D.4. Validation protocol and detailed results

The test split holds 18 simulations that training never uses: the nine schedules of test site 1 after the low-pressure rotational history and the nine schedules of test site 2 after the heavy continuous history with supplement. Training sees these two sites only after their other histories, so the test withholds combinations of site and history rather than whole sites. Each test forecast starts from the MEMS state at the end of 31 December 2025 and runs recursively for 3,652 days in 32-bit floating point with the recorded weather and management of the simulation. We selected the final checkpoint among the late training checkpoints by a selection score on these 18 test simulations, so the test results are optimistic by the noise of that choice. The last two checkpoints differ by up to 3.7 percentage points on the worst of these selection scores.

For output j, the relative error over the pooled test days is

$$\mathrm{RE}_j = 100\,\frac{\sum_{r,d}\left|\widehat{y}_{rjd} - y_{rjd}\right|}{\sum_{r,d}\left|y_{rjd}\right|}, \tag{D6}$$

summed over test simulations r and days d. The skill score against a reference forecast is $1 - \mathrm{MAE}/\mathrm{MAE}_{\mathrm{ref}}$.

Figure 3 shows one test simulation at each test site, the one with the median plant C error. In Figure D1, the plant errors do not grow with the forecast horizon, and the plant C MAE is 27.5 to 45.7 kilograms per hectare in each year. The SOC error grows from 0.043 Mg C ha^{-1} in the first year to 0.65 Mg C ha^{-1} in the seventh and stays at 1.3 to 2.3 percent of the site-climatology error in every year (Appendix Table D2). On the first forecast day,

TABLE D1. Ten-year accuracy of the surrogate against MEMS, 18 test simulations. The plant rows count the days of the grazing season (day of year 120 to 270), 27,180 days in all. The total SOC row counts one value per test simulation, the stock at the end of the 10 years. The relative error is the sum of the absolute errors divided by the sum of the absolute MEMS values. The skill score is 1 – MAE/MAE_{ref}. Persistence holds the MEMS value at the end of 1 January 2026, and Figure D1 defines the site climatology. The number in parentheses is the number of test simulations in which the surrogate MAE is lower than the reference MAE.

	MEMS			Relative	Skill score against	
Output	mean	MAE	Bias	error (%)	Persistence	Site climatology
Plant C (kg C ha^{-1})	459	36.5	+1.8	7.95	0.92 (18)	0.94 (18)
Plant N (kg N ha^{-1})	17.9	1.58	+0.17	8.84	0.91 (18)	0.93 (18)
Total SOC in year 10 (Mg C ha^{-1})	119	0.48	−0.41	0.403	0.98 (18)	0.99 (18)

one step from the MEMS state, the relative error is 1.26 percent for plant C, 0.654 percent for plant N, and 0.00131 percent for total SOC (Appendix Table D3). By the second day the plant errors reach 6.40 and 9.05 percent, so the first-day values do not measure the accuracy of a longer forecast. None of the 18 test forecasts contains a non-finite value or a negative value in the 411 nonnegative state values.

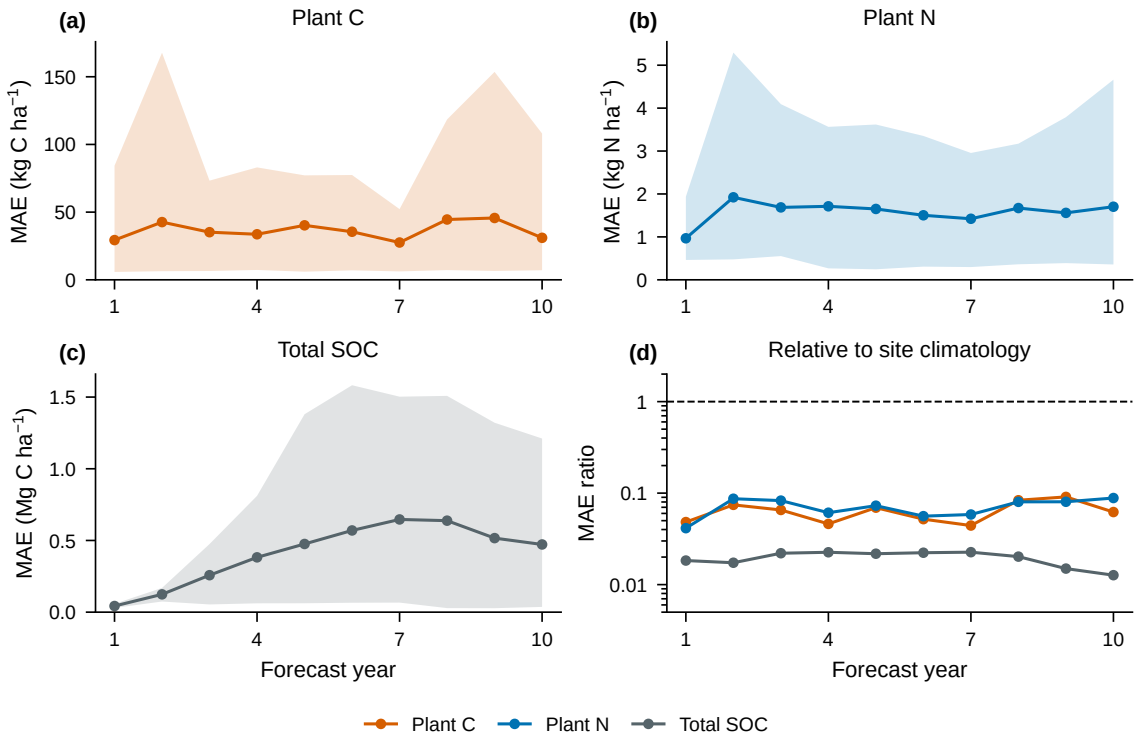


FIGURE D1. **Surrogate error by forecast year, 18 test simulations.** Panels (a) to (c) show the mean absolute error (MAE) of each output in each year, pooled over the test simulations, and the shaded band spans the 10th to 90th percentiles of the MAE of the individual test simulations. Plant C and N errors cover the days of the grazing season, and total SOC errors cover all days. Panel (d) divides the surrogate MAE by the MAE of the site climatology in the same year, on a logarithmic axis, and below the dashed line the surrogate is more accurate than the site climatology. The site climatology is the mean MEMS value for each day of the year over 2036 to 2055 across all training simulations at the test site. These simulations start from other SOC stocks, so for total SOC the site climatology adds their mean change to the MEMS value of the test simulation at the end of 1 January 2026.

Table D2 gives the MAE of each forecast year, and Table D3 gives the accuracy on the first forecast day. Figure D2 compares total SOC at the end of the tenth year on the 18 test simulations and on 417 training simulations, and the largest test error is 2.71 Mg C ha^{-1}. Across the seven checkpoints from step 2,500 to step 5,000 of the final training part, the ten-year MAE is 36.5 to 37.9 kilograms per hectare for plant C, 1.58 to 1.69 kilograms per hectare for plant N, and 0.43 to 0.51 Mg C ha^{-1} for total SOC. Because the model has one random seed, this range is the only available measure of training variation.

TABLE D2. Surrogate MAE in each forecast year, pooled over the 18 test simulations. The plant columns count the days of the grazing season. Total SOC counts all days of the year. Figure D1 shows the spread over the test simulations.

Forecast year	Plant C (kg C ha^{-1})	Plant N (kg N ha^{-1})	Total SOC (Mg C ha^{-1})
1	29.3	0.967	0.0428
2	42.6	1.92	0.124
3	35.2	1.69	0.258
4	33.6	1.71	0.382
5	40.2	1.65	0.475
6	35.5	1.50	0.570
7	27.5	1.42	0.647
8	44.6	1.67	0.638
9	45.7	1.56	0.516
10	31.0	1.70	0.472

TABLE D3. Accuracy of the surrogate against MEMS on the first forecast day (1 January 2026), 18 test simulations. The surrogate makes one step from the MEMS state at the end of 31 December 2025. Each output counts one value per test simulation. All test simulations of one site start from the same state, so this day has only 2 distinct MEMS values, one per site. The relative error is defined as in Table D1.

Output	MEMS mean	MAE	Relative error (%)
Plant C (kg C ha^{-1})	29.2	0.369	1.26
Plant N (kg N ha^{-1})	0.882	0.00577	0.654
Total SOC (Mg C ha^{-1})	136	0.00179	0.00131

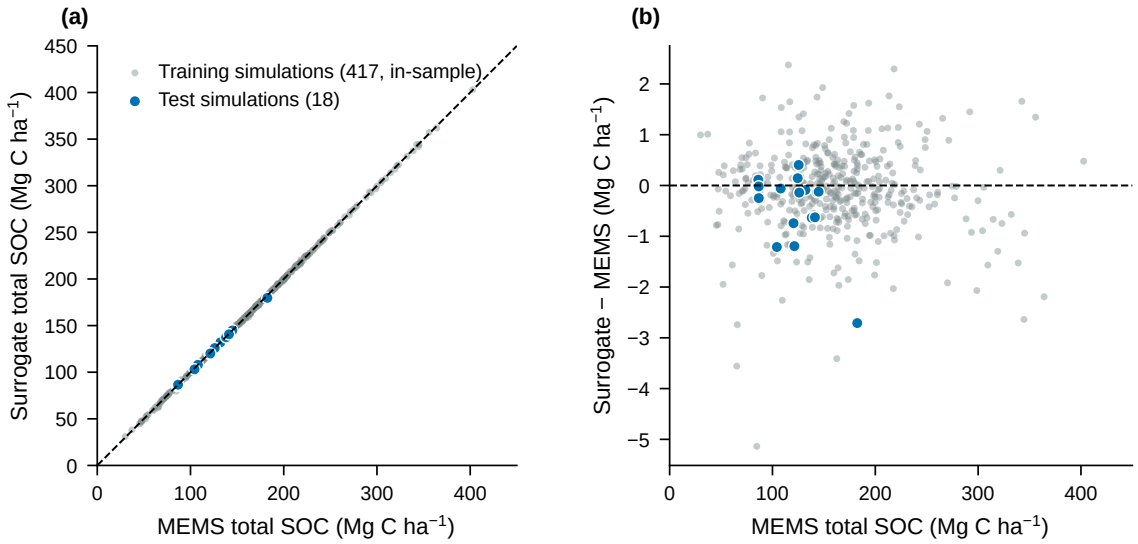


FIGURE D2. **Total SOC at the end of the tenth year, surrogate against MEMS.** Each point is one MEMS simulation. Panel (a) compares the two stocks against the dashed 1:1 line, and panel (b) plots the surrogate error against the MEMS stock. The training simulations are in-sample.

Appendix E. Additional results and limits

This appendix reports the results that Section 5 summarizes or omits and states the limits of the model. Figure E1 gives the variance shares of NPV and SOC change quoted in Sections 5.1 and 5.2, and Table E1 gives the highest-NPV management system of each start history at each planned stocking rate. Table E2 gives the main results of this appendix for each start history. Where this appendix lists five values, they belong to the heavy continuous, intensive rotational, low-pressure continuous, low-pressure rotational, and ungrazed histories, in this order.

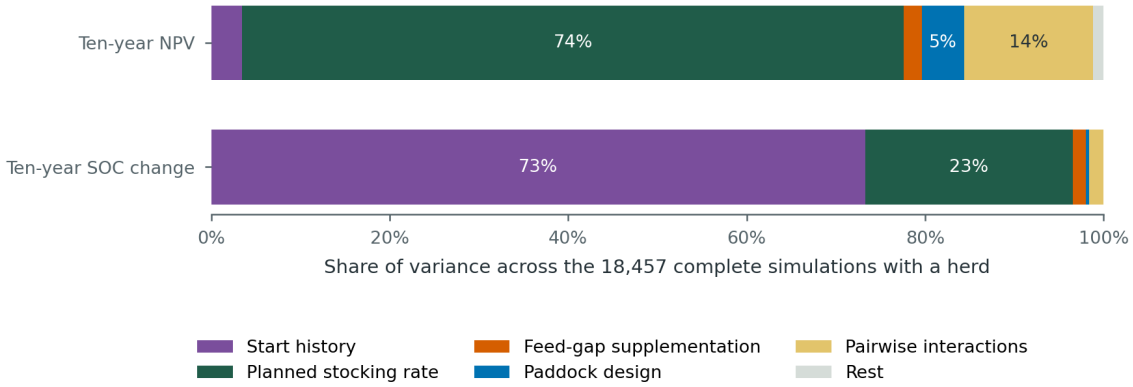


FIGURE E1. **Variance shares of NPV and SOC change.** Shapley shares of the variance of the ten-year NPV and the ten-year SOC change across the 18,457 complete simulations with a herd, for the start history, the planned stocking rate, the feed-gap supplementation, the paddock design (paddock count and grazing period together), all pairwise interactions, and the remainder. The Shapley share of a factor is its mean gain in R^2 when it joins a regression on indicator variables, averaged over all orders of entry. The five simulations without a herd have no paddock design or feed-gap supplementation and are excluded.

TABLE E1. **The highest-NPV management system at each planned stocking rate.** Each start history has 350 management systems at each planned stocking rate (SR, LSU ha^{-1}) from 0.2 to 3.2: 70 paddock designs at each of five levels of feed-gap supplementation ϕ. At SR 0, the farm keeps no herd and so has no paddock design or ϕ. A simulation is complete when it runs the ten years 2026 to 2035 without reaching an animal weight floor. For each start history and SR, the table gives the number of complete simulations and the complete management system with the highest ten-year NPV. CG is continuous grazing on one paddock, which has no grazing period. Feed-gap supplementation ϕ is the share of each grazing-season day's feed gap that hay on pasture fills. NPV is the ten-year net present value of the 60-hectare farm in $ thousands, at a 5 percent discount rate. ΔSOC is the change in farm mean total SOC from 1 January 2026 to 31 December 2035. Bold rows mark the profit optimum of each start history, the complete system with the highest NPV over all SR. Every SR has at least one complete simulation in every start history.

SR (LSU ha^{-1})	Complete simulations	Paddocks	Grazing period (d)	ϕ	NPV ($ thousand)	ΔSOC (Mg C ha^{-1})
Heavy continuous						
0	1	no herd			−10.3	52.62
0.2	280	1	CG	1	1.0	50.93
0.4	241	1	CG	1	12.4	49.48
0.6	220	1	CG	1	23.4	47.75
0.8	209	1	CG	1	33.4	45.31
1.0	199	2	30	1	42.4	43.74
1.2	184	1	CG	1	54.6	40.88
1.4	171	2	10	1	61.7	40.27
1.6	158	2	15	1	71.3	37.57
1.8	131	2	10	1	79.6	33.90
2.0	**117**	**2**	**6**	**1**	**81.7**	**30.82**
2.2	97	2	10	1	70.5	26.91
2.4	73	2	20	1	41.1	23.05
2.6	70	2	25	1	13.1	20.55
2.8	67	12	4	1	−21.5	23.46
3.0	65	12	6	1	−52.9	25.58
3.2	52	12	6	1	−87.4	23.56
Intensive rotational						
0	1	no herd			−10.3	13.28
0.2	350	1	CG	0	0.7	11.54
0.4	350	1	CG	0	11.9	9.94
0.6	350	1	CG	0	23.5	8.50

Continued on the next page

Table E1, continued

SR (LSU ha^{-1})	Complete simulations	Paddocks	Grazing period (d)	ϕ	NPV ($ thousand)	ΔSOC (Mg C ha^{-1})
0.8	350	1	CG	1	32.1	7.31
1.0	348	1	CG	1	45.1	5.47
1.2	342	1	CG	1	57.5	3.60
1.4	316	2	7	1	63.8	2.10
1.6	289	2	10	1	73.9	0.80
1.8	280	2	10	1	82.1	−0.82
2.0	**239**	**2**	**5**	**1**	**87.7**	**−3.38**
2.2	155	4	4	1	82.3	−4.94
2.4	96	4	6	1	69.8	−7.00
2.6	66	6	6	1	37.0	−8.12
2.8	65	6	7	1	−1.7	−9.32
3.0	45	12	4	1	−44.9	−8.65
3.2	34	12	6	1	−82.8	−5.02
Low-pressure continuous						
0	1	no herd			−10.3	8.26
0.2	350	1	CG	0	2.3	5.97
0.4	350	1	CG	0	14.2	3.88
0.6	350	1	CG	0	25.9	2.16
0.8	348	1	CG	1	36.3	−0.49
1.0	347	2	25	1	46.9	−2.32
1.2	346	1	CG	1	59.3	−5.35
1.4	337	1	CG	1	70.1	−7.87
1.6	329	2	5	1	80.1	−8.27
1.8	307	2	5	1	91.3	−10.40
2.0	298	2	5	1	99.7	−13.20
2.2	**275**	**2**	**5**	**1**	**106.9**	**−16.55**
2.4	205	4	3	1	105.8	−21.84
2.6	104	2	20	1	69.6	−27.34
2.8	67	12	4	1	36.3	−22.10
3.0	49	12	4	1	−6.7	−25.02
3.2	36	12	6	1	−42.1	−21.82
Low-pressure rotational						
0	1	no herd			−10.3	3.19
0.2	350	1	CG	0	2.3	0.81

Continued on the next page

Table E1, continued

SR (LSU ha^{-1})	Complete simulations	Paddocks	Grazing period (d)	ϕ	NPV ($ thousand)	ΔSOC (Mg C ha^{-1})
0.4	349	1	CG	1	13.3	−1.49
0.6	350	1	CG	1	25.5	−3.74
0.8	350	1	CG	1	38.3	−6.36
1.0	345	1	CG	1	47.9	−9.20
1.2	344	1	CG	0.75	60.5	−12.32
1.4	344	1	CG	1	70.5	−14.87
1.6	332	2	6	1	81.1	−15.04
1.8	322	2	5	1	91.9	−17.12
2.0	309	2	10	1	102.5	−21.12
2.2	**279**	**2**	**15**	**1**	**109.2**	**−24.48**
2.4	227	2	6	1	108.3	−27.90
2.6	108	4	7	1	82.1	−33.36
2.8	69	12	4	1	41.5	−29.43
3.0	54	12	4	1	1.1	−32.45
3.2	37	12	6	1	−38.6	−29.63
Ungrazed						
0	1	no herd			−10.3	1.26
0.2	350	1	CG	0	2.3	−1.29
0.4	350	1	CG	0	13.7	−3.78
0.6	350	1	CG	1	24.0	−6.20
0.8	350	1	CG	1	37.9	−9.15
1.0	346	1	CG	0.75	50.2	−12.12
1.2	344	1	CG	1	60.8	−14.72
1.4	344	2	10	1	70.0	−16.16
1.6	339	2	5	1	80.7	−17.72
1.8	325	2	10	1	91.7	−21.54
2.0	306	2	5	1	101.4	−23.71
2.2	**279**	**2**	**4**	**1**	**108.9**	**−27.38**
2.4	224	2	4	1	107.3	−31.78
2.6	109	4	7	1	82.4	−36.52
2.8	68	6	6	1	41.2	−38.58
3.0	58	12	4	1	2.6	−35.84
3.2	39	12	6	1	−37.6	−33.13

TABLE E2. **Results of the sweep by start history.** Each column gives one start history, in the order of the start SOC from lowest to highest. A simulation is complete when it runs the ten years 2026 to 2035 without reaching an animal weight floor. Each start history has 5,600 simulations with a herd, 350 at each planned stocking rate (SR, LSU ha^{-1}) from 0.2 to 3.2 and 1,120 at each level of feed-gap supplementation ϕ, the share of the daily feed gap that hay on pasture fills. The completion rows count these simulations and leave out the one simulation without a herd, which completes. The rotation rows give the percent complete of all calendar-based rotations with a grazing period (gp) of 2 to 7 days and of 50 to 75 days. A failed simulation is limited by the first-calf cohort when the first-calf cows reach their floor of 330 kilograms or when the bred heifers join the first-calf cohort at weaning below that floor. The floor of the mature cows is 385 kilograms. Over the five start histories, 4 other failed simulations end when the first-calf cows join the mature cohort at weaning at or below its floor. The animal rows give medians, over the complete simulations with a herd, of the ten-year mean of each simulation. The margin of the bred heifer is its weight when it joins the first-calf cohort minus 330 kilograms, and the margin of the mature cow is its weight on day 270 minus 385 kilograms. NPV is the ten-year net present value of the 60-hectare farm at a 5 percent discount rate, over the complete simulations with and without a herd. The supplementation rows pair each complete simulation at $\phi = 1$ with the complete simulation of the same start history, SR, paddock count N, and grazing period at $\phi = 0$, and give the medians of $\phi = 1$ minus $\phi = 0$. The heavy continuous history has no complete simulation at $\phi = 0$, so it has no matched pair (n/a). The longest run is the largest number of consecutive grazing days on which the occupied patch has no grazeable forage. The patch-use imbalance of a simulation is its largest ten-year patch-occupancy count divided by its smallest. The ratio divides the imbalance of a rotation with N paddocks by that of continuous grazing with the same start history, ϕ, and SR, over the matched complete simulations with a finite imbalance.

	Heavy continuous	Intensive rotational	Low-pressure continuous	Low-pressure rotational	Ungrazed
Completion, 5,600 simulations with a herd in each start history					
Complete simulations, *n* (%)	2,334 (41.7)	3,675 (65.6)	4,098 (73.2)	4,169 (74.4)	4,181 (74.7)
Complete at SR 0.2 (%)	80.0	100.0	100.0	100.0	100.0
Complete at SR 3.2 (%)	14.9	9.7	10.3	10.6	11.1
Complete at $\phi = 0$ (%)	0.0	52.4	62.5	64.5	65.1
Complete at $\phi = 1$ (%)	97.7	93.8	94.5	95.1	95.7
Rotations with gp 2 to 7 d (%)	45.0	70.8	78.5	79.5	79.7
Rotations with gp 50 to 75 d (%)	38.9	56.1	62.3	65.5	65.2
Failed simulations					
Failed simulations (*n*)	3,266	1,925	1,502	1,431	1,419
Ending in 2026 to 2029 (%)	99.9	92.4	89.1	87.2	87.8
Limited by the first-calf cohort (%)	71.3	94.8	97.9	98.4	98.7

Continued on the next page

Table E2, continued

	Heavy continuous	Intensive rotational	Low-pressure continuous	Low-pressure rotational	Ungrazed
First-calf cows at their floor (*n*)	1,408	1,495	1,202	1,163	1,176
Bred heifers join below that floor (*n*)	921	329	268	245	225
Mature cows at their floor (*n*)	937	100	30	23	17
Animals in complete simulations, medians					
Mature cow on day 270 (kg)	568.7	567.5	573.2	573.7	573.3
Calf at weaning (kg)	202.7	201.1	202.6	202.8	202.7
First-calf cow at first weaning (kg)	475.2	466.0	479.5	481.5	480.6
Mature cow BCS at calving	4.99	4.99	4.99	4.99	4.99
Mature cow conception rate	0.857	0.857	0.858	0.858	0.858
First-calf cow BCS at calving	4.39	4.31	4.50	4.53	4.52
First-calf cow conception rate	0.732	0.688	0.736	0.743	0.739
Bred heifer margin at joining (kg)	87.9	85.8	90.6	91.4	91.1
Mature cow margin on day 270 (kg)	183.7	182.5	188.2	188.7	188.3
NPV of complete simulations ($ thousand)					
Lowest	−243.0	−201.1	−179.6	−173.1	−178.1
Highest	81.7	87.7	106.9	109.2	108.9
Feed-gap supplementation at 2.2 LSU ha^{-1}, $\phi = 1$ minus $\phi = 0$, medians					
Matched pairs (*n*)	0	1	43	44	44
Hay on pasture cost ($ thousand)	n/a	24.3	15.3	13.2	11.9
First-calf cow BCS at calving	n/a	0.80	0.64	0.51	0.52
First-calf cow conception rate	n/a	0.128	0.133	0.117	0.119
NPV ($ thousand)	n/a	19.3	21.7	18.4	18.4
Longest run without grazeable forage, continuous grazing, median days					
Failed simulations	23.0	16.5	21.5	16.5	24.5
Complete simulations	20.0	1.5	3.0	3.5	2.5
Patch-use imbalance relative to continuous grazing, median ratio					
$N = 2$	0.46	0.45	0.35	0.37	0.39
$N = 3$	0.57	0.54	0.46	0.48	0.50
$N = 4$	0.35	0.29	0.22	0.23	0.24
$N = 6$	0.31	0.22	0.17	0.17	0.17
$N = 12$	0.16	0.10	0.07	0.07	0.07

E.1. Which simulations complete

Figure E2 shows the share of complete simulations by history, feed-gap supplementation, and stocking rate, and Table E2 gives the shares of each history. In every history, completion falls with stocking rate and rises with feed-gap supplementation. At 0.2 LSU per hectare, 80.0 percent of the heavy continuous simulations and all simulations of the other histories complete, and at 3.2 LSU per hectare, 14.9, 9.7, 10.3, 10.6, and 11.1 percent complete. At $\phi = 0$, 0.0, 52.4, 62.5, 64.5, and 65.1 percent of the simulations with a herd complete, and at $\phi = 1$, 97.7, 93.8, 94.5, 95.1, and 95.7 percent. In the ungrazed history without hay on pasture ($\phi = 0$), all 70 paddock designs complete up to 0.8 LSU per hectare, and at least one completes up to 2.4 LSU per hectare. The first designs to fail, at 1.0 LSU per hectare, are two paddocks with grazing periods of 50 and 55 days. With $\phi = 1$, all paddock designs complete up to 2.4 LSU per hectare, and at least one completes up to 3.2 LSU per hectare. The low-pressure continuous and low-pressure rotational histories complete about as often as the ungrazed history, in 4,098 and 4,169 of their simulations with a herd compared with 4,181, and the intensive rotational history completes less often, in 3,675 simulations. In every history, calendar-based rotations with short grazing periods complete more often than those with long ones. Of all rotations with a grazing period of 2 to 7 days, 45.0, 70.8, 78.5, 79.5, and 79.7 percent complete, against 38.9, 56.1, 62.3, 65.5, and 65.2 percent with a grazing period of 50 to 75 days, periods that only the designs with two paddocks and, at 50 days, with three paddocks use.

No simulation of the heavy continuous history completes at $\phi = 0$, even at 0.2 LSU per hectare, where all 70 simulations end in 2027 when the first-calf cows reach their floor. At $\phi = 0.25$, 120 simulations complete, 70 of them at 0.2 LSU per hectare, and the highest stocking rate with a complete simulation is 1.0, 1.8, 2.4, and 3.2 LSU per hectare at $\phi = 0.25$, 0.5, 0.75, and 1, respectively. This history starts with 84.1 Mg C per hectare of SOC, compared with 158.5 to 218.9 in the other histories, and its patches offer less forage. With twelve paddocks, a five-day grazing period, $\phi = 1$, and 0.8 LSU per hectare, the occupied patch holds a mean of 152 grams per square meter of grazeable forage on the grazing days of 2026, against 370 grams per square meter in the ungrazed history. Without hay at 0.8 LSU per hectare, every simulation of this history ends by 2027. In 34 of these 70 simulations, the bred heifers join the first-calf cohort at weaning in 2026 at 309 to 330 kilograms, at or below its floor, which ends the simulation. In 7 others, all with grazing periods of 10 to 50 days, the mature cows reach their floor between June and September 2026. In the 29 simulations that continue into 2027, the mature cows fall from 527 kilograms to 419 to 454 kilograms by weaning in 2026, and the bred heifers join the first-calf cohort at only 330 to 347 kilograms.

Failed simulations end early. The five histories have 3,266, 1,925, 1,502, 1,431, and 1,419 failed simulations, of which 99.9, 92.4, 89.1, 87.2, and 87.8 percent end in 2026 to 2029, and

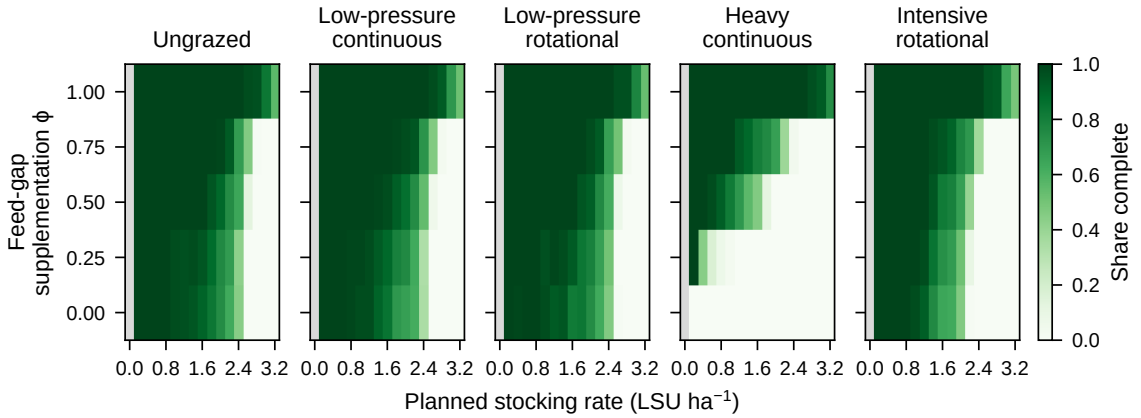


FIGURE E2. **Share of complete simulations.** Each tile is one planned stocking rate and level of feed-gap supplementation ϕ in one start history and covers its 70 paddock designs, and the shade gives the share of these simulations that complete the ten years. The gray column at a planned stocking rate of 0 marks the simulation without a herd, which has no paddock design or ϕ and completes in every history. A simulation ends early on the first day on which the mature cows reach 385 kilograms, the first-calf cows reach 330 kilograms, or a heifer cohort reaches 70 percent of its entry weight.

99.3 percent of the failed simulations of the heavy continuous history end in 2026 and 2027. The first-calf cohort is the limit in 71.3 percent of the failed simulations of the heavy continuous history and in 94.8 to 98.7 percent of those of the other histories, either because the first-calf cows reach their floor of 330 kilograms or because the bred heifers join the first-calf cohort at weaning below that floor. The mature cows reach their floor of 385 kilograms in 937 simulations of the heavy continuous history and in 17 to 100 simulations of each other history, and in 4 other simulations the first-calf cows join the mature cows at weaning and leave the mature cohort at or below this floor. No simulation ends at the heifer floor. On day 270 of the complete simulations of each history, the median bred heifer is 85.8 to 91.4 kilograms above the first-calf floor when it joins the cow herd, and the median mature cow is 182.5 to 188.7 kilograms above its own floor, so a shortage of forage reaches the first-calf floor first.

The selective-grazing rule contributes to these failures. Without hay on pasture, 62.5 percent of the continuous-grazing simulations of the ungrazed history complete, compared with 65.1 percent of its calendar-based-rotation simulations. The rule of equation (C1) scores a patch by its biomass and live-leaf share, but the animals eat only the forage above the 20-gram residual. A patch that is grazed to the residual and then grows new leaves has a high live-leaf share, and near the water attractor it can keep the highest score while it supplies no grazeable forage. In the continuous-grazing simulation of the ungrazed history at 1.4 LSU per hectare without hay, the herd stays on one patch for 30 days from day of year 239 of 2031. The patch has no grazeable forage on each of the last 10 days of

the stay, from day 259, and four of the twelve patches are not occupied in that year. On these days the mature and first-calf cows lose 3.0 and 2.5 kilograms per day, respectively, and the simulation ends on 5 April 2032, when the first-calf cows reach their floor. In the same history, continuous grazing without hay completes at 1.2 and 1.6 LSU per hectare. In four of the five histories, the longest run of grazing days without grazeable forage on the occupied patch separates failed from complete continuous grazing. Outside the heavy continuous history, its median is 16.5 to 24.5 days in the failed continuous-grazing simulations, which number 24 to 28 in each history, and 1.5 to 3.5 days in the complete ones, and 60.7 to 70.8 percent of the failed continuous-grazing simulations have a run of 10 days or more. Calendar-based rotation limits such runs because it moves the herd at the end of each grazing period. In the failed rotations of these four histories, the median longest run is 3 to 6 days at every paddock count, 3 or 4 days with grazing periods of 1 to 7 days, and 5 or 6 days with grazing periods of 10 days or more. In the heavy continuous history, every simulation has a run of 10 days or more, and the median longest run is 20 to 23 days in the failed and the complete simulations alike, so the run does not separate them. This rule therefore accounts for part of the completion difference between the grazing regimes (Appendix E.9).

E.2. Animal weights and condition

Figure E3 compares the simulated animals with the October weights of the Michigan State University research herd, converted to shrunk weight with a 4 percent shrink: 545.5 kilograms for 448 mature cows, 207.8 kilograms for 554 calves at weaning, and 462.4 kilograms for 143 two-year-old first-calf heifers. We chose the calf growth equation with the weaning weight of this herd as the target and fitted no parameter to these weights. Over the complete simulations with a herd of each history, the median mature cow weighs 567.5 to 573.7 kilograms on day 270, 4.0 to 5.2 percent above the research herd, and the median calf weighs 201.1 to 202.8 kilograms at weaning, 2.4 to 3.2 percent below it (Table E2). The median first-calf cow weighs 466.0 to 481.5 kilograms at its first weaning, 0.8 to 4.1 percent above the two-year-old heifers of the herd. In each of these comparisons, the intensive rotational history has the lightest animals. The 5th percentiles of the five histories are 541.1 to 552.5 kilograms for the mature cows, 193.2 to 194.8 kilograms for the calves, and 401.2 to 415.2 kilograms for the first-calf cows. The weights fall with stocking rate above 2.0 LSU per hectare.

Calf weight responds little to management, because milk is an obligation of the cow in the model and a shortage of feed lowers the weight of the cow rather than her milk (Appendix C.5). In each history, the median calf weight at weaning falls from 201.8 to 203.3 kilograms at 0.2 LSU per hectare to 193.6 to 195.1 kilograms at 3.2 LSU per hectare, so management changes calf revenue through the number of calves, which follows the

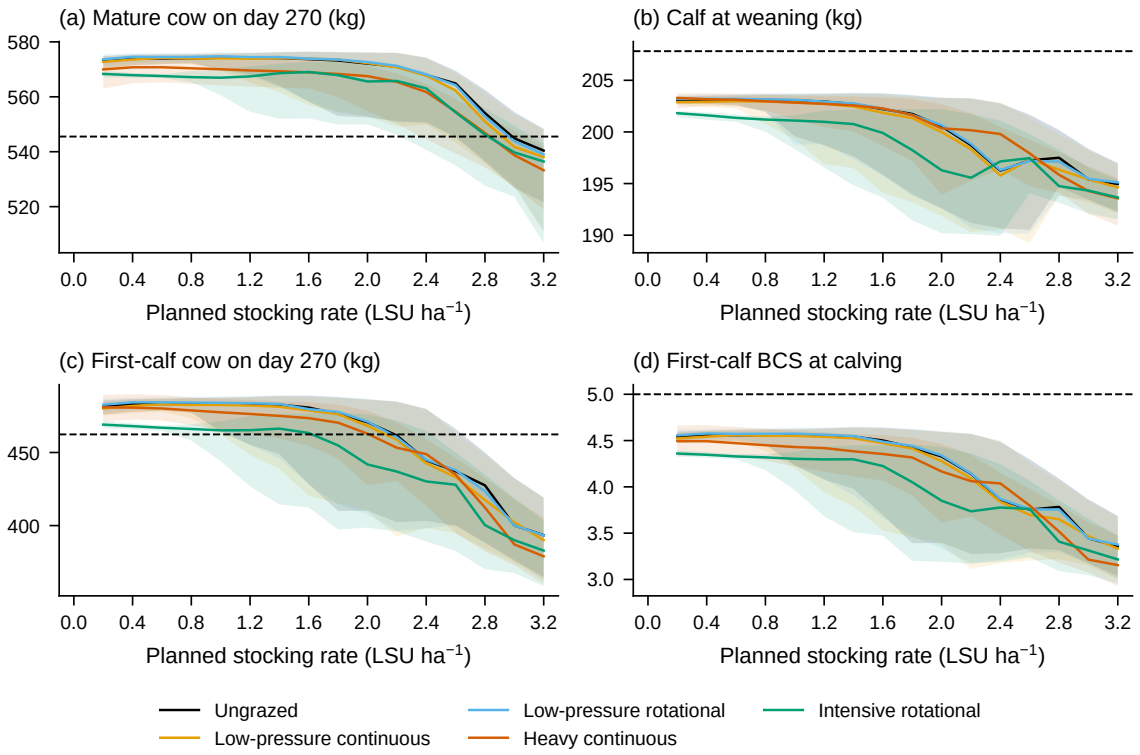


FIGURE E3. **Simulated animals compared with the Michigan State University research herd.** Lines are medians over the complete simulations of each start history at each planned stocking rate, bands span the 5th to 95th percentiles, and each simulation enters as its mean over the ten years. Panels (a) to (c) give shrunk body weights on day 270, the weaning date, and panel (c) gives the first-calf cows before they join the mature cows. The dashed lines give the October weights of the research herd with a 4 percent shrink, and in panel (c) the weight of its two-year-old first-calf heifers. Panel (d) gives the body condition score (BCS) of the first-calf cows on the calving date, day 75, and the dashed line marks BCS 5.

conception rate, and through a small change in calf weight. The winter ration returns the mature cows to BCS 5 before calving. In every history, their median score at calving is 4.99 (5th to 95th percentile 4.95 to 4.99), and their median conception rate is 0.857 or 0.858. The first-calf cows carry the effect of the grazing season into the next year. Their median score at calving is 4.39, 4.31, 4.50, 4.53, and 4.52, with 5th percentiles of 3.35 to 3.50, and their conception rate is 0.688 to 0.743 at the median and 0.507 to 0.534 at the 5th percentile. Between BCS 4 and 5, the conception rate of equation (B4) rises by 0.25 per score, compared with 0.07 between BCS 5 and 6. The mature conception rate varies only from 0.846 to 0.858 (5th to 95th percentile in any history), so the condition of the first-calf cows accounts for most of the difference in the calf crop between management systems. Panel (d) of Figure E3 shows that in every history the median first-calf score changes by at most 0.11 up to 1.4 LSU per hectare and then falls to 3.16 to 3.38 at 3.2 LSU per hectare.

E.3. Private-profit and soil-carbon trade-offs

Table E3 and Figure E4 report the highest-NPV management systems. Among the complete simulations, the ten-year NPV ranges from −$243,026 to $81,738 in the heavy continuous history, from −$201,123 to $87,707 in the intensive rotational history, from −$179,582 to $106,878 in the low-pressure continuous history, from −$173,085 to $109,205 in the low-pressure rotational history, and from −$178,138 to $108,906 in the ungrazed history. The low-pressure rotational history reaches $109,205, the highest NPV in the grid, with 2.2 LSU per hectare, two paddocks, a 15-day grazing period, and $\phi = 1$. The ungrazed history reaches $108,906 at the same stocking rate and feed-gap supplementation with two paddocks and a four-day grazing period. In the ungrazed and the two low-pressure histories, the highest NPV occurs at 2.2 LSU per hectare and $\phi = 1$. In the two histories with supplement, it occurs at 2.0 LSU per hectare and $\phi = 1$ and is $19,171 to $27,467 lower than in the low-pressure histories.

TABLE E3. Highest-NPV management system of each start history. SR is the planned stocking rate in LSU per hectare, NPV is in US dollars, and ΔSOC is the ten-year change of farm mean total SOC in Mg C ha^{-1}. Conception is the mean over the ten calving dates of both cow cohorts. A management system is nondominated when no other complete management system of the same history has both a higher NPV and a smaller SOC loss. The counts include the no-herd simulation at SR 0.

History	SR	ϕ	Paddock design	NPV	ΔSOC	Conception	Complete simulations	Nondominated systems
Ungrazed	2.2	1	$N = 2$, gp= 4	108,906	−27.38	0.842	4,182	194
Low-pressure continuous	2.2	1	$N = 2$, gp= 5	106,878	−16.55	0.843	4,099	187
Low-pressure rotational	2.2	1	$N = 2$, gp= 15	109,205	−24.48	0.841	4,170	186
Heavy continuous	2.0	1	$N = 2$, gp= 6	81,738	30.82	0.839	2,335	110
Intensive rotational	2.0	1	$N = 2$, gp= 5	87,707	−3.38	0.837	3,676	73

Up to the peak, NPV rises with stocking rate because calf and cull sales grow faster than costs (Figure E4a). Without a herd, the farm has no sales and still bears the fence cost, for an NPV of −$10,344. Winter feeding dominates the costs. In the highest-NPV management system, the undiscounted ten-year sales are $812,096 for calves and $186,449 for cull cows, and the feed cost is $535,795, of which winter hay accounts for $483,671, corn for $43,243, and hay on pasture for $8,881. Above the peak, NPV falls because fewer simulations complete, the first-calf condition and the conception rate decline, and the simulations that complete need more hay on pasture (Section E.4).

Panel (b) of Figure E4 shows that in every history the SOC change of the highest-NPV management system falls with stocking rate, from the system without a herd up to 2.6 LSU per hectare. Within each history, the least-loss management system keeps no herd. Among the systems with a herd, the one with the least SOC loss uses 0.2 LSU per hectare in every

history, with twelve paddocks in four histories and two paddocks with a 75-day grazing period in the intensive rotational history. Panel (c) shows the profit and SOC frontier of each history, which contains 73 to 194 systems and runs from the system without a herd to the highest-NPV one. In the four histories in which the system with a herd and the least SOC loss uses twelve paddocks, the system without a herd has both a higher NPV and a smaller SOC loss and so dominates it. In these histories, the frontier system with a herd and the least SOC loss uses six paddocks at 0.2 LSU per hectare, with an NPV of −$9,732 to −$8,635. In the ungrazed history, the frontier runs from an NPV of −$10,344 and an SOC change of 1.26 Mg C per hectare (no herd) to $108,906 and −27.38 Mg C per hectare. Its highest-NPV end gains $119,250 of NPV and loses 28.64 Mg C per hectare more SOC than its least-loss end, which on the 60-hectare farm amounts to 1,718 Mg C, or $69 of NPV per Mg C over the whole frontier.

E.4. Feed-gap supplementation

Figure E5 shows the ungrazed history at the five levels of feed-gap supplementation. Hay on pasture raises the highest NPV of every history, and every history reaches its highest NPV at $\phi = 1$. The highest ungrazed NPV is $93,659 at $\phi = 0$ and $108,906 at $\phi = 1$, 16.3 percent higher. In the four histories that complete without hay on pasture, the best management system at $\phi = 1$ earns 15 to 28 percent more than the best one at $\phi = 0$, and most of the gain comes from the first-calf cows. We pair each management system of these four histories at $\phi = 1$ with the system of the same history, stocking rate, paddock count, and grazing period at $\phi = 0$. At low stocking rates there is little feed gap to fill, and the median NPV difference is zero up to 0.6 LSU per hectare in the intensive rotational history, up to 0.8 in the two low-pressure histories, and up to 1.0 in the ungrazed history. At 1.0 LSU per hectare, it is $239, $109, $22, and $0 in these four histories. From 1.6 to 2.4 LSU per hectare, $\phi = 1$ has the higher NPV in 91.2 to 100 percent of the pairs of each history and stocking rate. Its median gain is $3,911 to $8,117 at 1.6 LSU per hectare and $22,002 to $27,638 at 2.4 LSU per hectare, where the intensive rotational history has no pair. At 2.2 LSU per hectare, the two low-pressure histories and the ungrazed history have 43 or 44 pairs each, and the intensive rotational history has one. In the first three, the median pair at $\phi = 1$ pays $11,908 to $15,337 for hay on pasture over ten years, raises the first-calf score at calving by 0.51 to 0.64 and the first-calf conception rate by 0.117 to 0.133, and has $61,176 to $72,758 higher calf sales and $21,277 to $25,702 lower cull sales. Its median NPV gain is $18,385 to $21,725. The one pair of the intensive rotational history pays $24,262 for hay and gains $19,329. Calf revenue is the number of calves sold times their weight and price. In each of the four histories, 62 or 63 percent of the added calf revenue comes from 31.9 to 43.1 more calves sold over ten years, as the farm culls 16.7 to 22.9 fewer cows and its herd holds 1.5 to 2.0 more cows, and the rest comes from calves that wean 5.5 to 7.4

kilograms heavier. Without hay on pasture, the first-calf cows calve below BCS 5, where the conception rate rises steeply with condition (Appendix E.2), so a small amount of hay keeps more cows in the herd. At 2.6 LSU per hectare, no simulation of the ungrazed history completes without hay, and the median complete simulation at $\phi = 1$ feeds 36.2 tonnes of hay per year on pasture. The heavy continuous history depends on hay: its highest NPV is \$19,751 at $\phi = 0.25$, \$70,102 at $\phi = 0.5$, \$73,949 at $\phi = 0.75$, and \$81,738 at $\phi = 1$.

E.5. When calendar-based rotation pays

Appendix Table E5 compares the highest-NPV continuous grazing with the highest-NPV calendar-based rotation at each stocking rate of the ungrazed history, each taken over the five levels of feed-gap supplementation, and gives the management system without a herd at a stocking rate of 0. Across the five histories at 0.2 to 1.4 LSU per hectare, the NPV of the best rotation minus that of the best continuous grazing ranges from −\$4,390 to \$4,800, and continuous grazing has the higher NPV in 30 of these 35 pairs of history and stocking rate. Continuous grazing has the higher NPV in all 15 pairs at 0.2 to 0.6 LSU per hectare, by \$951 to \$3,602. The best rotation has the higher NPV in all 25 pairs from 1.6 to 2.4 LSU per hectare, by \$318 to \$21,854, and from 2.6 LSU per hectare upward in all 16 pairs in which a continuous-grazing simulation completes, by \$24,197 to \$124,760. At 3.2 LSU per hectare, continuous grazing completes only in the heavy continuous history.

In the ungrazed history, three stocking rates show the mechanism. In this paragraph, costs and sales are undiscounted ten-year totals, and each lead is a difference in NPV, which discounts them and adds the terminal herd value. At 1.0 LSU per hectare, the two regimes have almost the same sales, and the rotation pays \$2,211 more for fencing and \$933 more for labor, so continuous grazing leads by \$2,867. At 2.2 LSU per hectare, the best management systems of both regimes use $\phi = 1$. The rotation, with two paddocks and a four-day grazing period, feeds 3.0 tonnes of hay per year on pasture compared with 8.1 tonnes under continuous grazing, a saving of \$12,312 over ten years. It raises the first-calf score at calving from 4.52 to 4.58, its calf sales are \$5,877 higher and its cull sales \$2,108 lower, and after \$2,211 more for fencing and \$5,379 more for labor it leads by \$7,100. At 2.6 LSU per hectare, continuous grazing completes only with $\phi = 1$, and its best system feeds 36.6 tonnes of hay per year on pasture at a ten-year cost of \$88,828. The best rotation, four paddocks with a seven-day grazing period, feeds 21.1 tonnes per year at a cost of \$51,202, raises the first-calf score at calving from 4.12 to 4.36, has \$27,068 higher calf sales, and leads by \$31,848. The best rotation of the ungrazed history also loses less SOC than the best continuous grazing at every stocking rate at which both regimes complete, by 0.11 to 2.48 Mg C per hectare up to 2.4 LSU per hectare and by 5.90 to 13.75 Mg C per hectare from 2.6 LSU per hectare upward.

E.6. How calendar-based rotation redistributes patch use

Figure E6 quantifies one spatial pathway of calendar-based rotation. On each grazing-season day the whole herd occupies one 5-hectare patch, so every complete simulation with a herd allocates 1,510 patch-occupancy days among the 12 patches. Patch-use imbalance is the largest patch count divided by the smallest, and a ratio of one means that all patches are occupied equally often. The statistic counts days of occupancy and carries no information on cattle-days or on the grazed area within a patch.

The paddock counts use different grazing periods: two paddocks use 1 to 75 days, three use 1 to 50 days, four use 1 to 35 days, six use 1 to 25 days, and twelve use 1 to 10 days. The share of complete simulations also differs by stocking rate, so a summary over all complete simulations would compare different sets of management systems. Figure E6 instead holds the history, the feed-gap supplementation, and the grazing period fixed, and each point is one simulation. Under continuous grazing, the stocking rate sets how unevenly the herd uses the patches. In the figure, continuous grazing leaves at least one patch unoccupied for the ten years at 0.2 to 0.6 LSU per hectare, which makes its imbalance infinite, and its imbalance falls from 68.2 at 0.8 LSU per hectare to 2.52 at 2.2 LSU per hectare. The intake response in the score of equation (C1) rises with the biomass of the patch, so at a low stocking rate a grazed patch keeps a high score and the herd stays on the preferred patches, while at a high stocking rate the herd depletes each patch faster and moves on. From 0.6 to 2.4 LSU per hectare, every rotation is more even than continuous grazing, and more paddocks give more even use: at 0.8 LSU per hectare, the imbalance is 36.5 with two paddocks, 19.4 with four, 8.69 with six, and 1.05 with twelve. Three paddocks are less even than two in this figure, with an imbalance of 54.4 at 0.8 LSU per hectare. At 0.4 LSU per hectare, the imbalance reaches 370 with two paddocks and 310 with four, and twelve paddocks give 1.05 at every stocking rate. From 2.6 LSU per hectare upward, every regime stays below 2.

Matched comparisons within each history show the same pattern. Relative to continuous grazing with the same history, feed-gap supplementation, and stocking rate, the median imbalance ratio of the five histories is 0.35 to 0.46 with two paddocks, 0.46 to 0.57 with three, 0.22 to 0.35 with four, 0.17 to 0.31 with six, and 0.07 to 0.16 with twelve, and the heavy continuous history has the highest ratio at every paddock count (Table E2). The rotation is more even in 84.4 to 100 percent of the pairs of each history and paddock count. Feed-gap supplementation has almost no effect: between $\phi = 1$ and $\phi = 0$ with all else equal, the median ratio is 1.00 in each of the four histories with complete simulations at $\phi = 0$, and in each of them 80 percent of the pairs lie between 0.82 and 1.02. A 7-day grazing period gives a median of 1.33 to 1.45 times the imbalance of a 1-day grazing period. The ranking of two and three paddocks depends on the grazing period and the stocking rate: three paddocks are more even than two in 17.6 to 28.4 percent of the matched pairs

of each history, and in 33.5 to 65.9 percent of the pairs at 1.4 LSU per hectare or less with grazing periods of 10 to 50 days. The comparisons exclude the complete simulations in which the herd leaves at least one patch unoccupied for the ten years, so that the ratio is infinite, which number 378, 225, 517, 514, and 501. All but three of them are at 0.2 to 0.6 LSU per hectare, where they make up 51.0, 21.4, 49.0, 49.0, and 47.6 percent of the complete simulations, and the other three, two in the low-pressure continuous history and one in the ungrazed history, use three paddocks with a 30-day grazing period at 0.8 LSU per hectare. No simulation with twelve paddocks leaves a patch unoccupied, because the calendar moves the herd to every patch. The outcomes combine forage, SOC, animal, and cost channels, so the figure cannot isolate the effect of the redistribution, and the simulated cattle locations are not validated against field observations.

E.7. Payment designs in detail

Table E4 and Figure E7 report the payment experiment of Section 3.4 for each farm. Under a two-sided outcome payment, the baseline determines which farms receive a payment but does not affect the management that a participant chooses. Under a practice payment or a calf premium, the qualifying rule determines whether the payment changes management.

Against the own baseline, the switch price of the outcome payment is $14 per Mg C for the heavy continuous farm, $44 for the intensive rotational farm, $36 for the low-pressure continuous farm, $26 for the low-pressure rotational farm, and $38 for the ungrazed farm. At $50 per Mg C, the heavy continuous farm moves from 2.0 to 1.6 LSU per hectare, the intensive rotational farm moves from 2.0 to 1.8 LSU per hectare, and the two low-pressure farms and the ungrazed farm move from 2.2 to 2.0 LSU per hectare with four paddocks and a three-day grazing period. The heavy continuous farm then adds 405 Mg C for a payment of $18,305, the intensive rotational farm 154 Mg C for $6,694, the low-pressure continuous farm 282 Mg C for $12,386, the low-pressure rotational farm 347 Mg C for $15,206, and the ungrazed farm 317 Mg C for $13,570. Section 5.4 gives the choice of each farm at $128 per Mg C and the price at which each farm keeps no herd. The profit optimum earns nothing under this baseline, so no payment is windfall.

The start-stock baselines pay mainly the heavy continuous farm, whose SOC rises in every complete simulation (Section 5). With annual gains, the heavy continuous farm receives $46,256 at $28 per Mg C for a move from 2.0 to 1.8 LSU per hectare that adds 184 Mg C, and 90 percent of its payment is windfall. At the same price, the intensive rotational and low-pressure continuous farms receive $3,046 and $178, respectively, for no change. At $128 per Mg C, the heavy continuous farm receives $329,432 for 1,181 Mg C with 58 percent windfall, the intensive rotational farm $21,681 for 129 Mg C with 64 percent windfall, and the low-pressure continuous farm $1,879 for 13 Mg C with 43 percent windfall. The cost per additional Mg C is $251 on the heavy continuous farm at $28 per Mg C, and at $128

per Mg C it is $279 on the heavy continuous farm, $168 on the intensive rotational farm, and $143 on the low-pressure continuous farm. The intensive rotational farm changes under the annual-gain rule from $110 per Mg C and joins the net rule only from $148 per Mg C, with no herd. The two low-pressure farms and the ungrazed farm lose SOC in their profit optima, and under the annual-gain rule they do not change below $106 per Mg C (low-pressure continuous), $324 per Mg C (low-pressure rotational), or $356 per Mg C (ungrazed). At the last two prices, the low-pressure rotational and ungrazed farms move directly to no herd. Under the net rule, these two farms do not join at any price up to $400 per Mg C, and the low-pressure continuous farm joins from $282 per Mg C with no herd. Within the observed prices, only the heavy continuous farm joins the net rule. A farm that joins the net rule chooses the same management system as under the own baseline at every rate, as equation (B24) implies, because the two baselines differ by a lump sum.

A practice payment for rotation alone changes no farm up to $300 per hectare per year, because every profit optimum already uses two paddocks. At the Michigan EQIP rates, it transfers $37,240 to $85,539 to each farm and adds no SOC. With the cap of 2.0 LSU per hectare, the heavy continuous and intensive rotational farms already qualify, so their whole payment is windfall. The low-pressure continuous, low-pressure rotational, and ungrazed farms move from 2.2 to 2.0 LSU per hectare at $16 per hectare per year ($14 for the low-pressure rotational farm) and add 201, 202, and 220 Mg C, respectively. At the EQIP Pasture Standard rate, this costs $185 per additional Mg C on each low-pressure farm and $169 on the ungrazed farm. With the cap of 1.6 LSU per hectare, all five farms move to 1.6 LSU per hectare with no windfall (Section 5.4). Above the switch levels, a higher rate adds only transfer.

A calf premium for rotation alone is a pure transfer of $1,779 to $1,991 per farm at 0.3 percent of the calf price and raises the stocking rate at higher shares. At switch levels of 1.65 to 2.85 percent, the two low-pressure farms and the ungrazed farm move from 2.2 to 2.4 LSU per hectare, because the premium rewards more calves, and they lose SOC (Section 5.4). With the cap of 2.0 LSU per hectare, the premium moves the low-pressure continuous, low-pressure rotational, and ungrazed farms from 1.2, 1.15, and 1.25 percent of the calf price, respectively, to the same management systems as the practice payment with this cap. At 3.0 percent, the buyer then pays the heavy continuous and intensive rotational farms $17,881 and $17,788 for no change, and it pays $90 per additional Mg C on each low-pressure farm and $82 on the ungrazed farm. With the cap of 1.6 LSU per hectare, the switch level is 2.2 percent for the heavy continuous farm, 2.95 percent for the intensive rotational farm, and 5.55 to 5.85 percent for the other three, so within the observed band the premium moves only the heavy continuous and intensive rotational farms. The outcome-verified premium changes no farm up to 10 percent: the profit optimum qualifies by definition, and a move to a qualifying management system with more calf revenue

would need a premium of at least 18 percent of the calf price.

Appendix Table E6 adds the threshold design of Rouhi Rad et al. (2026). A uniform threshold applies one rule to all five farms, so the table compares program totals, and its cost per Mg C is the total payment to the five farms divided by their total additional SOC. At $50 per Mg C, a threshold at the no-payment change of each farm buys 1,130 Mg C over the five farms for $36,420 ($32 per Mg C), and the best uniform threshold for a similar target, at $150 per Mg C, buys 1,047 Mg C for $62,504 ($60 per Mg C). For about 1,600 Mg C, the farm-specific threshold costs $48 per Mg C and the best uniform threshold $223 per Mg C. At $100 per Mg C, the farm-specific threshold buys 2,136 Mg C over the five farms for $137,707.

TABLE E4. **Each payment design at the lowest and highest observed levels, by farm.** Each start history is one 60-hectare farm that chooses the complete management system with the highest NPV plus the present value of the payment. SOC is the ten-year SOC change of the chosen system minus that of the profit optimum a^0, in Mg C per farm. Paid is the present value of the payment over the ten-year contract, in $ thousand per farm. "out" means that the farm does not join, so it keeps a^0 and is paid nothing. Outcome levels are the 2024 mean credit price and the highest farmer-program price, practice levels the Michigan EQIP rates, and premium levels the observed auction premiums (Section 3.4). The buyer of the calves pays the premium.

		Heavy continuous		Intensive rotational		Low-pressure continuous		Low-pressure rotational		Ungrazed	
	Level	SOC	Paid	SOC	Paid	SOC	Paid	SOC	Paid	SOC	Paid
Outcome payment ($ per Mg C)											
Own baseline	28	184	4.8	0	0.0	0	0.0	120	3.1	0	0.0
	128	1,181	139.7	567	61.9	1,489	164.7	1,660	183.5	1,718	189.3
Start stock, annual gains	28	184	46.3	0	3.0	0	0.2	0	0.0	0	0.0
	128	1,181	329.4	129	21.7	13	1.9	0	0.0	0	0.0
Start stock, net change	28	184	46.3	out	out	out	out	out	out	out	out
	128	1,181	329.4	out	out	out	out	out	out	out	out
Practice payment ($ ha^{-1} yr^{-1})											
Rotation	76.55	0	37.2	0	37.2	0	37.2	0	37.2	0	37.2
	175.84	0	85.5	0	85.5	0	85.5	0	85.5	0	85.5
Rotation, SR ≤ 2.0	76.55	0	37.2	0	37.2	201	37.2	202	37.2	220	37.2
	175.84	0	85.5	0	85.5	201	85.5	202	85.5	220	85.5
Rotation, SR ≤ 1.6	76.55	405	37.2	251	37.2	497	37.2	566	37.2	580	37.2
	175.84	405	85.5	251	85.5	497	85.5	566	85.5	580	85.5
Calf premium (% of the calf price)											
Rotation	0.3	0	1.8	0	1.8	0	2.0	0	2.0	0	2.0
	3.0	0	17.9	0	17.8	−317	21.6	−205	21.5	−264	21.6
Rotation, SR ≤ 2.0	0.3	0	1.8	0	1.8	0	0.0	0	0.0	0	0.0
	3.0	0	17.9	0	17.8	201	18.1	202	18.1	220	18.1
Rotation, SR ≤ 1.6	0.3	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
	3.0	405	14.4	251	14.2	0	0.0	0	0.0	0	0.0
SOC not below a^0	0.3	0	1.8	0	1.8	0	2.0	0	2.0	0	2.0
	3.0	0	17.9	0	17.8	0	19.9	0	19.8	0	19.9

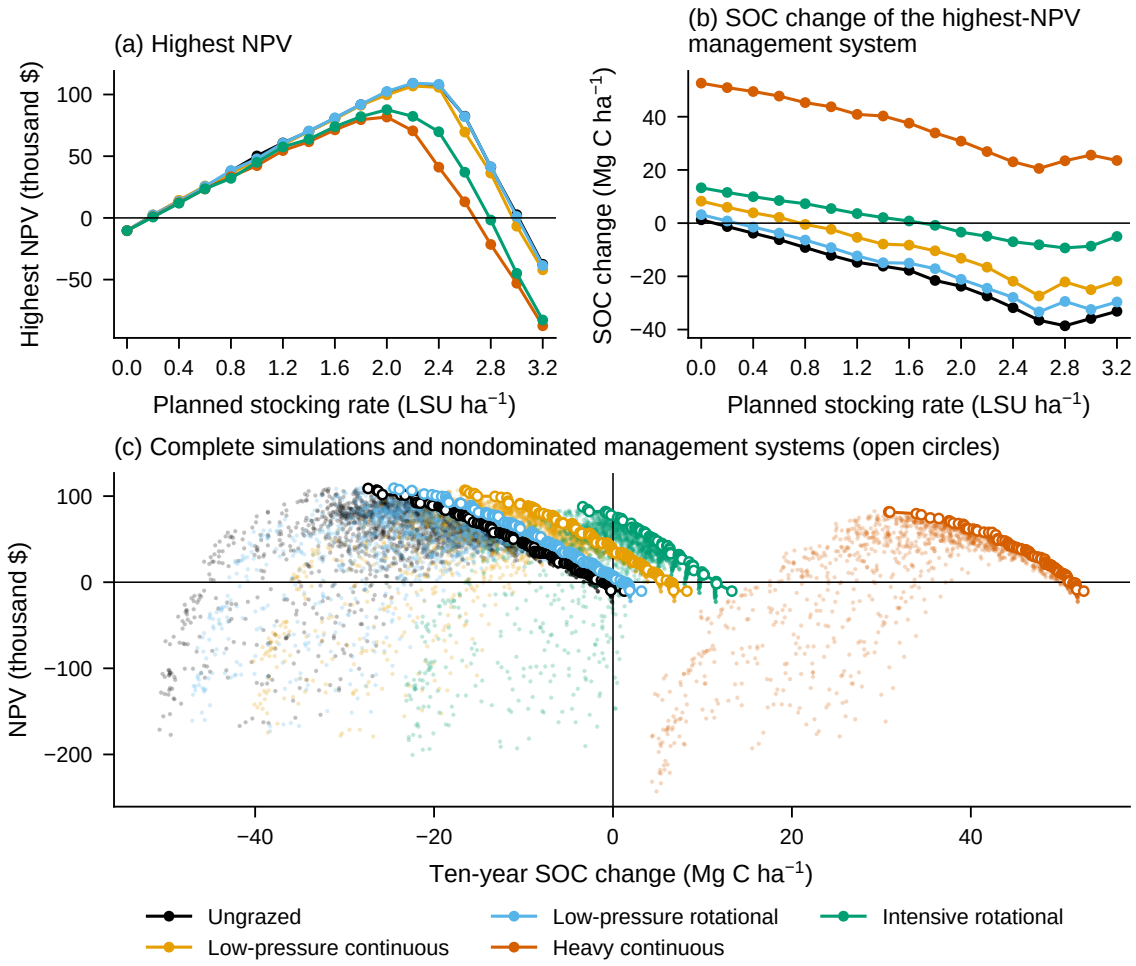


FIGURE E4. **Private NPV and soil-carbon change by start history.** Panel (a) gives the highest ten-year NPV among the complete simulations of each history at each planned stocking rate, over all paddock counts, grazing periods, and levels of feed-gap supplementation. Panel (b) gives the ten-year change in farm mean total SOC of the same management systems. At a planned stocking rate of 0, panels (a) and (b) give the simulation without a herd. Panel (c) plots every complete simulation, and open circles and lines mark the profit and SOC frontier of each history. The last open circle of each history, at an NPV of −$10.3 thousand and the largest SOC change of the history, is the simulation without a herd. The histories start from different SOC stocks, so SOC is compared within a history.

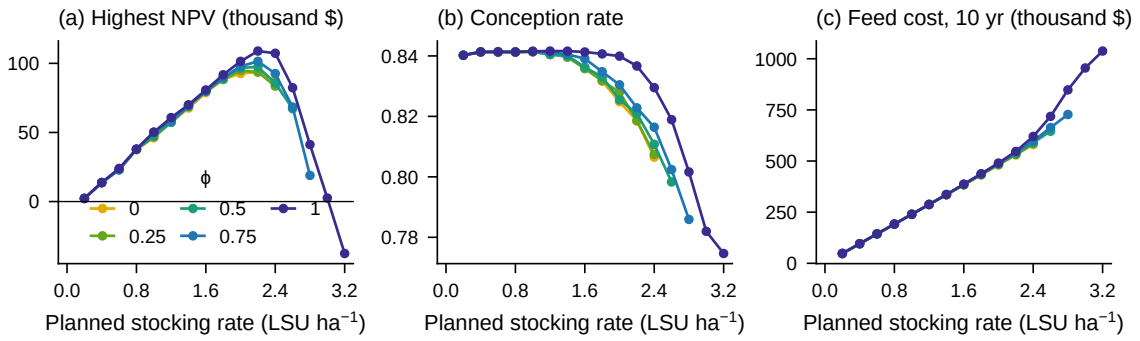


FIGURE E5. **Effect of feed-gap supplementation in the ungrazed start history.** Panel (a) gives the highest ten-year NPV among the complete simulations at each planned stocking rate and feed-gap supplementation ϕ. Panels (b) and (c) give the medians over the complete simulations of the mean conception rate and of the undiscounted ten-year feed cost (winter hay, hay on pasture, and corn). The lines start at 0.2 LSU per hectare because the simulation without a herd has no ϕ, and each line ends at the highest stocking rate with a complete simulation.

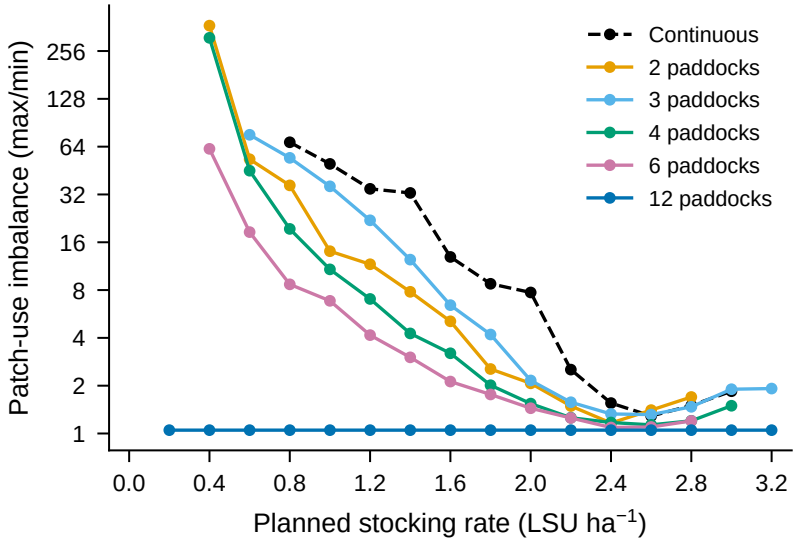


FIGURE E6. **Patch-use imbalance by stocking rate and paddock count.** Each point is one complete simulation of the ungrazed history with $\phi = 1$, and every rotation uses a 3-day grazing period, which all paddock counts share. The outcome is the ratio of the largest to the smallest ten-year patch-occupancy count, where one denotes equal use, on a base-2 logarithmic vertical axis. A point is missing when the simulation ends before ten years or when the simulation completes but the herd leaves at least one patch unoccupied, so that the ratio is infinite. In the ungrazed history, 501 complete simulations have an infinite ratio, 500 of them at 0.2 to 0.6 LSU per hectare. In this figure, they are continuous grazing at 0.2 to 0.6 LSU per hectare, three paddocks at 0.2 and 0.4 LSU per hectare, and two, four, and six paddocks at 0.2 LSU per hectare. Under calendar-based rotation the calendar sets the active management paddock, and under both regimes the selective-grazing rule sets the occupied patch within the eligible area.

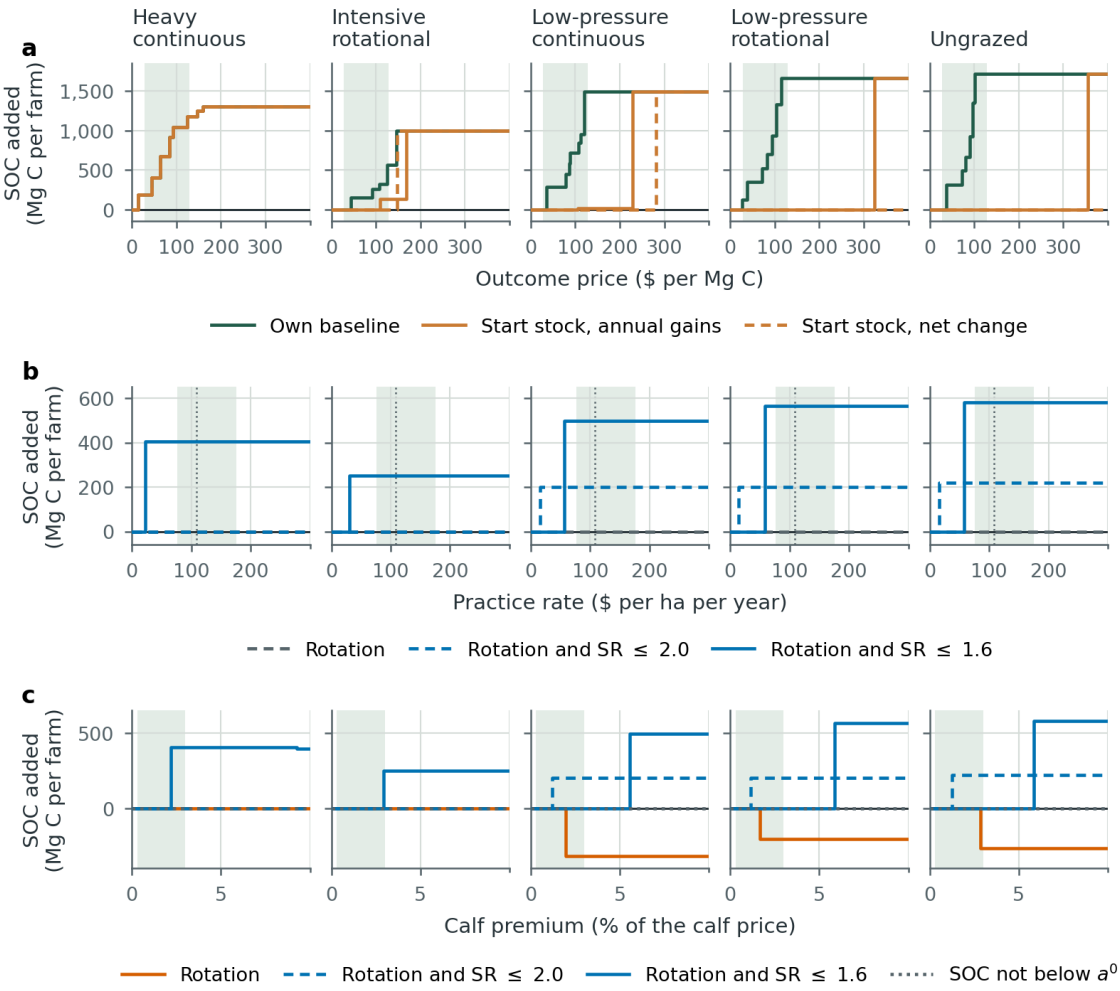


FIGURE E7. **Response of each farm to every payment design.** Each column is one start history, that is, one 60-hectare farm that chooses the complete management system with the highest NPV plus the present value of the payment. Lines give the additional SOC of the farm, the ten-year SOC change of the chosen management system minus that of the farm's profit optimum, times 60 hectares. (a) Outcome payments against the own baseline and against the start stock, with annual gains or with the net change. (b) Practice payments for rotation alone and for rotation with a cap of 2.0 or 1.6 LSU per hectare. (c) Calf premiums with the same three rules and with the outcome-verified rule, a ten-year SOC change not below that of a^0. Shaded bands mark the observed levels: \$28 to \$128 per Mg C in panel (a), the Michigan EQIP rates of \$76.55 to \$175.84 per hectare per year in panel (b), and 0.3 to 3.0 percent of the calf price in panel (c). The dotted line in panel (b) marks the CSP enhancement E528R rate of \$108.25 per hectare per year.

E.8. Comparison tables

Table E5 compares continuous grazing (CG) with the highest-NPV calendar-based rotation at every stocking rate of the ungrazed start history. Each regime takes its highest NPV over the five levels of feed-gap supplementation, and the table gives the feed-gap supplementation ϕ of that management system and, for the rotation, the paddock count N and the grazing period gp in days. The difference column is the rotation NPV minus the CG NPV, which is negative when continuous grazing has the higher NPV.

TABLE E5. Highest-NPV continuous grazing (CG) and highest-NPV calendar-based rotation by planned stocking rate, ungrazed start history, each over the five levels of feed-gap supplementation ϕ. SR is the planned stocking rate in LSU per hectare, NPV is in US dollars, and ΔSOC is the ten-year change of farm mean total SOC in Mg C ha^{-1}. The difference is the rotation NPV minus the CG NPV. n/a marks a stocking rate at which no continuous-grazing simulation completes. At SR 0 the farm keeps no herd, and the CG columns give the NPV and ΔSOC of the no-herd simulation.

	Continuous grazing (CG)			Highest-NPV rotation				
SR	ϕ	NPV	ΔSOC	Paddock design	ϕ	NPV	ΔSOC	Difference
0 (no herd)		−10,344	1.26					
0.2	0	2,256	−1.29	N = 2, gp= 60	0	−349	−1.05	−2,605
0.4	0	13,718	−3.78	N = 2, gp= 50	0	11,254	−3.46	−2,464
0.6	1	23,991	−6.20	N = 2, gp= 65	0	23,040	−6.09	−951
0.8	1	37,949	−9.15	N = 2, gp= 30	0.75	35,256	−8.75	−2,693
1.0	0.75	50,201	−12.12	N = 2, gp= 20	1	47,335	−11.43	−2,867
1.2	1	60,752	−14.72	N = 2, gp= 30	0.75	58,023	−13.92	−2,729
1.4	1	65,431	−16.64	N = 2, gp= 10	1	70,011	−16.16	4,581
1.6	0.75	79,248	−19.51	N = 2, gp= 5	1	80,750	−17.72	1,502
1.8	1	91,423	−22.44	N = 2, gp= 10	1	91,741	−21.54	318
2.0	1	94,006	−25.32	N = 2, gp= 5	1	101,444	−23.71	7,438
2.2	1	101,806	−28.49	N = 2, gp= 4	1	108,906	−27.38	7,100
2.4	1	93,462	−34.26	N = 2, gp= 4	1	107,310	−31.78	13,847
2.6	1	50,523	−42.42	N = 4, gp= 7	1	82,370	−36.52	31,848
2.8	1	−19,883	−47.13	N = 6, gp= 6	1	41,228	−38.58	61,111
3.0	1	−108,622	−49.59	N = 12, gp= 4	1	2,556	−35.84	111,178
3.2	n/a	n/a	n/a	N = 12, gp= 6	1	−37,585	−33.13	n/a

Table E6 reports the threshold variant of the outcome payment (Appendix B.13). The payment is $pA\max(0, \Delta\text{SOC} - \theta)$ for the ten-year SOC change and is paid at the end of 2035. We evaluate uniform thresholds from −30 to 40 Mg C per hectare in steps of 2.5, thresholds at the no-payment change of each farm plus $k \in \{-5, -2.5, 0, 2.5, 5\}$ Mg C per hectare, and the rates \$10, \$25, \$50, \$75, \$100, \$150, \$200, \$300, and \$400 per Mg C. For each band of total additional SOC over the five farms, the table gives the design of each kind with the smallest payment. For small targets, a high uniform threshold reaches only the

heavy continuous farm and costs as little as the farm-specific threshold. For larger targets, a uniform threshold either pays the heavy continuous farm for SOC that it gains without a change or sets a bar that the farms with SOC loss cannot reach.

TABLE E6. Outcome designs with a threshold on the ten-year SOC change, paid at the end of the contract: the design of each kind with the smallest payment for each band of total additional SOC over the five farms. A uniform threshold θ applies to every farm, and a history threshold is the no-payment SOC change of the farm plus k, both in Mg C ha^{-1}. Payment is the present value over the five farms in US dollars, and cost is the payment per additional Mg C.

Threshold	θ or k	Rate ($/Mg C)	Additional SOC (Mg C)	Payment	Cost per Mg C
History	0.0	25	184	2,973	16
Uniform	32.5	50	184	2,705	15
History	5.0	100	694	25,383	37
Uniform	35.0	100	694	28,570	41
History	0.0	50	1,130	36,420	32
Uniform	37.5	150	1,047	62,504	60
History	0.0	75	1,599	77,296	48
Uniform	−27.5	75	1,599	357,304	223
History	5.0	150	4,934	361,063	73
Uniform	−17.5	150	3,541	669,615	189

E.9. Limits of the animal, grazing, and payment models

The animal-demand rules and the herd lifecycle of the model are not validated. The simulated weights are compared with a Michigan research herd but not calibrated to it: at weaning, the median mature cows of the five start histories are 4.0 to 5.2 percent heavier and the calves 2.4 to 3.2 percent lighter than the herd, and the first-calf cows are 0.8 to 4.1 percent heavier than its two-year-old heifers. The model calves are born on day 75 and weaned at 195 days of age, about 20 days older than the calves of the herd at their October weighing, so they gain about 0.84 kilograms of shrunk weight per day, compared with 0.97 kilograms per day in the herd. Every management system feeds the same winter program, which returns the mature cows to BCS 5 by calving, so the management comparison covers only the grazing season. Gain follows the energy supply, and a protein shortage does not limit it. Milk is an obligation of the cow, so the calf weight at weaning responds little to management and a shortage of feed falls on the cows.

In the grazing model, the selective-grazing rule scores a patch by its biomass and live-leaf share and not by the forage above the grazing residual, so the herd can stay on a grazed-down patch with no grazeable forage while other patches have forage. The rule contributes to the lower completion of continuous grazing and of long grazing periods, and a rule that scores the grazeable forage, or that moves the herd when the occupied

patch has none, would show how much of the comparison between the grazing regimes depends on it. The crude protein of the plant forage eaten falls outside the training band of the surrogate on 22.6 percent of the grazing days and is clipped into that band. The patch-use measure counts only the days on which each patch is the location of the herd. It omits the number of animals on that patch, the area grazed within it, and behavioral differences among cattle, and the model moves the herd between management paddocks by the fixed calendar, not in response to forage.

In the payment experiment, the own baseline assumes that the program knows the no-payment management of each farm, which makes it a full-information benchmark. A real program must estimate the no-payment path with a model, and that estimate carries the error of the model. The profit optima lie at 2.0 to 2.2 LSU per hectare. A replay in MEMS of 1,095 farm simulations from an earlier specification of the experiment, with the same daily actions, shows that the surrogate overpredicts the forage of the grazed patch and understates the SOC loss, and that the error grows with the stocking rate. At 2.0 to 2.2 LSU per hectare, the farm SOC of the surrogate at the end of 2035 exceeds the MEMS value by a median of 3.9 Mg C per hectare. The median difference is 2.3 Mg C per hectare at 1.6 to 1.8 LSU per hectare, where the cap of 1.6 lies, and −0.3 Mg C per hectare at 0.8 to 1.0. We conjecture that the surrogate therefore understates the additional SOC of a move to a lower stocking rate and overstates its private cost, which would make the switch levels too high. The replay covers 0.8 to 2.8 LSU per hectare, so it does not test the lower stocking rates and the no-herd system that the outcome payment chooses at high prices.

The experiment omits transaction, measurement, and verification costs, the demand for qualifying calves, and SOC loss after the contract ends. The farm without a herd has no land rent or other use of its land, and because no management system buys or sells the start herd, the no-herd system does not receive the sale value of that herd. We expect a land rent or a herd sale to lower the price at which a farm keeps no herd. The model also represents only one farm, so if other farms raise the calves that a farm without a herd no longer supplies, their SOC change and emissions are a leakage that the experiment does not count.